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TELEGRAPHIC MEASUREMENT

OF

DIFFERENCES OF LONGITUDE

BY

OFFICERS OF THE U. S. NAVY

IN

1878 and 1879.

GREENWICH—LISBON—FUNCHAL—PORTO GRANDE—PERNAMBUCO—BAHIA—
RIO DE JANEIRO—MONTE VIDEO—BUENOS AYRES.

PERNAMBUCO—PARA.

WASHINGTON:
BUREAU OF NAVIGATION, NAVY DEPARTMENT.
1880.

Astr. 1608.80



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TELEGRAPHIC DETERMINATION

OF

LONGITUDES

ON THE

EAST COAST OF SOUTH AMERICA

EMBRACING THE MERIDIANS OF

LISBON, MADEIRA, ST. VINCENT, PERNAMBUCO, BAHIA, RIO DE JANEIRO, MONTEVIDEO, BUENOS AYRES, AND PARA,

WITH THE

LATITUDE OF THE SEVERAL STATIONS,

BY

LIEUTENANT COMMANDERS F. M. GREEN AND C. H. DAVIS,
AND LIEUTENANT J. A. NORRIS, U. S. N.,

IN

1878 and 1879.

With the compliments of the United States Hydrographic Office.
J.C.P. de Krafft, Captain U.S. Navy,
Hydrographer to the Bureau of Navigation.
Washington, D.C. Nov. 20, 1880.

PUBLISHED BY ORDER OF COMMODORE WM. D. WHITING, U. S. N
Chief of the Bureau of Navigation,
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1880.

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Gilbert

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INTRODUCTION.

In continuation of the work of establishing secondary meridians of longitude at all important points where the extension of telegraphic communication might afford the necessary facilities, it was decided by the Navy Department, after the completion of the West India work, in 1877,* to take advantage of the recently completed telegraph cables to measure from some well determined point in Europe by way of Madeira and the Cape Verde Islands to the ports on the east coast of South America. The cable of the Eastern Telegraph Company extends from the Land's End, in England, to Carcavellos, at the mouth of the Tagus River, where it joins the cable of the Brazilian Submarine Telegraph Company, connecting Europe with Madeira, Porto Grande in the Cape Verde Islands, and Pernambuco, on the coast of Brazil. Along the coast of South America, from Para on the north to Buenos Ayres on the south, the lines of the Western and Brazilian Telegraph Company unite the ports of Para, Maranham, Pernambuco, Bahia, Rio de Janeiro, Monte Video, and Buenos Ayres.

The longitudes of points on the east coast of South America have always been in dispute and have been very uncertain until recently, the results obtained by apparently trustworthy methods differing by almost incredibly large quantities.

In 1867 the *Connaissance de Temps* stated the longitude of Fort Villegagnon, as determined by M. Liais, the director of the Imperial Observatory, as $2^h\,52^m\,2^s.9$. In 1878 the same authority, from observations made by M. Penaud, stated it as $2^h\,52^m\,30^s.8$, while the labors of Admiral Mouchez gave the results $2^h\,52^m\,35^s.9$.

An exhaustive discussion was made by Admiral Mouchez of all attainable results from the earliest time,† and, as will be seen, his conclusions differ but slightly from the result as determined by this expedition.

An exact and trustworthy determination of the longitude of Buenos Ayres also adds greatly to the value of the extensive chain of meridian distances measured from the National Observatory at Cordova by the director, Dr. B. A. Gould. This chain extends from Asuncion, in Paraguay, on the north, to Buenos Ayres, on the south, and from Colonia, on the east, to Santiago de Chile, on the west, and embraces more than twenty stations.

* Report on the telegraphic determination of differences of longitude in the West Indies and Central America, by Lieut. Commander F. M. Green, U. S. N. U. S. Hydrographic Office Publication No. 65. Washington, 1877.

† See *Annales Hydrographiques* for 1867.

The publication of the West India work was not completed till August, 1877, and it was not till the 29th of October, 1877, that the expedition was enabled to sail from New York for Lisbon in the sailing store-ship Guard, which had been assigned for the purpose. In consideration of this delay, it was determined to commence the work by measuring between Lisbon and Madeira.

Strange as it may appear, the difference of longitude between the observatories of Lisbon and Greenwich or Paris had never been satisfactorily determined, but as French astronomers had recently been turning their attention to the telegraphic method of determining longitudes, the Chief of the Bureau of Navigation decided to request the French *Bureau des Longitudes* to co-operate in our work by measuring from Paris to Lisbon, thus affording a trustworthy starting-point for the measurement to South America. Lieutenant-Commander Charles H. Davis, U. S. N., who had succeeded Mr. Miles Rock in charge of one observatory, was therefore directed by the Secretary of the Navy to proceed to Paris and to endeavor to effect this desired arrangement. Furnished with the proper credentials, he arrived in Paris on November 24, 1877, and being placed, through the United States legation, in official communication with the *Bureau des Longitudes*, was invited to attend a meeting of that body, on November 28, and explain the desires of the United States Government. Having done so, it was understood that the *Bureau des Longitudes* would co-operate with the United States officers, but up to the present time the measurement between Paris and Lisbon has not been made.

After a very boisterous passage, having experienced a constant succession of heavy gales, the Guard arrived at Lisbon on the 30th of November, and Lieutenant-Commander Davis rejoined the ship on the 6th of December—one day after the departure of the monthly steamer for Madeira.

On visiting the director of the Royal Observatory, Capt. F. A. Oom, Royal Portuguese Navy, he offered every assistance, placing a small detached observatory at our disposal, and promising to have a line established from the observatory to connect with the line to Carcavellos. Permission from the government being obtained, the instruments were landed on the 14th of December, and were immediately set up. Upon visiting the superintendent of the telegraph station at Carcavellos, Franklin Harwood, esq., it was found that it would not be consistent with safety to the cables to join the land line from Lisbon to the cable leading to Madeira. Instructions had been received from the office of the telegraph company in London to afford us every facility for the work, and this was done in the most liberal and courteous manner by Mr. Harwood and his staff. It was decided to establish a transmitting station at Carcavellos, which was placed under charge of Lieut. S. Belden.

By the first steamer leaving Lisbon for Madeira, Lieutenant-Commander Davis and Lieutenant Norris sailed on January 5, and on their arrival at Funchal, after obtaining permission from the proper authorities, established their observatory at Fort St. Jago.

Bad weather at one end or the other of the line, and other unavoidable causes, interfered to prevent the commencement of the proposed work till the 20th of January, when, the weather being good at both stations and instruments in working order, work

was begun, but was unluckily interrupted by an accident at Funchal, where a sudden gust of wind, during the progress of the observations, blew off the roof of the observatory, precipitating part of it on to the transit instrument, which was upset and seriously injured. Fortunately, each party was provided with one of the transit instruments belonging to the Transit of Venus Commission, to be used in case of accident, and this spare instrument was mounted the next day, only delaying the work one night.

At Lisbon everything worked admirably. Every difficulty was overcome by the good offices of Captain Oom and his assistants, Captain Oom taking personal charge of the telegraphic connections and aiding us in every possible way. By his kindness also the transit instrument broken at Madeira was repaired at the Royal Observatory. Time-signals were exchanged as usual through the cable between Carcavellos and Madeira, the chronometers at Lisbon and Carcavellos being automatically compared both before and after the cable-signals.

Satisfactory measurements of the difference of longitude between the Lisbon and Madeira stations were made on the 21st, 22d, 23d, 26th, and 27th of January, 1878. The latitude of the Lisbon station was not determined, that having been satisfactorily done by Captain Oom.

On the 1st of February the instruments were dismounted and re-embarked, and on the 3d the Guard sailed from Lisbon for Porto Grande, C. V. I.* It was intended to call at Madeira on the way, but when within a few miles of the anchorage a southwest gale sprung up, and at its close, the wind being fair for Porto Grande, it was determined to push on at once. On reaching the vicinity of the Cape Verde Islands the air was so full of dust and fine sand from the African shore that the land, although over 7,000 feet high, could not be seen till the ship was within less than two miles of it.

On the 21st of February the ship anchored in Porto Grande harbor, and having obtained permission from the authorities, the observatory was landed and put up on the 23d in an excellent position within the inclosure of the telegraph company's station. Bricks, cement, and trowels having been brought from the United States, a handy bricklayer was found among the ship's company and a pier was soon finished in a workmanlike manner. The cable between Madeira and Porto Grande had a serious fault about half way between the stations, so that only a very small battery power could be used, but, as our galvanometer mirrors were very delicately suspended, this caused no trouble. While the ship was proceeding from Lisbon to Porto Grande, Messrs. Davis and Norris had carefully determined the latitude of

* The officers of the Guard, on leaving Lisbon, were as follows:
Lieut. Commander Francis M. Green, commanding.
Lieut. Commander Charles H. Davis, supernumerary for astronomical work.
Lieut. Samuel Belden, executive officer.
Lieut. J. A. Norris, supernumerary for astronomical work.
Master C. P. Rees, navigator.
Master D. L. Wilson.
Ensign J. C. Colwell.
Ensign W. H. Nostrand.
Ensign E. J. Dorn.
Passed Assistant Surgeon J. C. Boyd.
Assistant Paymaster C. I. Lewis.
Clerk to commanding officer, Louis Chauvenet.

the Madeira station by zenith telescope observations and had made numerous observations for instrumental constants. On February 28 and March 2, 4, and 5, the measurement of longitude between Madeira and Porto Grande was satisfactorily completed and Lieutenant-Commander Davis, with his party and instruments, rejoined the ship at Porto Grande, by the Portuguese mail-steamer, on the 15th.

As the healthy season had not fairly set in on the Brazilian coast, it was decided to wait a short time before dispatching a party to Pernambuco for the next measurement. A less interesting place than Porto Grande to pass two or three weeks could not be found. The island of St. Vincent is merely a heap of cinders and its only importance arises from its convenience as a coaling station for steamers between ports in the South Atlantic and Europe. The time was used by Lieutenant Norris in determining with great care the latitude of the station.

On the 5th of April, Lieutenant-Commander Davis and Lieutenant Norris sailed from Porto Grande for Pernambuco by the royal mail-steamer, taking their observatory, instruments and a seaman as observatory attendant. Arriving at Pernambuco on the 11th of the same month, permission was requested and cordially granted to erect the observatory within the grounds of the arsenal. On the 17th, 18th, 20th, and 23d good measurements were effected in spite of the faulty running of the Bond chronograph, which had given considerable trouble in Madeira and which finally broke down. As, however, the precaution had been taken to bring a chronograph belonging to the Transit of Venus Commission as a spare instrument, no serious delay was experienced.

On the 25th of April, the Guard sailed from Porto Grande for Rio de Janeiro, and while waiting for her arrival Lieutenant Norris determined the latitude of the Pernambuco station by zenith telescope observations.

The Guard, a very dull sailer, did not arrive at Rio de Janeiro till May 31, when, to our great disappointment, we were met by the news of a break in the cable south of Pernambuco.

After inquiring closely into the general condition of the cable and consulting with J. R. France, esq., superintendent of the Western and Brazilian Telegraph Company, it was decided that there was no immediate prospect of its successful repair, and Messrs. Davis and Norris were directed by the government land line of telegraph to rejoin the ship with their instruments. Attempts were made to use this land line for exchange of time signals, but the expedition had not proper instruments for receiving signals over it and it was constantly out of repair. By the kind permission of M. E. Liais, director of the imperial observatory, Lieutenant-Commander Green mounted his instruments on June 7 on the terrace of the Imperial Observatory, a most admirable position, combining all desired advantages. The cable running south from Rio de Janeiro to Montevideo and Buenos Ayres being in good working order, no time was lost in beginning these measurements, and the officers from Pernambuco arriving at Rio de Janeiro on the 15th of June, proceeded to Montevideo by the same steamer on the 18th, arriving on the 23d. Until the 1st of July stormy weather prevailed, but on the 1st, 5th, and 6th satisfactory exchanges took place between Rio de Janeiro and Montevideo, followed by a series of cloudy nights in Rio de Janeiro. On July

12, the observatory and instruments at Montevideo were transferred to Buenos Ayres in order to make an attempt to exchange signals between that station and Rio de Janeiro. The attempt was made but was unsuccessful. While at Rio de Janeiro, His Majesty the Emperor of Brazil, who had manifested much interest in the work, visited Lieutenant-Commander Green's observatory and was present during an exchange of time-signals between Rio de Janeiro and Montevideo.

Finding it necessary to connect Buenos Ayres with the chain of longitudes already measured, by a direct measurement with Montevideo, Lieutenant-Commander Green, with Mr. D. L. Wilson and an observatory attendant, proceeded with the necessary instruments by the Pacific mail steamer to Montevideo, on July 19, arriving on the 23d, and at once occupying the station just left by Lieutenant-Commander Davis. On the 26th, 27th, and 28th of the same month, measurements between Montevideo and Buenos Ayres were effected and both parties, with all the instruments, returned to the ship at Rio de Janeiro, by the royal mail-steamer of July 31, arriving August 8.

In the mean time earnest efforts had been made by the manager of the cable company to repair the link between Rio de Janeiro and Bahia, it being confidently asserted that its reopening for business would be effected within a very few days. It was, therefore, decided to establish one party at Rio de Janeiro and the other at Bahia, to take immediate advantage of the first chance to work. As Rear-Admiral Nichols, commanding the South Atlantic station, wished to delay the Guard to take home a quantity of stores, Lieutenent-Commander Green, with his party, proceeded to Bahia by mail-steamer August 13, leaving orders with Lieutenant Belden to follow him with the ship to Bahia as soon as practicable, Lieutenant-Commander Davis remaining at Rio de Janeiro in charge of the observatory and occupying a part of the time while waiting in determining its latitude. After establishing the observatory at Bahia, the officers waited as patiently as possible in almost hourly expectation of hearing that the break in the cable had been found and repaired. On the 2d of September the Guard arrived at Bahia from Rio de Janeiro.

It became evident at last that it would be useless to wait for the repair of the cable, and the event justified this conclusion, it not being ready for work for several months afterward. An attempt was made to utilize the Brazilian Government lines to complete our work between Rio de Janeiro and Pernambuco, but was unsuccessful, principally because we had no telegraph instruments similar to those in use. Lieutenant-Commander Davis and his party were, therefore, reluctantly summoned to rejoin the ship, which they did on the 27th of September, and on the 1st of October the Guard sailed for Norfolk, arriving on the 19th of November, after a pleasant and uneventful passage.

Yellow fever had prevailed to such an extent in the Southern States during the summer that great solicitude was felt and expressed lest the Guard, coming from a country almost proverbial for its pernicious fevers, should introduce the seeds of the disease in discharging her cargo of stores, and the ship was, therefore, hastily dispatched to be frozen out at Portsmouth, N. H. Sailing from Hampton Roads, November 28, she arrived at Portsmouth December 4, and on the 15th of December, 1878, was put out of commission.

Although the expedition had not been completely successful, everything within the bounds of possibility had been accomplished by the officers engaged. It was promptly decided by the Secretary of the Navy to complete the work as soon as the season should open, by sending the same officers to measure from Greenwich to Lisbon and from Pernambuco to Rio de Janeiro, guarding against any delay from breaks in the cables by furnishing telegraph instruments which could be used on the land-lines.

The early months of 1879 were occupied in the preliminary computation of the observations already made, and on the 17th of May Lieut. Commanders F. M. Green and C. H. Davis, Lieut. J. A. Norris, and Assist. Paymaster A. K. Michler, with two seamen as observatory attendants, and the same observatories and instruments previously used, sailed from Philadelphia for Liverpool in the steamship Indiana, under orders from the Secretary of the Navy to thoroughly complete the work already commenced.

The party arrived in Liverpool May 28, and Lieutenant Norris was at once dispatched to Penzance with one set of instruments, the other being shipped direct to Lisbon.

The telegraphic connections between Greenwich and Lisbon consist of about 300 miles of land-line from Greenwich to Porthcurnow, about 4 miles from the Land's End, 828 miles of cable from Porthcurnow to Carcavellos, at the mouth of the Tagus, and 12 miles of land-line from Carcavellos to the Royal Observatory at Lisbon. As it was manifestly impracticable to connect the land-lines and cable together so that signals might be sent through, it was at first designed to measure the differences of longitude between Greenwich and Porthcurnow and between Porthcurnow and Carcavellos. As soon as practicable after arriving in London, an interview was had by appointment with Sir George Airy, the astronomer royal, who cordially offered every assistance, but strongly recommended that Porthcurnow should be occupied as a comparing or transmitting station. At first there seemed to be several objections to this plan, but it was really the only one which afforded any chance of success, as during a stay of three weeks at the Land's End hardly a star was seen, owing to the persistent rain and fog which prevailed.

After ordering from Messrs. Siemens Brothers two polarized relay ink-writers, for use in case of necessity in the Brazilian work, the party dispersed in London, Lieutenant-Commander Davis and Lieutenant Norris sailing for Lisbon in the royal mail-steamer of June 9, and Lieutenant-Commander Green and Mr. Michler proceeding to Penzance, the nearest town to Porthcurnow.

On arriving at Porthcurnow, the superintendent of the Eastern Telegraph station, G. Spratt, esq., to whom we were indebted for many kindnesses, gave permission to establish the observatory in a convenient place on the telegraph company's premises, and the instruments were at once set up; but there was absolutely no chance for astronomical observations, and it was soon evident that the plan proposed by Sir George Airy was much the most likely to prove successful.

Through the kind offices of W. H. M. Christie, esq., first assistant to the astronomer royal, and H. Eaton, esq., superintending engineer of post-office telegraphs, telegraphic communication was readily established between the observatory at Porth-

curnow and the Royal Observatory at Greenwich, Mr. Christie, in the absence of the astronomer royal, undertaking to send second-signals from the standard clock and to furnish its error and rate for every night on which signals were exchanged.

Arriving at Lisbon on June 13, Lieutenant-Commander Davis, after a slight delay caused by necessary formalities, established himself in exactly the same place occupied during the year before, during the Lisbon-Madeira measurement, and sent Lieutenant Norris to Carcavellos.

Permission having been formally obtained from the proper authorities, the instruments were set up at Lisbon and Carcavellos, and on June 25, 26, 27, and 29 signals were successfully exchanged throughout the line, by which accurate comparisons were made of the Lisbon and Greenwich time-keepers.* This measurement was in every respect successful. Mr. Christie very promptly redeemed his promise by furnishing a full and exact record both of clock-errors and time-signals, and the accordance of the various nights' work amply proves its accuracy, although it gives the somewhat startling result that the longitude of the observatory at Lisbon has up to the present time been in error more than two miles

Upon the completion of this measurement, Lieutenant-Commander Green, as directed in the orders from the Navy Department, turned over the charge of the remainder of the work to Lieutenant-Commander Davis and returned to the United States. On being joined at Lisbon, on July 13, by Assistant Paymaster A. K. Michler, with the instruments and observatory from Porthcurnow, Lieutenant-Commander Davis with the whole party sailed for Brazil the same day, on the royal mail-steamer Elbe. On the 27th of July, Lieutenant Norris and Mr. Michler landed at Bahia with their instruments, Lieutenant-Commander Davis proceeding to Rio de Janeiro. At Bahia, Lieutenant Norris at once proceeded to occupy the station selected the year before in Fort San Antonio, finding the brick pier then built in good condition. (See Frontispiece.)

Lieutenant-Commander Davis, on arrival at Rio de Janeiro, July 30, found the cable in good working order and occupied the old station at the Imperial Observatory. The instruments were mounted and ready for work on August 8, but a faulty insulation of the line between the observatory and telegraph office delayed the commencement of the measurement till the 11th.

On the 12th, 13th, 15th, and 18th of August, excellent exchanges and observations were effected, giving a satisfactory measurement between Rio de Janeiro and Bahia. There being still one unmeasured link between Bahia and Pernambuco, Lieutenant-Commander Davis embarked with his instruments on the royal mail-steamer Minho, August 24, arriving at Pernambuco on the 29th. Here the old site in the arsenal was again occupied, and on September 7, 8, 10, and 11 satisfactory exchanges of time signals were effected, completing the long measurement from Greenwich to Buenos Ayres.

The only remaining work to be done was to measure from Pernambuco to Para, but the condition of the cables would not permit signals to be sent through between these two stations without repetition at Maranham. Lieutenant Norris was therefore

* For a detailed account of the method used, see page 26.

directed to proceed to Para by steamer with his observatory, which he did, arriving there on the 26th of September, while Mr. Michler was sent with a break-circuit chronometer and chronograph to Maranham, where a transmitting-station was established, and on September 28 and 30 exchanges were effected, completing all the work assigned. The whole party then returned to the United States in the steamer City of Para, arriving safely in Washington October 28, 1879.

Throughout the work every possible aid and courtesy was extended to the officers of the expedition both by the government officials of every country visited and by the officers of the telegraph cable companies. Among the government officers to whom we were especially indebted for courteous assistance are W. H. M. Christie, esq., first astronomical assistant at the Royal Observatory, Greenwich; Capt. F. A. Oom and Lieutenant Alvesdoria, of the Royal Observatory, Lisbon; M. E. Liais, director of the Imperial Observatory at Rio de Janeiro; Dr. E. Capanema, director of the imperial telegraphs, Rio de Janeiro; Thomas Adamson, esq., United States consul-general, Rio de Janeiro; and H. W. Dimon, esq., United States consul, Lisbon.

Without the zealous co-operation of the officers in charge of the telegraph stations, our labors would have been of no avail. While we were cordially greeted and assisted everywhere, we feel under special obligations to F. Harwood and E. V. Wyse, esqs., of the Eastern Telegraph Company's station at Carcavellos; J. Howe, esq., superintendent at Pernambuco of the Brazilian Submarine Telegraph Company; J. R. France, esq., general superintendent, A. Watson, Alfred Peters, and W. King, esqs., of the Western and Brazilian Telegraph Company; J. Oldham, esq., superintendent of the River Plate Telegraph Company; and G. Spratt, esq., station superintendent of the Eastern Telegraph Company at Porthcurnow.

DESCRIPTION OF STATIONS.

GREENWICH.

The station at Greenwich was the Royal Observatory, the initial point of the measurement being the meridian of the meridian circle.

The land-line of telegraph, from Greenwich to Porthcurnow, is about 300 miles in length.

PORTHCURNOW.

The station at Porthcurnow was not used for astronomical observations, but merely for exchanging time-signals with the observatory at Greenwich and the station at Carcavellos. The observatory hut was for this purpose set up in the grounds of the Eastern Telegraph Company's station and connected by temporary wires with the cable to Carcavellos and the land-line to Greenwich.

The cable from Porthcurnow to Carcavellos is about 828 miles long.

From information kindly furnished by Colonel Cooke, R. E., director of the Ordnance Survey, the station at Porthcurnow was found to be in—

	°	′	″
Lat. N.	50	2	44
Lon. W.	5	39	18

LISBON.

The instruments were mounted at Lisbon in a small observatory close to the main building of the Royal Observatory, which is in the Tapada or Royal Park, about two miles west of the city. This small building, kindly placed at our disposal by the director of the observatory, was circular, about 15 feet in diameter, with a revolving tower roof, and was admirably adapted for the purpose. A circular stone pier of the proper height was erected on the brick foundation already prepared, and on it the transit was mounted, while small tables served to support the chronograph, chronometer, and telegraph instruments. By the kindness of the Portuguese officials a telegraph wire was extended south from the observatory and joined to the government line leading to Carcavellos, 11 miles distant.

The center of the pier was 136.8 feet south and 44.95 feet east of the center of the dome of the Royal Observatory.

The latitude of the center of the dome of the observatory, as communicated by Captain Oom, is 38° 42′ 31″.25.

CARCAVELLOS.

The station here was, like Porthcurnow, merely a transmitting-station, no astronomical instruments being mounted. The chronograph and chronometers were set up in the instrument-room of the Eastern Telegraph Company.

MADEIRA.

The site occupied by the observatory at Madeira was the upper platform or rampart of Fort St. Jago, a foundation of solid masonry, at an elevation of about 35 feet above the sea, and situated about 50 yards west of the cable-landing. The fort is on a point of rocks at the eastern end of the beach, in front of the town of Funchal.

Angles taken from the center of the pier, which was built of brick and capped with stone, gave the following bearings:

Loo Rock light staff, S. 68° 37′ W., distant 5,597 feet.

Vane on tower of cathedral, N. 78° 10′ W.

Church of N. S. do Monte, E. Tower, N. 6° 28′ W.

Brazen Head, edge of cliff, S. 76° 0′ E.

An insulated copper wire, about 50 yards long, was stretched from the observatory to the cable-house, and connected with the end of the cable each evening previous to exchange of signals

The flag-staff of Fort St. Jago was 37 feet south and 32½ feet east of the center of the pier.

PORTO GRANDE, ST. VINCENT, CAPE VERDE ISLANDS

The observatory was placed in the southwest corner of the inclosure surrounding the Brazilian Submarine Company's office.

From the center of the pier, the flag-staff in front of the company's office was 38 feet north and 125 feet east

The center of the pier was 25 feet east of the inner face of the western wall and 42 feet 5 inches north of the inner face of the south wall of the inclosure.

The telegraph instruments in the observatory were connected by two insulated wires to the line and earth in the company's office, about 150 feet distant.

By scraping away about three inches of loose red soil, a hard bed of disintegrated rock was uncovered, forming an excellent foundation for the pier, which was built of brick and cement.

PERNAMBUCO.

The observatory at Pernambuco was placed in the grounds of the Naval Arsenal, close to the water's edge, between a small octagonal building, called the Casa do Orden, and a slip for hauling up boats and small vessels The observatory was close to the stone landing stairs of the arsenal, and was connected with the office of the Brazilian Submarine Telegraph Company, in the town of Recife, by an insulated wire about 500 yards long. This wire was carried over the walls of the arsenal and suspended from the roof of the office of the captain of the port, from the trees of the esplanade, and from the balconies of the intervening houses.

The pier was built of brick and capped with stone.

The center of the pier was 7 feet 8 inches south, and 12 feet 8 inches east of the angle of the sea-wall formed by the slip before mentioned. From the center of the pier the center of the Casa do Orden bore south 30° 6′ west (true), distant 151.2 feet; the flag-staff of Fort Picaõ bore north 41° 42′ 30″ east (true), distant 2,453.6 feet, and the light-house (also at Fort Picaõ) bore north 40° 11′ 30″ east (true) 2,779.5 feet.

The light-house is a much better reference point than the flag-staff, as the latter is very rotten and likely to fall, Fort Picaõ not having been occupied as a military post since about the year 1818. It will never in all probability be used again as a battery.

BAHIA.

The station occupied by the observatory at Bahia was in the old dismantled fort at San Antonio Point, where the light-house is situated, about 2½ miles from the center of the city. (See Frontispiece.) No suitable location could be found in the town where a wire could be conveniently run to the telegraph office, and as the cable is landed at this point and as the situation was healthy, comparatively speaking, and afforded an excellent foundation for the instruments, it was preferred. A brick pier was built on the solid masonry terreplein of the fort, and the officers and attendants were lodged in the deserted barracks, sometimes occupied during the hot season by people who come to the point for bathing. An adjacent cove afforded a quiet landing-place for the instruments, and the horse-railway, which terminates at the point, afforded a passably convenient access to the city.

From the center of the pier the center of the light-house bore south 62° 32′ 30″ east, distant 59.74 feet. The observatory was connected with the cable house about 500 yards distant by an insulated wire, which was joined to the cables leading to Rio de Janeiro or to Pernambuco, as required.

RIO DE JANEIRO.

The site occupied at Rio de Janeiro was at the Imperial Observatory at Morro do Castello.

The buildings sheltering the instruments at this observatory stand on top of a massive stone structure intended, when begun, for a convent, but never completed as such. The terrace or platform on top of these walls affords a firm support and an unobstructed view. Our observatory hut was placed exactly south of the dome of the equatorial, the center of which marks the meridian of Rio de Janeiro. The center of the transit pier was 74 feet south of the center of the dome, which is north 65° 31′ 51″ west, distant 5,327.5 feet from the flag-staff on Fort Villegagnon, the point to which the longitude of Rio de Janeiro has generally been referred.

MONTEVIDEO.

The observatory at Montevideo was placed in the grounds of the English church (*Templo Inglez*), between the church and the water,

The center of the transit pier was 27 feet 9 inches south and 17 feet east of the southeast angle of the church building.

From the center of the pier the southeast tower of the cathedral (*Matriz*) bore north 6° 16′ 40″ (true), distant 1,088 feet.

An insulated wire 150 yards long connected the observatory with the landing hut of the cable.

BUENOS AYRES.

The site occupied by the observatory at Buenos Ayres was within the inclosure of the government building, and between its wall and the iron fence surrounding it. The railway lies directly beneath the embankment on the edge of which the fence is built, but as the trains ceased running early in the evening no inconvenience was experienced from them.

The center of the pier was 43 feet 5 inches south and 4 feet east of the northeast angle of the government building, and was situated north 39° 18′ east (true), distant 228 feet from the spot occupied by Dr. Gould in the measurement between Buenos Ayres and Cordova. This spot is now covered by the new post-office building.

From the center of the pier the vane on the cupola of the custom-house bore south 87° 2′ 30″ east (true), distant 304 feet.

An insulated wire connected the observatory with the telegraph office in the railroad station, about 400 yards distant, and thence the connection was through one of the local lines with the office of the River Plate Telegraph Company, in the Calle Piedad.

MARANHAM.

This station was only used for transmitting comparisons of the chronometers at Para and Pernambuco, the chronograph and chronometer used for that purpose being set up in the office of the Western and Brazilian Telegraph Company.

PARÁ.

The position occupied was on an open piece of ground between the custom-house and the sea-wall. The observation spot, previously occupied by the officers of the French survey, was the centre of the roof of the portico on the north side of the custom-house, and this point bore from the center of our transit pier south 29° 5′ west, distant 225.4 feet.

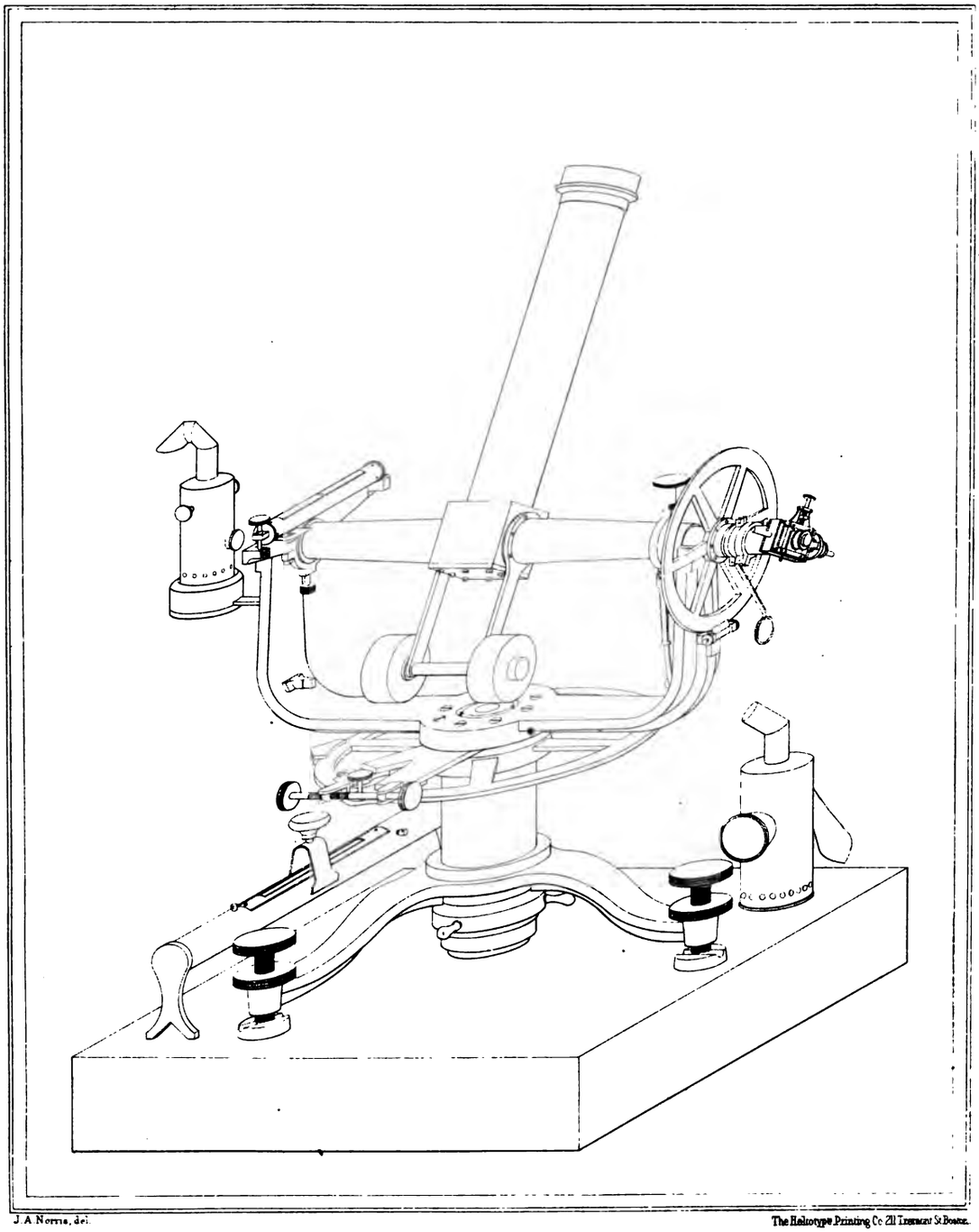
Commander Selfridge, U. S. N., marks the position used by him as an initial point for measuring longitudes in his survey of the Amazon and Madeira Rivers, in 1878, as 500 yards west of the custom-house.

DESCRIPTION OF INSTRUMENTS.

1. Transit instruments.
2. Chronometers.
3. Chronographs.
4. Batteries.
5. Telegraph instruments.
6. Observatories.

TRANSIT INSTRUMENTS.

For the determination of time and latitude, a combination of the transit instrument and zenith telescope was designed, in 1873, by Mr. J. A. Rogers, with especial reference to this work, and under his supervision two of these instruments, precisely



J. A. Norris, del.

The Heliotype Printing Co. 211 Tremont St. Boston.

PORTABLE TRANSIT INSTRUMENT
AND
ZENITH TELESCOPE

U. S. HYDROGRAPHIC OFFICE - 1880

alike, were constructed by Mr. Edward Kahler. (See Plate.) They are of 30 inches focal length, with object glasses of $2\frac{1}{2}$ inches clear aperture, and are so constructed that the eye-piece is at one end of the horizontal axis, a prism at the junction of the telescope tube and axis reflecting at a right angle the rays from the object-glass, thereby enabling the observer to direct the instrument to stars of any zenith distance without changing his position. The tripod and vertical cylinder above it are of cast iron; the rest of the metal is bronze.

The transverse arms which support the ends of the telescope axis are firmly screwed to the top of a vertical and slightly conical axis which revolves in the cylinder surmounting the tripod. A small universal level over the head of this axis is useful in leveling the instrument approximately. The horizontal circle is fourteen inches in diameter, has two verniers, and reads to 10″. It is furnished with a clamp and double-headed tangent screw for slow azimuthal motion, and with two sliding stops, which, when fastened at the proper points on the circle by clamp screws, enable the telescope to be turned exactly 180° in azimuth for zenith-telescope observations for latitude. A powerful clamp at the lower end of the vertical axis fixes it firmly in its seat when the instrument is adjusted in the meridian or prime vertical. By means of the horizontal circle the telescope can be readily turned 90° in azimuth from the meridian to the prime vertical—a feature of great importance when, from absence of a list to select stars from, or from any other cause, the zenith-telescope method of determining latitude is not practicable. The horizontal axis has a vertical finding-circle 8 inches in diameter, reading to minutes, and on the opposite end of the axis is a transverse level for indicating any change in inclination of the telescope.

This level, with a filar micrometer of the usual form attached to the eye-piece, forms the zenith-telescope attachment for the determination of latitude.

The eye-pieces are of the Ramsden pattern, magnifying about 50 and 80 diameters, respectively. In practice, the less powerful one was used.

By means of a screw of long pitch with a milled head, a parallactic motion is given to the eye-piece, so that stars may be accurately observed on every line of the reticule.

The bearings of the Ys which support the axis are segmental in shape and have no micrometric attachment for either level or azimuthal adjustment.

A small spot two-tenths of an inch in diameter ground on the reflecting face of the prism permits the transmission of rays of light from the lamp at the end of the axis to illuminate the lines of the reticule. These lines are seventeen in number, ruled with great accuracy on glass diaphragms by Prof. W. A. Rogers, of Harvard Observatory. In addition to these transit lines, diagonal lines were ruled across the glass plate at equal distances, to use instead of the micrometer for determining differences of zenith distances of stars should the micrometer threads be broken.

The end of the tube which carries the micrometer-box fits accurately against a movable collar on the axis in such a way that when the instrument is adjusted to stellar focus the collar can be screwed up against this shoulder, preventing disturbance of this adjustment.

The striding levels were of the usual form, having divisions on the glass tube about one millimeter apart, equal to 0″.94 of arc or 0^s.063 of time.

Each of these instruments is packed compactly in two boxes, and as, when mounted, the instruments only weigh 125 pounds each, they are easily transported.

The results of the observations have demonstrated that reversal of the axis is almost inevitably attended with a slight change of azimuth, and that a correction must always be introduced for flexure of the axis. As will be seen hereafter, the error arising from change of azimuth in reversing has been guarded against, and the flexure of the axis has been carefully determined, its effects being eliminated in reducing the observations.

In the much heavier instruments of a somewhat similar pattern, constructed for the Transit of Venus Commission, one of which was used by Lieutenant-Commander Davis, at Madeira, both flexure of the axis and disturbance of azimuth by reversal were very decidedly manifested, and are probably unavoidable in portable instruments of this pattern.

The pivots of the horizontal axis have been carefully and repeatedly examined and measured by reversals of the level and by an ingenious applicat on of the spherometer, devised by Prof. W. Harkness, U. S. N., for this purpose, and have been found to be practically equal and regular.*

For a description of the transit instrument used by Lieutenant-Commander Davis at Madeira, and belonging to the Transit of Venus Commission, the reader is referred to the publications of the Transit Commission.

CHRONOMETERS.

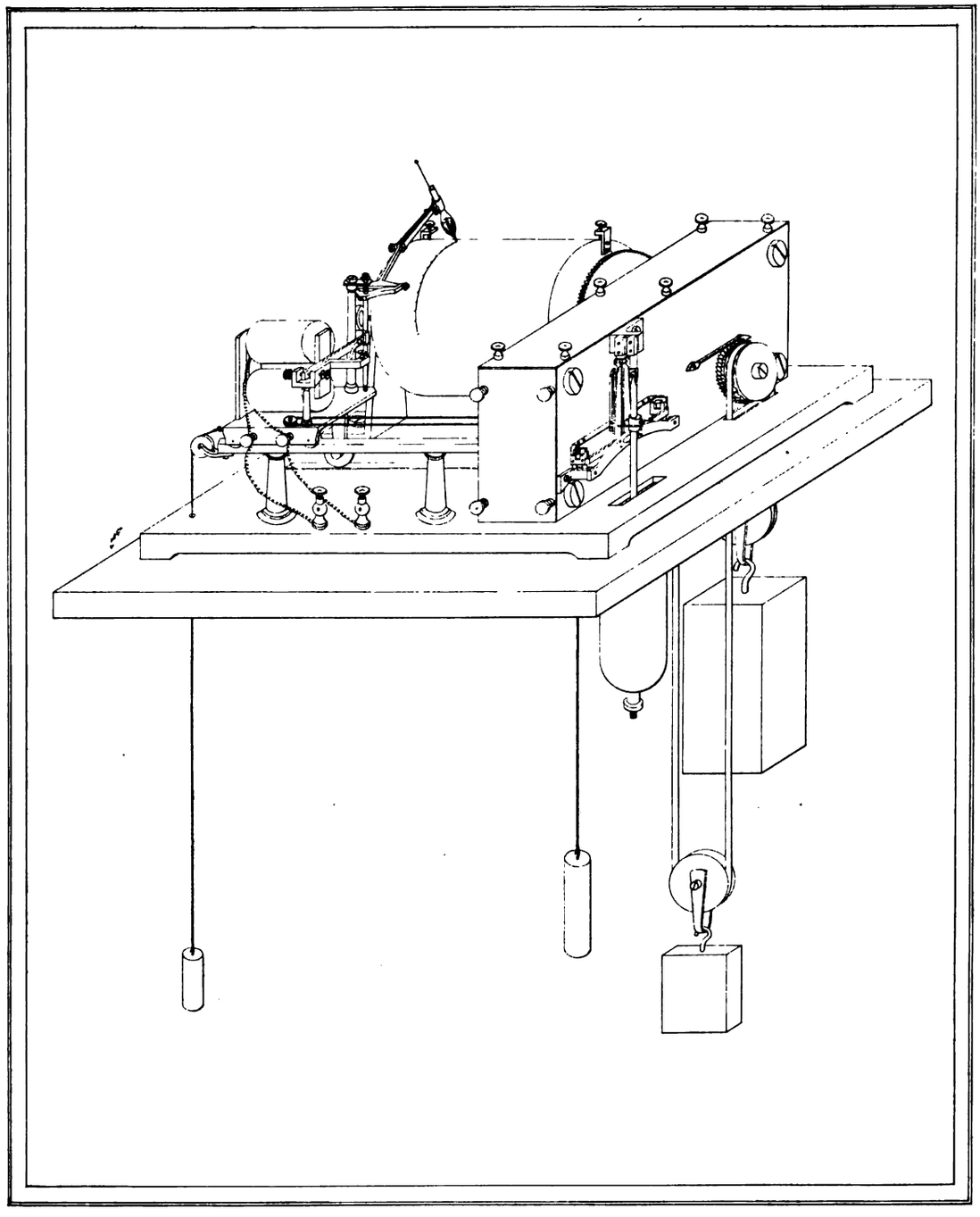
Four break-circuit sidereal chronometers were furnished to the expedition. One of these chronometers was used in each observatory as an astronomical clock, and the others were used at transmitting-stations or were kept as spare instruments in case of accident.

In these instruments the wheelwork is made to break an electric circuit at the end of each second (except that one which completes the even minute) by means of a toothed wheel acting on a jewel pallet attached to a light steel spring. In this steel spring is a platinum screw, which, when the circuit is complete, touches the point of another platinum screw connecting through a fixed insulated stud with one of the terminals on the outside of the chronometer. The spring on which the toothed wheel acts is connected with the other terminal. The electric circuit is complete through the chronometer, except when the teeth of the wheel acting on the jewel pallet separate the platinum points. The circuit is opened for about one-fortieth of a second and is closed during the remaining thirty-nine fortieths.

One tooth in the break-circuit wheel is omitted in order to show on the chronograph the commencement of each minute.

To diminish the injury to the platinum points by deflagration, on breaking circuit, in case too powerful a battery should be used, convenient tinfoil condensers are provided, connected by insulated wires with the terminals on the outside of the chronometers and neatly packed away in the top of the chronometer-case.

* For an account of Professor Harkness's application of the spherometer to the measurement of pivots, see vol. 38, page 487, of the monthly notices of the Royal Astronomical Society.



BOND'S PORTABLE CHRONOGRAPH

U. S. HYDROGRAPHIC OFFICE - 1875.

The rates of the chronometers do not seem to be affected in breaking circuit or by the passage of the galvanic current. These admirable instruments in every way answered the desired purpose.

CHRONOGRAPHS.

The registering of the times of transit of stars or of the receipt of time signals was performed by electric cylinder chronographs. Of these, two were made by Messrs. William Bond & Son, of Boston, expressly for this work, the other one, which was taken as a spare instrument, being made by Messrs. Alvan Clark & Sons, for the Transit of Venus Commission.* This spare instrument was much heavier and of a different pattern, its motion being governed by a vibrating spring. In the instruments made by Messrs. Bond & Son (see plate) a train of wheelwork driven by a weight causes a cylinder about 6½ inches in diameter and 7 inches long to make exactly one revolution in a minute. When in use this cylinder is covered with a sheet of paper easily put on and taken off, and fastened when in place by small set-screws.

A little carriage, to which a pen of peculiar construction is attached, moves upon wheels along the cylinder in the direction of its length, about one-tenth of an inch for each revolution of the cylinder, so that the pen records a spiral line round the cylinder. The pen is so mounted as to have a slight lateral movement, and is so attached to an electro-magnet that when the electric circuit in which it is placed is broken every second by the chronometer, which, with a small battery, is included in the same circuit, the mark made on the chronograph paper, instead of being a straight line, will be broken at regular intervals.

By means of a break-circuit key in the hands of the observer, and included in the same circuit, the electric current may be interrupted, causing the pen to make a similar mark on the occurrence of any event, such as the passage of a star across any of the wires of the telescope.

When the sheet on the cylinder becomes full it is taken off, and by means of a finely-divided scale the positions of these arbitrary marks, with reference to the nearest second marks, may be accurately and easily determined. A screw at the bottom of the pendulum affords an easy method of regulating the clockwork either for faulty running or for change of latitude.

For a description of the peculiar spring-governor escapement of these chronographs, see Brünnow's astronomical notices, page 164, and Volume I, Annals of the Cambridge Observatory.

These instruments were easily transported, but are quite delicate and require great care both in setting up and in packing for transportation. The constant vibration of the governor-spring is very apt to break it, unless it has been tempered with great care. Spare springs should therefore be provided before taking these instruments into the field.

The Siemens' ink-writers procured for use on land-lines make very good chronographs in case of need. In these instruments an ink-line is made on a continuous fillet of paper, the breaking of the circuit by either the chronometer or break-circuit key making short breaks in the ink-line, which can easily be read off with a scale.

* For a detailed description of this instrument, see the publications of the Transit of Venus Commission.

containing twelve Le Clanché cells was supplied to each party, of which two cells were used for the chronograph circuit, and as many of the remainder as necessary were used to establish communication through the cables between the observatories. At first this battery gave no trouble, but the cylinders did not retain their strength very well after setting up and taking down several times. After trying several battery-cells, the conclusion has been reached that for this or similar work, where the battery is to be used for a short time and then taken down and transported, a gravity battery in gutta-percha or vulcanite cells is the best. The form adopted is a vulcanite cell, about 4 inches square and 6 inches deep, with a few ounces of sulphate of copper in the bottom and nearly filled with water. One terminal is formed by a strip of copper extending across the bottom of the cell and up one side, and the other is attached to a wheel-shaped disc of zinc suspended just under the top of the water, from a wooden stick across the top of the jar. Sulphate of copper, zinc, and sheet-copper can be procured everywhere, and the jars, unlike glass, are not easily broken.

Each observatory was furnished with a set of cable instruments made by Messrs. Elliott Brothers, of London, consisting of a double-current key of the usual form, a switch for cutting out the galvanometer when sending signals, a reflecting galvanometer or speaking instrument, and a lamp.

Upon overland lines, the time-signals received can be recorded automatically by connecting the chronograph with the telegraph circuit by means of a relay magnet and a small local battery, but with long submarine cables the electric impulse transmitted is not strong enough to actuate the electro-magnets of the relay.

For telegraphing with weak impulses over submarine lines, a very beautiful device was invented by Sir William Thomson and is now in general use. To a delicately suspended magnet, surrounded by one or more coils of fine insulated wire, a small mirror is attached. From this mirror a beam of light from a lamp is reflected on a sheet of white paper, so that, the room being darkened, when no currents are passing through the cable the spot of light on the paper remains at rest, but when at the sending station either of the two keys is pressed, a positive or negative current, as the case may be, is sent through the cable and through the coil surrounding the magnet, causing the mirror to turn on its vertical axis and to deflect the ray of light to the right or left.

When the signal is perceived the observer touches his chronograph key, thus recording the time of its arrival.

In case of two longitude stations being connected by overland lines, these telegraph instruments would not be suitable, and to use in such contingency two of Messrs. Siemens Brothers' polarized inkwriters were procured. These instruments can be used as chronographs in an emergency. For a detailed and illustrated description of them the reader is referred to Prescott's Electricity and Electric Telegraph, page 517.

OBSERVATORIES.

Portable wooden observatories (see frontispiece) were designed by Mr. J. A. Rogers for sheltering the instruments and observers. They were ingeniously and strongly made so as to be easily and quickly set up and taken down, and, after being established at thirty-four stations during the West India and South American work, are, with but slight repair, ready for the field again. Each observatory was furnished with a hood of painted canvas to draw over the roof in rainy weather.

Brick piers were erected at each station for the transit instruments. For the Hydrographic Office instruments, the piers were made 22 inches square and 27 inches high above the observatory floor. The Transit of Venus instruments, being lower and having a broader foundation, require a pier 24 inches by 22 and 32 inches high above the floor.

In addition to the above instruments each party was furnished with a 3-inch theodolite, a sextant and artificial horizon, and a surveyor's chain, and with a supply-box containing carpenter's tools, wire, battery material, plaster of Paris, lanterns, oil, &c.

DETERMINATION OF CONSTANTS AND PERSONAL EQUATION.

The value of a division of the striding level and of the transverse zenith-telescope levels of the Hydrographic Office transits was determined by applying the levels to the telescope of the mural circle at the Naval Observatory, and noticing the change in reading of the levels consequent upon slight movements of the telescope upon its horizontal axis, carefully measured by the microscopes of the circle. The values thus found were afterwards verified by the use of a level-trier belonging to Prof. W. Harkness, U. S. N., kindly loaned for the occasion. By the latter instrument the level values of the Transit of Venus telescope used by Lieutenant-Commander Davis at Madeira were determined.

Repeated sets of these observations gave for the striding levels the following values:

Hydrographic Office transit No. 1, 1 division, $= 0''.94$ or $0^s.063$
Hydrographic Office transit No. 2, 1 division, $= 0''.94$ or $0^s.063$
Transit of Venus transit No. 1505, 1 division, $= 1''.112$ or $0^s.074$

For the zenith-telescope levels:

Hydrographic Office transit No. 1, 1 division, $= 1''.000$
Hydrographic Office transit No. 2, 1 division, $= 1''.006$
Transit of Venus transit No. 1505, 1 division, $= 1''.067$

The pivots of the Hydrographic Office transits were perfectly cylindrical and equal in size, as indicated by repeated measurements with Professor Harkness's spherometer. For the Transit of Venus instrument No. 1505, a correction of $\left\{\begin{matrix}-\\+\end{matrix}\right\}$ $0^s.152$ $\left\{\begin{matrix}\text{west}\\\text{east}\end{matrix}\right\}$ must be applied for inequality of pivots, the correction for irregularity of pivots being inappreciable.

The equatorial intervals of the transit threads or lines were determined by repeated observations of complete transits of circumpolar stars. Readjustment of the stellar focus necessitated new determinations for each season's work. The following values are for circle east:

HYDROGRAPHIC OFFICE TRANSITS—EQUATORIAL INTERVALS.*

Lines.	1878.		1879.	
	No. 1, 42 obs.	No. 2, 52 obs.	No. 1, 24 obs.	No. 2, 61 obs.
	s	s	s	s
B^1	− 27.88	− 28.23	− 27.86	− 28.25
B^2	23.88	24.19	23.85	24.22
B^3	19.90	20.17	19.90	20.18
C^1	7.95	8.06	7.95	8.07
C^2	3.95	− 4.02	3.95	4.00
C^3	− 0.01	+ 0.05	− 0.02	− 0.02
C^4	+ 4.00	4.02	+ 3.95	+ 4.00
C^5	7.97	8.07	7.95	8.07
D^1	19.88	20.16	19.92	20.17
D^2	23.85	24.18	23.85	24.22
D^3	+ 27.85	+ 28.22	+ 27.88	+ 28.26

TRANSIT OF VENUS TRANSIT, 1878.

	s
I	− 30.14
II	14.98
III	4.61
IV	+ 0.07
V	4.70
VI	14.91
VII	+ 29.99

In making a preliminary reduction of some of the first observations of time-stars with the Hydrographic Office transits, it was found that, after reducing the error of collimation to a minimum by repeated observations and adjustments with a horizontal collimator, the clock corrections deduced from stars observed with the circle end of the axis east differed from those resulting from observations made with the axis in a reversed position by an amount varying from one second to one second and two-tenths. It was at once perceived that this difference was due to flexure of the axis, which of course must be manifested in all instruments of a similar construction. To determine the exact amount of flexure, careful observations were made in the following manner: A large number of stars were observed on the meridian at various distances north and south of the zenith in both positions of the instrument, and from each star the chronometer correction was deduced, corrected for the level, collimation, and azimuth errors of the instrument.

* The outer groups of three wires on each side of the center were not used, those from B^1 to D^3, inclusive, being the only ones observed upon.

The difference between each chronometer correction as thus obtained and the true chronometer correction obtained by comparing the chronometer with the standard clock of the Naval Observatory was assumed after reducing to the equator, by dividing by the secant of the star's declination, to be due to the flexure at the zenith distance of that star expressed in seconds of time.

On obtaining the ratio of the flexure corrections at the several zenith distances by dividing each one by the value at the zenith, they were found to follow almost exactly the law of the cosine of the zenith distance, having a maximum value when the telescope was pointed to the zenith, and a zero value when it was horizontal.

The Transit of Venus instruments, although having shorter and thicker axes, were also found to show a decided flexure of the axis, less, however, in amount. Its value was determined by Professor Harkness, U. S. N., by observations made upon a pair of collimators, one of which was horizontal, the other vertical.

From these observations, the resulting corrections for flexure were found to be—

Hydrographic Office transit No. 1, zenith value, $f = 0^s.55$
Hydrographic Office transit No. 2, zenith value, $f = 0^s.54$
Transit of Venus transit No. 1505, zenith value, $f = 0^s.275$

Hence the flexure correction for any star is—

Hydrographic Office No. 1, for circle $\left\{\begin{matrix}\text{east} + \\ \text{west} -\end{matrix}\right\}$ $0^s.55 \cos z \sec \delta$

Hydrographic Office No. 2, for circle $\left\{\begin{matrix}\text{east} + \\ \text{west} -\end{matrix}\right\}$ $0^s.54 \cos z \sec \delta$

Transit of Venus No. 1505, for circle $\left\{\begin{matrix}\text{east} + \\ \text{west} -\end{matrix}\right\}$ $0^s.275 \cos z \sec \delta$

The values of a revolution of the micrometer-screws were determined by numerous observations of circumpolar stars at elongation.

From all these observations, the following values have been deduced:

Hydrographic Office No. 1, 1 revolution $= 65''.80$
Hydrographic Office No. 2, 1 revolution $= 66''.83$
Transit of Venus No. 1505, 1 revolution $= 69''.70$

PERSONAL EQUATION.

The problem of determining the correction to be applied to the results of a series of longitude measurements through a chain of submarine cables for the personal equation of the observers is a complicated one. To introduce arbitrarily in each link of the measurement a correction deduced from observations made either previously or subsequently to the measurements is not at all satisfactory, as, under the varied climatic and physiological conditions inevitably experienced in a chain of positions extending over nearly 90° in latitude, such a correction must at times vary considerably from the truth. The most trustworthy and accurate way of eliminating this source of error would doubtless be to exchange stations after each measurement, but considerations of time and expense make this course manifestly impracticable.

In longitude measurements made over land-lines, where the clocks are compared automatically, the only personal equation involved is that due to the observers' difference in the observation of star transits; but over submarine cables this course is generally impracticable, and the chronometers being compared by means of manual signals, as described on page 25, another source of personal error is introduced, which may increase or diminish the correction for star transits, and which doubtless varies in signals over different cables.

By means of the repeated use of the personal-equation machine of Prof. J. R. Eastman, at the Naval Observatory, it was found that the habitual errors of the various observers engaged in these measurements had all the same sign; that is, they all habitually observed the transit of a star a few hundredths of a second after its actual occurrence, but their respective differences were so small that it seemed evident that to introduce results so minute as corrections would not increase the trustworthiness of the result.

By repeatedly exchanging manual signals through an artificial cable, and comparing the results with those derived from automatic comparison of the chronometers obtained by making them beat on the same chronograph, it was found that an exact comparison within two-hundredths of a second was obtained by manual signals and the Thomson galvanometer.

The probable error, therefore, in any one measurement due to personal habit of observing would certainly be very small.

It was, after much consideration, decided not to introduce any arbitrary correction for personal equation, because such a correction, under the circumstances, must be to a certain degree problematical; but, in order to eliminate as far as possible any such source of error, one observer was, commencing at Lisbon, placed alternately east and west of the other.

While it cannot be expected in so long a chain of measurements to escape all sources of error and uncertainty, it is felt that in this case all practicable effort has been made to attain trustworthy results.

The change of plan in measuring from Greenwich to Lisbon involved the uncertainty of the personal equation of Lieutenant-Commander Davis, the observer at Lisbon, with the observers at Greenwich—an uncertainty which, however small it may be, affects the deduced longitude of the Lisbon station, and, consequently, to an equal amount the other positions of the chain.

METHODS OF OBSERVATION.

In establishing temporary astronomical stations for determining latitude and longitude, after obtaining the requisite permission, great care was exercised in selecting a suitable site for the observatory, and the following desirable conditions were always borne in mind, viz, a firm foundation for the transit pier, an unobstructed north and south view, facility of measurement to some permanent landmark to which the latitude and longitude can be referred, proximity to the telegraph office, so that too much wire should not be required to connect with the observatory, and seclusion from crowds

of inquisitive idlers. These conditions are all of the first importance, and a neglect of any of them would be very likely to cause serious embarrassment.

After selecting a site, an approximate meridian line for use in setting up the observatory and pier was laid out with a theodolite.

After setting up the observatory and building the pier at a station, two or three days were allowed for the cement to harden, and the transit instrument, chronograph, and telegraph instruments were then set up and carefully adjusted.

During the first ensuing clear evening, the transit was placed accurately in the meridian by repeated observations of zenith and circumpolar stars. The telegraph instruments being connected in each place with the cable at the telegraph office or at its landing-hut, where no transmitting stations intervened, so that telegraphic communication was established between the observatories, and everything being in readiness, the routine was as follows:

About 6 p. m. messages were exchanged as to clearness of weather and an understanding was arrived at as to the probable time that the business of the office would be finished, so as to make the cable available for time-signals. About two hours before this time, the night's work was begun by observing the transits of six or eight time-stars, if possible, between the zenith and the equator, and two or three circumpolar stars for instrumental corrections. The time of using the cable varied very much on different parts of the line, it not being clear between Porthcurnow and Carcavellos till 2 o'clock a. m., while between Rio de Janeiro and Montevideo signals could almost always be exchanged by 8 o'clock in the evening. As soon as practicable the observatories were placed in telegraphic communication with each other by connecting the wires leading from the observatories to the telegraph office at each place with the main line. At each observatory the other end of the wire, after including the telegraph instruments in the circuit, was put to earth, care being taken to have the earth plates at both observatories of the same material.

One of the telegraph clerks then came to the observatory at each place to assist in sending messages, and communication being established, the chronometers at the two stations were compared in the following manner:

At ten seconds before the completion of a minute by his chronometer, the senior observer sent a rattle or preparatory signal by tapping his key several times in quick succession, then exactly at the even minute pressed his key again for about a quarter of a second and repeated this signal at intervals of five seconds till the completion of the next even minute. The hour and minute when the first signal was sent were then telegraphed to the receiving station and repeated to insure correctness.

The time of arrival of these signals, marked by the deflection of the ray of light from the galvanometer, was recorded by the chronograph at the receiving station and five similar sets were sent from each observatory to the other, making sixty-five comparisons sent and received by each observer.

After these exchanges of signals, the transits of five or six more time stars and two or three circumpolars were observed, completing the night's work.

The position of the axis of the transit instrument was reversed every night, and the level was applied and read after nearly every star.

In exchanging time signals positive currents were always used to guard against any possible difference of velocity as compared with negative currents. This course had the additional advantage that only one sending-key was used and the deflection of the ray of light at the receiving station took place always in the same direction.

In the cases where a transmitting station was interposed between the two terminal stations, the comparison of time-pieces became somewhat more complicated though equally exact. This was the case between Greenwich and Lisbon, between Lisbon and Madeira, and between Para and Pernambuco, the circumstances and consequently the method of proceeding in each case being different.

There is no source of danger more carefully guarded against by the officials having charge of submarine cables than any connection with land-lines which may admit the heavy-battery power used in their working. Between Greenwich and Lisbon the telegraphic connection consisted of the land-line between Greenwich and Porthcurnow, the cable between Porthcurnow and Carcavellos, and the land-line between Carcavellos and Lisbon. The clock at Greenwich was, therefore, compared automatically with the chronometer at Porthcurnow, each time-piece being made successively, by an arrangement of relays, to mark seconds on the chronograph at the other station, and the chronometers at Porthcurnow and Carcavellos were then compared by cable signals in the manner described above, the chronometers at Carcavellos and Lisbon being automatically compared both before and after the exchange, through the cable. As it would have been difficult to keep the lines open for a subsequent exchange between Greenwich and Porthcurnow, and as the rates of the time-pieces at those two stations were very small, only one set of exchanges each way was attempted.

In this manner a very exact and satisfactory comparison between the Greenwich and Lisbon time-pieces was effected.

Between Lisbon and Madeira the connection was made by a land-line between Lisbon and Carcavellos and the cable from Carcavellos to Madeira. To effect the comparison between the Lisbon and Madeira clocks, an automatic exchange was made both ways between Lisbon and Carcavellos; then the comparison was made in the usual manner through the cable between Carcavellos and Madeira, and immediately on its completion another automatic comparison was exchanged between Lisbon and Carcavellos.

Between Para and Pernambuco the cable was in a faulty condition, so that signals could not safely be sent through, and a transmitting station was therefore established at Maranham, with a break-circuit chronometer and chronograph, in charge of Assistant Paymaster Michler. The comparison of the Para and Maranham chronometers was effected in the usual manner; then the Maranham and Pernambuco time-pieces were compared in the same way, this exchange being followed by another comparison between Para and Maranham.

Before commencing the observations at each station, observing lists for time and circumpolar stars were prepared, the stars being arranged in the order of their right ascension, and the list showing the star's magnitude, zenith distance when on the meridian, and the approximate chronometer time of its culmination.

In preparing to observe for latitude, pairs of stars, generally not more than 20°

from the zenith and not differing in zenith distances when on the meridian more than ten minutes of arc, were selected from the British Association Catalogue. It was found desirable that the stars of a pair should be not less than one and not more than fifteen minutes of right ascension apart. An observing list was prepared showing the available pairs of stars in order of right ascension, the zenith distance north or south of each, and the approximate chronometer time of culmination.

The telescope being in the meridian, one of the movable stops on the horizontal circle was moved up against one side of the tangent-clamp, and there fixed by its clamp-screw. The tangent-clamp was then loosened and the instrument was turned 180° on its vertical axis. This was generally done by turning off approximately 180° by the horizontal circle and then adjusting the telescope in azimuth by a circumpolar star on the middle transit-thread at culmination. When the telescope was adjusted in its new position, the other stop was fixed against the other side of the tangent clamp. It could then be turned exactly 180° in azimuth, bringing up against the stops when in the meridian line.

Being set at the mean zenith distance of the first pair of stars, the bubble of the transverse level was brought to the middle of its tube, and as soon as the first star of the pair arrived on the middle transit-thread it was bisected by the middle micrometer-thread, and the time by chronometer noted by the assistant, who also recorded the micrometer and level reading. The telescope was then turned in azimuth 180°, the tangent-clamp bringing up against its stop. On the appearance of the second star it was bisected on the middle transit-thread by the same micrometer-wire, and the time, micrometer, and level again read and recorded. This proceeding was repeated for each pair of stars, from ten to twenty pairs being generally observed. From two to four evenings were generally devoted to latitude, the observations being in all cases made by Lieutenant Norris.

REDUCTION OF OBSERVATIONS.

CHRONOMETER CORRECTIONS.

The reduction of the time observations has been performed by Miles Rock, esq., Lieut. Commander C. H. Davis, and Lieut. J. A. Norris.

Mr. Rock has computed the observations made by Lieutenant-Commander Green, and Lieutenant-Commander Davis and Lieutenant Norris have reduced their respective observations, the system followed by all computers being the same.

A preliminary reduction of each night's work was first made by applying to the mean recorded time of the transit of each star the corrections for missed threads, flexure of axis, level, azimuth, collimation, diurnal aberration, and hourly rate. In this preliminary reduction the instrumental constants for azimuth and collimation were approximately determined by the formulæ

$$a = \frac{(\alpha' - \alpha) - (t' - t)}{A' - A}$$

$$c = \frac{(\alpha' - \alpha) - (t' - t)}{\sec \delta' + \sec \delta}$$

The difference between the reduced meridian passage of each star and its right ascension gave a value for the clock correction at the epoch to which all observa-

tions of that night were reduced. In cases where this value differed considerably from the mean, the star was examined, and if still found anomalous was rejected in the final reduction. Each star then furnished an equation of condition for the final reduction to determine the corrections to be applied to the assumed values of the azimuth and collimation constants and the mean clock-correction. These equations were solved by the method of least squares, and with the corrected values of the azimuth and collimation constants new clock-corrections were obtained. The mean of these was the final accepted clock-correction.

As it was found that a reversal of the axis invariably caused a change in the azimuth of the instrument, the equations of condition were divided into two or more groups, each involving a separate value of the azimuth constant.

The weight unity (p) was assigned to all clock-stars whose declination did not exceed 40°, and which were observed on the normal number (11) of threads. For all stars whose declination exceeded 40° the weight assigned to a complete observation of 11 threads was derived from the formula

$$p = \frac{1.3}{\sqrt{1 + 0.3}} \sec^2 \delta$$

which was deduced by Prof. T. H. Safford from a large number of observations (See Report of Chief of Engineers United States Army for 1873, p. 1243.)

Each conditional equation was further weighted for missed threads according to the formula

$$p' = \frac{1 + \frac{3}{N}}{1 + \frac{3}{n}}$$

N denoting the normal number (11) of threads and n the number of threads observed. (See Chauvenet's Astronomy, Vol. II, p. 198.)

The hourly rate was obtained by interpolation, or graphically, from the differences between the mean preliminary clock-corrections of the several nights at a station.

The middle time of the signals sent and received at both stations was the epoch of comparison. The middle time of observation of clock-stars was the epoch to which the reduction was usually made, and the clock-correction was reduced to the epoch of comparison by applying the hourly rate.

The right ascensions of all stars observed at stations north of the equator were taken from the American Nautical Almanac, or from the Berlin list of 539 stars. The places in the American Nautical Almanac have been corrected by the constant $-0^s.024$ + the correction given by Professor Newcomb in Appendix III, Washington Observations, 1870. At stations south of the equator, the places were taken partly from a list furnished by Dr. B. A. Gould, director of the Argentine National Observatory, and partly from the American Nautical Almanac, Berlin List, and *Connaissance des Temps* In taking the apparent right ascensions, the differences of longitude between the meridians of the Ephemerides and the several stations have been taken into account whenever the effect was sensible.

In the column marked "Chronometer corrections" the quantities in brackets are the apparent corrections derived from circumpolar stars, and do not enter into the final determination of the chronometer corrections.

To guard against clerical errors, the chronograph sheets were always read twice and collated with the record.

DEDUCED LONGITUDE.

As the rate of the chronometers is so small that the difference between the times shown by their faces is practically the same throughout the time of exchanging signals, the middle time between the beginning and end of the exchange of time-signals is used as the time to which the chronometer corrections are reduced.

If we call this time T_0 and let ΔT_e and ΔT_w = the corrections to the eastern and western chronometers at T_0

T' and T'' the differences between the chronometers by signals sent from east to west and from west to east.

w = wave and armature time, or the time required for the electric impulse to pass through the conductor and to overcome the inertia of the receiving instrument.

λ' = the difference of longitude deduced from signals sent from east to west.

λ'' = the difference of longitude deduced from signals sent from west to east.

λ = the true difference of longitude;

Then—

$$\lambda' = (\Delta T_e - \Delta T_w) + T' + w$$
$$\lambda'' = (\Delta T_e - \Delta T_w) + T'' - w$$
$$\lambda = \tfrac{1}{2}(\lambda' + \lambda'')$$
$$w = \tfrac{1}{2}(\lambda' - \lambda'')$$

The details of the deduced longitudes for each night will be found on pp. ———.

REDUCTION OF LATITUDE OBSERVATIONS.

As the resulting latitudes would be directly affected by any error in the declinations of the stars observed, great care was necessary to use only those places which had been accurately determined. For latitudes in the northern hemisphere, only such stars were used as could be found in the American, English, and French Nautical Almanacs, the Berlin Star List, or the various Greenwich catalogues. For southern latitudes, where no suitable stars could be found in the above list, Dr. B. A. Gould, director of the observatory at Cordova, very kindly determined the places with great exactness of such B. A. C. stars as were necessary to make up an observing-list.

We were also much indebted to E J. Stone, esq., lately Her Majesty's astronomer at the Cape of Good Hope, for star places from his general list, not then published.

The reductions to apparent place, as well as the latitude reductions, were made by Lieutenant Norris. In reducing the latitude-work, the following notation was adopted:

Let—

φ = the latitude of the station.

δ_n and δ_s = the apparent declinations of the northern and southern stars.

z_n and z_s = the zenith distance of the northern and southern stars.
z_0 = the zenith distance corresponding to the zero-reading of the micrometer.
m_n and m_s the micrometer-reading for northern and southern stars in seconds of arc.
M_n and M_s = the micrometer-reading for northern and southern stars in micrometer revolutions.
l_n and l_s = the state of the level for northern and southern stars in seconds of arc.
L_n and L_s = the state of the level for northern and southern stars in level divisions.
r_n and r_s = the value of mean refraction at z_n and z_s.

In these observations either one of two cases may occur.

CASE I.—When the circle is east for the northern star and west for the southern star, the micrometer-reading increases as the zenith distance decreases.

CASE II.—When the circle is east for the southern star and west for the northern star, the micrometer-reading increases as the zenith distance increases.

Hence in CASE I—

$$z_n = z_0 - m_n + l_n + r_n$$
$$z_s = z_0 - m_s + l_s + r_s$$
$$z_s - z_n = (m_n - m_s) - (l_n - l_s) - (r_n - r_s)$$

And in CASE II—

$$z_n = z_0 + m_n - l_n + r_n$$
$$z_s = z_0 + m_s - l_s + r_s$$
$$z_s - z_n = -(m_n - m_s) + (l_n - l_s) - (r_n - r_s)$$

Now, since—

$$\varphi = \delta_s + z_s = \delta_n - z_n = \tfrac{1}{2}(\delta_n + \delta_s) + \tfrac{1}{2}(z_s - z_n)$$

Then in CASE I—

$$\varphi = \tfrac{1}{2}(\delta_n + \delta_s) + \tfrac{1}{2}(m_n - m_s) - \tfrac{1}{2}(l_n - l_s) - \tfrac{1}{2}(r_n - r_s)$$

And in CASE II—

$$\varphi = \tfrac{1}{2}(\delta_n + \delta_s) - \tfrac{1}{2}(m_n - m_s) + \tfrac{1}{2}(l_n - l_s) - \tfrac{1}{2}(r_n - r_s)$$

$$\tfrac{1}{2}(m_n - m_s) = \tfrac{1}{2}(M_n - M_s) \times \begin{cases} 65''.80 \text{ for transit No. 1.} \\ 66''.83 \text{ for transit No. 2.} \\ 69''.70 \text{ for transit No. 1505.} \end{cases}$$

$$\tfrac{1}{2}(l_n - l_s) = \tfrac{1}{2}(L_n - L_s) \times \begin{cases} 1.00 \text{ for transits No. 1 and 2} \\ 1.07 \text{ for transit No. 1505.} \end{cases}$$

$$\tfrac{1}{2}(r_n - r_s) = \tfrac{1}{2}(m_n - m_s) \times \begin{cases} 0.0167 \text{ for } z = 0°. \\ 0.0200 \text{ for } z = 20°. \\ 0.0225 \text{ for } z = 30°. \end{cases}$$

These values for differential refraction are taken from Bessel's tables, as given in Chauvenet's Astronomy. The stars were all observed on the meridian or so nearly so as to require no correction on that account, otherwise the correction $6.1347\ \tau^2 \sin 2\ \delta$ should be applied where τ = the star's hour angle in seconds of time.

DATA OF REDUCTION AND RESULTS.

The following table of clock-errors entering into the Greenwich–Lisbon measurement was received from the Astronomer Royal:

ROYAL OBSERVATORY, GREENWICH.

ERRORS AND RATES OF THE STANDARD SIDEREAL CLOCK.

Date.	Clock slow.	Observer.	No. of stars.	Adopted correction slow at 16 hrs. Sid. Time.	Adopted daily rate.
1879.	s			s	s
June 25....	6.80	C.	2	6.81	— 0.26
	6.82	T.	5		
26....	6.50	A. D.	1	6.57	— 0.17
	6.59	H. P.	7		
27....	6.50	T.	2	6.48	— 0.20
	6.47	C.	5		
28....	6.15	A. D.	11	6.16	— 0.19
	6.19	B. D.	2		
29....	6.08	T.	4	6.09	— 0.20
	6.11	C.	2		

Transits of stars observed at Lisbon, Portugal, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	
1879.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
June 25	48 Cephei, S. P. . .	W.	10	10 44 20.24	+1.10	−0.22	−5.40	−1.99	+0.04	13.77	3 5 2.33	+4 20 [48.56]	.
	1 H. Ursæ Minoris	.	11	10 52 29.44	−1.25	+0.43	+1.70	+1.07	0.04	31.43	15 13 18.70	47.27	0.1
	μ' Bootis	.	10	10 59 10.76	0.68	0.20	−0.03	0.51	0.03	10.79	15 19 58.40	47.61	0.1
	ν Bootis pr.	.	11	11 5 50.56	0.72	0.24	+0.08	0.54	0.03	50.73	15 26 38.13	47.40	0.2
	α Coronæ Borealis .	.	11	11 8 49.92	0.60	0.20	−0.30	0.46	0.03	49.71	15 29 37.14	47.43	0.2
	ϕ Bootis	.	11	11 12 44.38	−0.71	0.27	+0.06	+0.53	0.03	44.56	15 33 32.06	47.50	0.2
	5 H. Camelop., S. P.	.	10	11 16 55.02	+0.56	−0.22	−3.81	−1.34	0.02	50.23	3 37 37.77	[47.54]	. .
	ϵ Serpentis	.	11	11 24 3.74	−0.45	+0.17	−0.74	+0.41	0.02	3.15	15 44 50.59	47.44	0.1
	ζ Ursæ Minoris . .	W.	8	11 27 36.47	−2.03	+0.70	+4.09	+1.98	+0.02	41.23	15 48 28.97	+4 20 [47.74]	.
	θ Draconis	E.	11	11 38 51.54	+0.98	+0.45	+1.12	−0.90	+0.01	53.20	15 59 40.57	+4 20 [47.37]	.
	Groom. 2320 . . .	.	11	11 45 12.86	1.26	0.64	2.21	1.17	+0.01	15.81	16 6 3.15	[47.34]	. .
	19 Ursæ Minoris . .	.	11	11 53 28.68	1.79	1.03	+4.28	1.83	0.00	33.95	16 14 21.54	[47.59]	.
	λ Ophiuchi	.	11	12 4 5.58	0.44	0.25	−1.00	0.44	0.00	4.83	16 24 52.35	47.52	0.2
	σ Herculis	.	11	12 9 27.22	0.73	0.38	+0.16	0.59	0.00	27.90	16 30 15.44	47.54	0.2
	ζ Herculis	.	11	12 15 59.06	0.63	0.33	−0.24	0.51	−0.01	59.26	16 36 46.87	47.61	0.2
	η Herculis	.	11	12 18 0.20	0.70	0.36	+0.02	0.56	0.01	0.71	16 38 48.18	47.47	0.2
	49 Herculis	.	11	12 25 50.82	0.51	0.28	−0.69	0.45	0.01	50.46	16 46 37.79	47.33	0.2
	κ Ophiuchi	E.	11	12 31 13.10	+0.48	+0.27	−0.83	−0.44	−0.02	12.56	16 51 59.97	+4 20 47.41	0.2

$a' = -1^s.321$ (circle W.); $a'' = -1^s.673$ (circle E.); $c = 0^s.421$ (−with circle E.).

Chronometer No. 1254, at $12^h\ 0^m$ chron. time, $4^h\ 20^m\ 47^s.48 \pm 0^s.018$ slow, gaining $0^s.032$ per hour.

$14^h\ 40^m$ $4^h\ 20^m\ 47^s.39$

June 26	2 H. Ursæ Minoris	E.	11	10 34 55.27	+1.20	+0.27	+0.81	−0.92	+0.03	56.66	14 55 43.33	+4 20 [46.67]	.
	3 Serpentis	.	11	10 48 27.24	0.45	0.08	−0.39	0.37	0.02	27.03	15 9 13.84	46.81	0.1
	1 H. Ursæ Minoris	.	11	10 52 30.57	1.25	0.22	+0.90	0.98	0.02	31.98	15 13 18.66	[46.68]	
	u' Bootis	.	11	10 59 11.26	0.68	0.15	−0.01	0.47	0.02	11.63	15 19 58.39	46.76	0.2
	β Coronæ Bor. . .	.	11	11 2 6.67	0.61	0.14	0.13	0.43	0.02	6.88	15 22 53.64	46.76	0.2
	θ Coronæ Bor. . .	.	11	11 7 19.37	+0.63	+0.14	0.10	−0.43	0.01	19.62	15 28 6.12	46.50	0.2
	5 H. Camelop., S. P.	.	11	11 16 53.02	−0.56	−0.13	2.02	+1.04	0.01	51.36	3 37 37.83	[46.47]	
	β Serpentis . . .	E.	11	11 19 52.86	+0.52	+0.12	−0.28	−0.38	+0.01	52.85	15 40 39.53	+4 20 46.68	0.2
	γ Serpentis	W.	11	11 30 8.82	−0.52	+0.06	−0.34	+0.35	0.00	8.37	15 50 55.25	+4 20 46.88	0.1
	Groom. 2320 . .	.	11	11 45 15.81	1.26	0.07	+1.11	0.91	−0.01	16.63	16 6 3.14	[46.51]	.
	γ Herculis	.	11	11 55 52.07	0.54	0.02	−0.29	0.36	0.01	51.61	16 16 38.32	46.71	0.2
	ω Herculis	.	11	11 59 7.08	0.51	0.02	−0.36	0.35	0.02	6.56	16 19 53.23	46.67	0.2
	A Draconis	.	11	12 7 29.43	1.30	0.04	+1.19	0.95	0.02	30.29	16 28 17.19	[46.90]	
	ζ Herculis	.	11	12 16 0.44	0.63	+0.04	−0.12	0.40	0.03	0.10	16 36 46.88	46.78	0.1
	49 Herculis	.	11	12 25 51.65	0.51	0.00	−0.35	0.35	0.03	51.11	16 46 37.79	46.68	0.2
	ϵ Ursæ Minoris . .	W.	10	12 37 39.49	−2.90	+0.08	+4.28	+2.52	−0.04	43.43	16 58 30.29	+4 20 [46.86]	.

$a' = -0^s.697$ (circle E.); $a'' = -0^s.837$ (circle W.); $c = 0^s.352$ (−with circle E.).

Chronometer No. 1254, at $11^h\ 30^m$ chron. time, $4^h\ 20^m\ 46^s.72 \pm 0^s.022$ slow, gaining $0^s.034$ per hour.

$14^h\ 25^m$ $4^h\ 20^m\ 46^s.62$

Transits of stars observed at Lisbon, Portugal, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
June 27	μ' Bootis	W.	11	10 59 12.73	—0.68	—0.12	—0.01	+0.60	+0.03	12.55	15 19 58.39	+4 20 45.84	0.05
	ν Bootis pr.	.	11	11 5 52.44	0.72	0.15	+0.04	0.63	0.03	52.27	15 26 38.11	45.84	0.05
	α Coronæ Borealis	.	11	11 8 51.61	—0.60	—0.14	—0.16	+0.54	0.03	51.28	15 29 37.13	45.85	0.04
	5 H. Camelop., S. P.	.	11	11 16 54.59	+0.56	+0.16	2.05	—1.56	0.02	51.72	3 37 37.89	[46.17]	. .
	κ Serpentis	.	9	11 22 35.48	—0.54	—0.17	—0.26	+0.50	0.02	35.03	15 43 20.92	45.89	0.00
	ζ Ursæ Minoris . .	.	11	11 27 40.52	2.03	0.70	+2.20	2.32	0.02	42.33	15 48 28.84	[46.51]	. .
	Groom. 2320. . .	.	11	11 45 16.88	1.26	0.58	+0.94	1.28	+0.01	17.27	16 6 3.10	[45.83]	. .
	γ Herculis	.	11	11 55 53.06	0.54	0.38	—0.25	0.50	0 00	52.39	16 16 38.32	45.93	0.04
	ω Herculis	W.	11	11 59 8.07	—0.51	—0.41	—0.33	+0.49	0.00	7.31	16 19 53.23	+4 20 45.92	0.03
	Groom. 2373. . .	E.	10	12 15 6.55	+1.97	+0.23	+3.18	—2.38	—0.01	9.54	16 35 56.32	+4 20 [46.78]	. .
	49 Herculis	.	11	12 25 52.22	0.51	0.13	—0.45	0.53	0.01	51.87	16 46 37.79	45.92	0.03
	κ Ophiuchi	.	11	12 31 14.41	0.48	0.14	0.53	0.52	0.02	13.96	16 51 59.98	46.02	0.13
	ε Herculis	.	11	12 34 56.88	0.62	0.22	0.17	0.59	0.02	56.94	16 55 42.99	46.05	0.16
	d Herculis	.	11	12 36 25.74	0.65	0.24	0.11	0.61	0.02	25.89	16 57 11.50	45.61	0.28
	60 Herculis	.	11	12 39 4.04	0.50	0.20	—0.48	0.52	0.02	3.72	16 59 49.59	45.87	0.02
	ζ Draconis	.	11	12 47 42.66	+1.18	+0.40	+1.21	—1.24	0.03	44.18	17 8 29.70	[45.52]	. .
	Groom. 966, S. P.	E.	11	13 2 51.74	—0.84	—0.34	—3.81	+1.83	—0.03	48.55	5 23 34.74	+4 20 [46.19]	. .

$a' = -0^s.708$ (circle W.); $a'' = -1^s.079$ (circle E.); $c = 0^s.492$ (— with circle E.).
Chronometer No. 1254, at $12^h\ 0^m$ chron. time, $4^h\ 20^m\ 45^s.89 \pm 0^s.023$ slow, gaining $0^s.033$ per hour.
$15^h\ 30^m$ $4^h\ 20^m\ 45^s.77$

ine 29	3 Serpentis	E.	11	10 48 29.48	+0.45	+0.04	—0.16	—0.53	+0.02	29.30	15 9 13.83	+4 20 44.53	0.04
	1 H. Ursæ Minoris	.	8	10 52 33.29	1.25	0.17	+0.37	1.39	0.02	33.71	15 13 18.59	[44.88]	. .
	μ' Bootis	.	11	10 59 13.63	0.68	0.16	—0.01	0.66	0.02	13.82	15 19 58.36	44.54	0.05
	ι Draconis	.	11	11 1 32.59	0.99	0.26	+0.20	1.03	0.02	33.03	15 22 17.52	[44.49]	. .
	θ Coronæ Borealis	.	11	11 7 21.46	+0.63	+0.16	—0.04	—0.62	0.02	21.61	15 28 6.11	44.50	0.01
	5 H. Camelop., S. P.	.	11	11 16 54.76	—0.56	—0.25	0.82	+1.61	0.01	54.75	3 37 40.28	[45.53]	. .
	π Serpentis	.	11	11 22 37.60	+0.40	+0.22	0.19	—0.52	0.01	37.52	15 43 21.94	44.42	0.07
	ε Serpentis	.	11	11 24 6.13	+0.45	+0.26	0.16	—0.53	+0.01	6.16	15 44 50.57	44.43	0.06
	Groom. 750, S. P.	E.	11	11 38 25.24	—3.63	—1.56	—2.85	+6.31	0.00	23.51	3 59 4.90	+4 20 [41.41]	. .
	19 Ursæ Minoris . .	W.	11	11 53 34.79	—1.79	—0.21	+1.61	+2.06	0.00	36.46	16 14 21.37	+4 20 [44.91]	. .
	η Draconis (seq.) .	.	11	12 1 39.93	1.05	0.05	+0.52	1.04	—0.01	40.38	16 22 24.74	[44.36]	. .
	λ Ophiuchi	.	11	12 4 8.22	0.44	0.01	—0.38	0.49	0.01	7.87	16 24 52.37	44.50	0.01
	ζ Herculis	.	11	12 16 2.47	0.63	—0.02	0.09	0.58	0.01	2.30	16 36 46.88	44.58	0.09
	49 Herculis	.	11	12 25 53.65	0.51	0.00	0.26	0.51	0.02	53.37	16 46 37.80	44.43	0.06
	κ Ophiuchi	.	11	12 31 15.94	0.48	—0.05	0.31	0.50	0.02	15.58	16 51 59.97	44.39	0.10
	ε Herculis	.	11	12 34 58.64	—0.62	—0.11	0.10	+0.57	0.02	58.36	16 55 42.99	44.63	0.14
	19 H. Camelop., S. P.	W.	11	12 41 58.30	+1.33	+0.42	—2.95	—2.60	—0.02	54.48	5 2 39.42	+4 20 [44.94]	. .

$a' = -0^s.285$ (circle E.); $a'' = -0^s.632$ (circle W.); $c = 0^s.508$ (— with circle E.).
Chronometer No. 1254, at $11^h\ 45^m$ chron. time, $4^h\ 20^m\ 44^s.49 \pm 0^s.017$ slow, gaining $0^s.025$ per hour.
$14^h\ 20^m$ $4^h\ 20^m\ 44^s.43$

Transits of stars observed at Lisbon, Portugal, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Jan. 21	63 Aurigæ	E.	10	2 32 12.76	+0.71	—0.12	+ 0.08	—0.96	—0.08	12.39	7 3 18.66	+4 31 6.27	0.03
	δ Draconis, S. P. .	.	11	2 41 30.28	—0.40	+0.06	—10.17	+1.94	0.08	21.63	19 12 27.68	[6.05]	. .
	ι Geminorum . . .	.	11	2 47 6.60	+0.61	—0.10	0.85	—0.85	0.08	5.33	7 18 11.54	6.21	0.03
	β Canis Minoris . .	.	11	2 49 30.56	0.48	0.08	2.07	0.75	0.07	28.07	7 20 34.32	6.25	0.03
	α Geminorum . . .	.	11	2 55 46.26	0.64	0.10	0.55	0.88	0.07	45.30	7 26 51.57	6.27	0.03
	α Canis Minoris . .	.	11	3 1 53.49	0.46	0.08	2.24	0.75	0.07	50.81	7 32 57.11	6.30	0.03
	κ Geminorum . . .	.	11	3 6 2.55	0.59	0.10	1.08	0.82	0.06	1.08	7 37 7.57	6.49	0.13
	ϕ Geminorum . . .	.	11	3 14 59.44	+0.60	—0.10	0.91	—0.84	0.06	58.13	7 46 4.47	6.34	0.14
	ϵ Draconis, S. P. .	E.	11	3 17 33.20	—0.51	+0.08	—11.23	+2.17	—0.06	23.65	19 48 30.20	+4 31 [6.55]	. .
	η Cancri	W.	11	3 54 36.66	—0.56	—0.14	— 1.34	+0.76	—0.04	35.34	8 25 41.64	+4 31 6.30	0.14
	δ Cancri	.	11	4 6 42.72	0.54	0.14	1.48	0.75	0.04	41.27	8 37 47.47	6.20	0.10
	ι Cancri	.	11	4 8 15.88	0.62	0.16	0.77	0.82	0.04	15.11	8 39 21.39	6.28	0.12
	ζ Hydræ	.	7	4 17 54.60	0.47	0.12	2.19	0.72	0.03	52.51	8 48 58.91	6.40	0.11
	a^2 Cancri	.	11	4 20 46.56	0.50	0.13	— 1.84	0.73	0.03	44.79	8 51 51.08	6.29	0.08
	1 Draconis	.	9	4 48 19.27	2.84	0.72	+19.57	5.04	—0.02	40.30	9 19 47.03	[6.73]	. .
	9 Draconis	W.	11	5 53 31.27	—1.84	—0.47	+10.47	+3.02	+0.01	42.46	10 24 48.66	+4 31 [6.20]	.

$a' = -4^s.061$ (circle E.); $a'' = -4^s.052$ (circle W.); $c = 0^s.729$ (— with circle E.).
Chronometer No. 1295, at $5^h\ 34^m.7$ chron. time, $4^h\ 31^m\ 6^s.300 \pm 0^s.015$ slow, losing $0^s.027$ per hour.

Jan. 22	γ Geminorum . . .	W.	11	1 59 35.37	—0.53	—0.07	—0.08	+0.70	—0.08	35.31	6 30 42.13	+4 31 6.82	0.21
	15 Monocerotis . . .	.	11	2 3 10.86	0.49	0.06	0.10	0.68	0.08	10.81	6 34 17.79	6.98	0.17
	ϵ Geminorum . . .	.	11	2 5 21.20	0.59	0.08	—0.05	0.74	0.08	21.14	6 36 27.99	6.85	0.13
	51 Cephei	.	5	2 12 6.03	—7.55	—0.96	+3.04	+13.87	0.08	14.35	6 43 21.63	[7.28]	.
	50 Draconis, S. P. .	.	11	2 19 7.79	+0.88	+0.11	—0.70	—2.63	0.07	5.38	18 50 12.22	[6.84]	.
	ζ Geminorum . . .	.	11	2 25 47.84	—0.56	—0.07	—0.06	+0.72	0.07	47.80	6 56 54.76	6.96	0.22
	63 Aurigæ	.	11	2 32 11.89	0.71	0.09	0.00	0.87	0.07	11.89	7 3 18.66	6.77	0.23
	λ Geminorum . . .	.	11	2 40 0.34	0.53	0.07	—0.08	0.70	0.06	0.30	7 11 7.25	6.95	0.21
	δ Geminorum . . .	W.	11	2 41 45.74	—0.57	—0.07	—0.06	+0.72	—0.06	45.70	7 12 52.65	+4 31 6.95	0.25
	ι Cancri	E.	8	4 8 14.60	+0.62	—0.02	—0.05	—0.80	—0.04	14.31	8 39 21.40	+4 31 7.09	0.13
	a^2 Cancri	.	11	4 20 44.60	0.50	0.02	0.13	0.72	0.03	44.20	8 51 51.09	6.89	0.20
	θ Hydræ	.	11	4 36 56.62	0.44	0.02	0.17	0.70	0.02	56.15	9 8 3.16	7.01	0.15
	38 Lyncis	.	11	4 40 11.20	0.69	0.02	0.01	0.88	0.02	10.96	9 11 17.72	6.76	0.24
	40 Lyncis	.	11	4 42 33.25	0.67	0.02	—0.02	0.85	0.02	33.01	9 13 39.90	6.89	0.17
	1 Draconis	.	11	4 48 40.85	+2.84	—0.10	+1.37	—4.95	0.01	40.00	9 19 47.10	[7.10]	.
	β Cephei, S. P. . .	E.	11	4 55 52.81	—0.52	+0.02	—0.79	+2.05	—0.01	53.56	21 27 0.42	+4 31 [6.86]	
	γ^1 Leonis	E.	11	5 42 10.40	+0.56	+0.09	—0.14	—0.75	0.00	10.16	10 13 16.95	+4 31 6.79	0.17
	9 Draconis	E.	11	5 53 41.67	+1.84	+0.30	+1.11	—2.97	+0.01	41.96	10 24 48.73	+4 31 [6.77]	

$a' = -0^s.196$ (circle W.); $a'' = -0^s.284$ (circle E. to β Cephei); $a''' = -0^s.429$ (circle E. after β Cephei); $c = 0^s.685$ (— with circle E.).
Chronometer No. 1295, at $5^h\ 33^m.7$ chron. time, $4^h\ 31^m\ 6^s.906 \pm 0^s.018$ slow, losing $0^s.023$ per hour.

Transits of stars observed at Funchal, Madeira, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Jan. 21	β Cancri	E.	7	4 11 6.47	+0.26	+0.18	+0.19	+0.18	+0.04	7.32	8 9 56.20	+3 58 48.88	0.07
	κ Cephei, S.P. . .	.	6	4 14 0.46	−0.43	−0.41	2.07	−0.95	0.04	0.78	20 12 50.06	[49.28]	. .
	30 Monocerotis . .	.	7	4 20 46.24	+0.22	+0.16	+0.29	+0.17	0.03	47.11	8 19 36.00	48.89	0.06
	Groom. 1446 . .	.	7	4 27 24.06	0.84	0.46	−1.17	0.63	0.02	24.84	8 26 13.89	[49.05]	. .
	ι Cancri	.	6	4 40 31.65	0.31	0.17	+0.03	0.20	0.01	32.37	8 39 21.40	49.03	0.08
	57 Cancri med. . .	.	7	4 48 0.85	0.32	0.22	0.02	0.20	+0.01	1.62	8 46 50.54	48.92	0.03
	α^2 Cancri . . . . ,	E.	7	4 53 1.32	+0.26	+0.25	+0.17	+0.18	0.00	2.18	8 51 51.23	+3 58 49.05	0.10
	θ Hydræ	W.	7	5 9 14.18	−0.24	+0.47	−0.03	−0.21	−0.01	14.16	9 8 3.16	+3 58 49.00	0.05
	40 Lyncis	.	7	5 14 50.91	0.34	0.66	+0.00	0.25	0.01	50.97	9 13 39.89	48.92	0.03
	η Leonis	.	7	6 1 54.64	0.28	0.03	−0.02	0.22	0.05	54.10	10 0 43.02	48.92	0.03
	32 Ursæ Majoris . .	.	7	6 10 26.73	0.56	0.15	+0.08	0.51	0.05	25.84	10 9 14.24	[48.40]	. .
	9 Draconis	.	7	6 26 0.23	0.84	0.50	+0.18	0.89	0.06	59.12	10 24 48.68	[49.56]	. .
	41 Leonis Minoris .	.	7	6 38 0.48	0.30	0.24	−0.01	0.23	0.07	0.11	10 36 49.10	48.99	0.04
	42 Leonis Minoris .	W.	7	6 40 18.56	−0.32	+0.27	−0.00	−0.24	−0.08	18.19	10 39 7.12	+3 58 48.93	0.02

$a' = +0^s.483$ (circle E.); $a'' = -0^s.060$ (circle W.); $c = 0^s.191$ (+ with circle E.).
Chronometer No. 1254, at $5^h\ 0^m$ chron. time, $3^h\ 58^m\ 48^s.95 \pm 0^s.013$ slow, gaining $0^s.045$ per hour.
$5^h\ 36^m.1$ $3^h\ 58^m\ 48^s.92$

an. 22	66 Orionis	E.	7	1 59 44.86	+0.24	+0.02	+0.25	+0.17	+0.04	45.58	5 58 33.64	+3 58 48.06	0.01
	ν Orionis	.	7	2 1 49.79	0.27	0.03	+0.16	+0.18	0.04	50.47	6 0 38.57	48.10	0.03
	22 H. Camelop. . .	.	7	2 6 40.89	+0.62	+0.11	−0.89	+0.49	0.04	41.26	6 5 28.98	[47.72]	. .
	δ Ursæ Min., S. P.	.	5	2 12 22.98	−2.27	−0.65	+7.72	−2.94	0.03	24.87	18 11 15.18	[50.31]	. .
	23 H. Camelop. . .	.	7	2 26 44.88	+1.05	+0.08	−2.14	+0.97	0.02	44.86	6 25 32.86	[48.00]	. .
	15 Monocerotis . .	.	7	2 35 29.02	0.26	−0.02	+0.20	0.18	0.02	29.66	6 34 17.78	48.12	0.05
	θ Geminorum . . .	.	7	2 45 58.92	0.33	0.17	−0.02	0.21	+0.01	59.28	6 44 47.46	48.18	0.11
	δ Geminorum . . .	E.	7	3 14 4.13	+0.29	−0.07	+0.10	+0.19	−0.01	4.63	7 12 52.64	+3 58 48.01	0.06
	κ Geminorum . . .	W.	7	3 38 19.85	−0.30	+0.15	+0.05	−0.23	−0.03	19.49	7 37 7.58	+3 58 48.09	0.02
	ϕ Geminorum . . .	.	7	3 47 17.15	0.31	−0.21	0.04	0.24	0.03	16.40	7 46 4.48	48.08	0.01
	6 Cancri	.	7	3 57 16.76	0.31	0.17	+0.03	0.24	0.04	16.03	7 56 4.08	48.05	0.02
	3 Ursæ Majoris . .	.	7	4 1 58.68	0.62	0.30	−0.55	0.58	0.04	56.59	8 0 45.20	[48.61]	. .
	Groom. 1446 . .	.	7	4 27 28.61	0.75	0.02	−0.81	0.76	0.06	26.21	8 26 13.90	[47.69]	. .
	δ Cancri	.	7	4 39 0.03	0.28	−0.04	+0.09	0.22	0.07	59.51	8 37 47.48	47.97	0.10
	ρ Ursæ Majoris . ,	W.	7	4 52 50.83	−0.60	0.00	−0.52	−0.56	−0.08	49.07	8 51 37.01	+3 58 [47.94]	. .

$a' = +0^s.524$ (circle E.); $a'' = +0^s.335$ (circle W.); $c = 0^s.192$ (+ with circle E.).
Chronometer No. 1254, at $3^h\ 0^m$ chron. time, $3^h\ 58^m\ 48^s.07 \pm 0^s.014$ slow, gaining $0^s.043$ per hour.
$5^h\ 35^m.2$ $3^h\ 58^m\ 47^s.86$

Transits of stars observed at Lisbon, Portugal, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1878.				h. m. s.	s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
Jan. 23	δ Ursæ Min., S. P.	E.	5	1 40 7.87	−5.36	+1.27	−7.99	+13.05	−0.05	8.79	18 11 15.29	+4 31 [6.50]	. .
	μ Geminorum . . .	.	11	1 44 30.56	+0.57	−0.13	−0.17	− 0.84	0.05	29.94	6 15 37.13	7.19	0.10
	23 Camelopardalis .	.	3	1 54 26.68	2.32	0.55	+2.13	4.32	0.05	26.21	6 25 32.80	[6.59]	. .
	γ Geminorum . . .	.	11	1 59 35.34	0.53	0.13	−0.23	0.81	0.05	34.65	6 30 42.13	7.48	0.19
	15 Monocerotis . . .	.	11	2 3 11.29	0.49	0.12	0.28	0.78	0.05	10.55	6 34 17.78	7.23	0.06
	ε Geminorum . . .	.	11	2 5 21.34	0.59	0.14	−0.15	0.85	0.05	20.74	6 36 27.98	7.24	0.05
	51 Cephei	.	1	2 12 14.40	7.55	1.78	+9.01	16.02	0.04	13.12	6 43 21.48	[8.36]	. .
	ζ Geminorum . . .	.	11	2 25 48.04	0.56	0.13	−0.19	0.83	0.04	47.41	6 56 54.76	7.35	0.06
	63 Aurigæ	.	11	2 32 11.94	0.71	0.17	+0.01	1.00	0.04	11.45	7 3 18.66	7.21	0.08
	Piazzi vii. 67 . .	E.	11	2 47 9.13	+1.31	−0.31	+0.80	− 2.13	−0.03	8.77	7 18 16.16	+4 31 [7.39]	. .
	α Canis Minoris . .	W.	11	3 1 50.19	−0.46	+0.08	−0.78	+ 0.74	−0.03	49.74	7 32 57.12	+4 31 7.38	0.09
	κ Geminorum . . .	.	11	3 6 0.21	0.58	0.11	0.38	0.82	0.03	0.15	7 37 7.58	7.43	0.14
	φ Geminorum . . .	.	11	3 14 57.16	−0.60	+0.11	0.32	+ 0.83	0.02	57.16	7 46 4.47	7.31	0.02
	ε Draconis, S. P. .	.	3	3 17 28.71	+0.51	−0.09	3.93	− 2.16	0.02	23.02	19 48 30.22	[7.20]	. .
	η Cancri	.	11	3 54 34.60	−0.56	+0.11	0.47	+ 0.79	0.01	34.46	8 25 41.65	7.19	0.10
	δ Cancri	.	11	4 6 40.37	0.54	0.10	0.52	0.78	−0.01	40.18	8 37 47.49	7.31	0.02
	ι Cancri	.	11	4 8 14.19	0.62	0.11	0.27	0.85	0.00	14.26	8 39 21.41	7.15	0.14
	ζ Hydræ	.	11	4 17 52.06	0.47	0.08	0.76	0.75	0.00	51.66	8 48 58.93	7.27	0.02
	a^2 Cancri	.	11	4 20 44.12	0.50	0.09	−0.65	0.76	0.00	43.82	8 51 51.10	7.28	0.01
	1 Draconis	W.	7	4 48 30.14	−2.83	+0.52	+6.87	+ 5.24	+0.01	39.95	9 19 47.16	+4 31 [7.21]	. .

$a' = -0^s.580$ (circle E.); $a'' = -1^s.422$ (circle W.); $c = 0^s.757$ (— with circle E.).
Chronometer No. 1295, at $4^h\ 23^m.2$ chron. time, $4^h\ 31^m\ 7^s.289 \pm 0^s.017$ slow, losing $0^s.019$ per hour.

Jan. 26	π_4 Orionis	E.	11	0 13 37.60	+0.46	−0.02	− 0.60	− 1.37	−0.02	36.05	4 44 44.26	+4 31 8.21	0.02
	π_5 Orionis	.	11	0 16 49.04	0.44	0.02	0.65	1.36	0.02	47.43	4 47 55.54	8.11	0.08
	ι Tauri	.	11	0 24 43.21	+0.56	−0.02	0.35	− 1.45	0.02	41.93	4 55 50.17	8.24	0.05
	ε Ursæ Min., S. P.	.	8	0 27 15.01	−2.09	+0.08	6.91	+10.02	0.02	16.09	16 58 24.40	[8.31]	. .
	β Tauri	.	11	0 47 29.94	+0.61	−0.02	− 0.22	− 1.54	0.01	28.76	5 18 36.95	8.19	0.00
	Groom. 966 . . .	.	11	0 52 23.66	1.71	0.06	+ 2.48	5.22	0.01	22.56	5 23 31.37	[8.81]	. .
	ϕ^1 Orionis	.	10	0 57 2:57	0.48	0.02	− 0.54	1.37	0.01	1.11	5 28 9.32	8.21	0.02
	ζ Tauri	.	11	0 59 16.39	+0.56	−0.02	0.35	− 1.45	0.01	15.12	5 30 23.38	8.26	0.07
	ω Draconis, S. P. .	.	11	1 6 28.48	−0.46	+0.02	2.86	+ 3.75	0.01	28.92	17 37 37.32	[8.40]	. .
	130 Tauri	.	8	1 9 14.58	+0.54	−0.02	− 0.41	− 1.42	0.01	13.26	5 40 21.46	8.20	0.01
	ν Aurigæ	.	11	1 11 57.51	+0.71	−0.02	+ 0.01	− 1.75	0.01	56.45	5 43 4.52	8.07	0.12
	ψ Draconis, S. P. .	E.	5	1 12 54.23	−0.64	+0.02	− 3.33	+ 4.43	−0.01	54.70	17 44 2.96	+4 31 [8.26]	. .
	θ Aurigæ	W.	11	1 20 17.57	−0.69	−0.08	− 0.06	+ 1.66	−0.01	18.39	5 51 26.66	+4 31 8.27	0.08
	η Geminorum . . .	.	11	1 36 24.53	−0.57	−0.06	0.57	+ 1.43	0.01	24.75	6 7 33.08	8.33	0.14
	δ Ursæ Min., S. P.	.	8	1 40 51.38	+5.36	+0.59	−25.95	−22.33	−0.01	9.04	18 11 15.80	[6.76]	. .
	23 Camelopardalis .	.	11	1 54 13.30	−2.32	−0.25	+ 6.90	+ 7.40	0.00	25.03	6 25 32.70	[7.67]	. .
	γ Geminorum . . .	.	11	1 59 33.85	0.53	0.06	− 0.74	1.38	0.00	33.90	6 30 42.13	8.23	0.04
	15 Monocerotis . . .	W.	11	2 3 9.66	−0.49	−0.05	− 0.92	+ 1.34	0.00	9.54	6 34 17.77	+4 31 8.23	0.04
	ε Geminorum . . .	W.	11	2 5 19.32	−0.59	+0.12	− 0.43	+ 1.46	0.00	19.88	6 36 27.97	+4 31 8.09	0.10
	51 Cephei	.	5	2 11 26.13	7.55	1.51	+25.92	27.43	0.00	73.44	6 43 21.06	[7.62]	. .
	ζ Geminorum . . .	.	11	2 25 46.20	0.56	0.11	− 0.55	1.41	0.00	46.61	6 56 54.76	8.15	0.04
	63 Aurigæ	W.	11	2 32 9.40	−0.71	+0.14	+ 0.03	+ 1.72	0.00	10.58	7 3 18.66	+4 31 8.08	0.11

$a' = -1^s.090$ (circle E.); $a'' = -1^s.884$ (circle W. to 15 Monocerotis); $a''' = -1^s.669$ (circle W. after 15 Monocerotis); $c = 1^s.339$ (— with circle E.).
Chronometer No. 1295, at $2^h\ 30^m.0$ chron. time, $4^h\ 31^m\ 8^s.195 \pm 0^s.013$ slow, losing $0^s.008$ per hour.
$5^h\ 25^m.7$ $4^h\ 31^m\ 8^s.218$

Transits of stars observed at Funchal, Madeira, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Jan. 23	9 Camelopardalis .	E.	7	0 43 11.39	+0.56	+0.16	−0.84	+0.76	+0.08	12.11	4 41 59.18	+3 58 [47.07]	. .
	ι Tauri	.	7	0 57 2.24	0.29	0.11	+0.13	0.33	0.06	3.16	4 55 50.20	47.04	0.08
	11 Orionis	.	7	0 58 49.79	0.27	0.11	0.19	0.32	0.06	50.74	4 57 37.87	47.13	0.01
	τ Orionis	.	7	1 12 54.63	0.21	0.10	0.39	0.31	0.05	55.69	5 11 42.88	47.19	0.07
	β Tauri	.	7	1 19 48.90	+0.31	0.15	0.03	+0.35	0.05	49.79	5 18 37.00	47.21	0.09
	δ Ursæ Min., S. P.	.	7	2 12 26.50	−2.27	1.10	+9.04	−5.79	+0.01	28.59	18 11 15.35	[46.76]	. .
	23 H. Camelop. . .	E.	7	2 26 46.04	+1.05	+0.44	−2.51	+1.72	0.00	46.74	6 25 32.84	+3 58 [46.10]	. .
	ζ Geminorum . . .	W.	7	2 58 8.20	−0.29	+0.03	+0.14	−0.37	−0.02	7.69	6 56 54.76	+3 58 47.07	0.05
	Piazzi vii, 67 . .	.	7	3 19 30.77	0.61	0.20	−1.09	0.95	0.03	28.29	7 18 16.15	[47.86]	. .
	ρ Geminorum . . .	.	7	3 22 32.02	0.32	0.11	+0.01	0.40	0.04	31.38	7 21 18.46	47.08	0.04
	φ Geminorum . . .	.	7	3 47 17.73	0.31	0.28	0.07	0.39	0.05	17.33	7 46 4.49	47.16	0.04
	6 Cancri	.	7	3 57 17.35	0.31	0.33	+0.06	0.39	0.06	16.98	7 56 4.09	47.11	0.01
	Groom. 1408 . .	.	7	4 5 33.53	0.83	0.97	−1.87	1.43	0.07	30.30	8 4 19.05	[48.75]	. .
	ζ Hydræ	.	7	4 50 12.00	0.25	0.25	+0.29	0.35	0.10	11.84	8 48 58.93	47.09	0.03
	σ^8 Ursæ Majoris . .	W.	7	5 0 58.56	−0.59	+0.47	−0.98	−0.90	−0.11	56.45	8 59 43.77	+3 58 [47.32]	. .

$a' = +0^s.614$ circle E.); $a'' = +0^s.653$ (circle W.); $c = 0^s.325$ (+ with circle E.).
Chronometer No. 1254, at $2^h\ 30^m$ chron. time, $3^h\ 58^m\ 47^s.12 \pm 0^s.013$ slow, gaining $0^s.042$ per hour.
$4^h\ 24^m.7$ $3^h\ 58^m\ 47^s.04$

Jan. 26	μ Eridani	E.	7	0 40 40.89	+0.22	+0.19	+0.30	+0.15	+0.05	41.80	4 39 25.90	+3 58 44.10	0.16
	9 Camelopardalis .	.	7	0 43 14.41	0.57	0.50	−0.70	0.37	0.05	15.20	4 41 59.11	[43.91]	. .
	π^4 Orionis	.	7	0 45 59.44	0.25	0.21	+0.24	0.15	0.05	0.34	4 44 44.26	43.92	0.02
	π^5 Orionis	.	7	0 49 10.77	0.24	0.20	+0.26	0.15	0.05	11.67	4 47 55.53	43.86	0.08
	ι Aurigæ	.	6	0 50 20.40	+0.33	+0.28	0.00	+0.18	0.05	21.24	4 49 5.00	43.76	0.18
	ε Ursæ Min., S. P.	.	7	0 59 39.24	−0.86	−0.70	+3.44	−1.38	0.04	39.78	16 58 24.44	[44.66]	. .
	β Orionis	.	7	1 9 57.40	+0.21	+0.15	+0.34	+0.15	0.03	58.28	5 8 42.30	44.02	0.08
	Groom. 966 . . .	E.	7	1 24 46.04	+0.78	+0.58	−1.33	+0.58	+0.02	46.67	5 23 31.42	+3 58 [44.75]	. .
	δ Ursæ Min., S. P.	W.	6	2 12 17.63	+2.27	+1.10	+8.32	+2.55	−0.01	31.86	18 11 15.86	+3 58 [44.00]	. .
	23 H. Camelop. . .	.	7	2 26 53.74	−1.05	−0.52	−2.31	−1.05	0.02	48.79	6 25 32.79	[44.00]	. .
	γ Geminorum . .	.	7	2 31 58.66	0.28	0.13	+0.16	0.20	0.02	58.19	6 30 42.08	43.89	0.05
	15 Monocerotis . . .	.	7	2 35 34.20	0.26	0.12	0.22	0.19	0.02	33.83	6 34 17.77	43.94	0.00
	ζ Geminorum . . .	.	6	2 58 11.23	0.29	0.20	0.12	0.20	0.03	10.63	6 56 54.76	43.13	0.19
	λ Geminorum . . .	.	7	3 12 23.83	−0.28	−0.16	0.16	−0.20	0.05	23.30	7 11 7.26	43.96	0.02
	τ Draconis, S. P. .	.	7	3 19 1.44	+0.26	+0.14	1.87	+0.52	0.05	4.18	19 17 48.10	[43.92]	. .
	ρ Geminorum . . .	W.	7	3 22 35.38	−0.32	−0.16	+0.01	−0.22	−0.06	34.63	7 21 18.46	+3 58 43.83	0.11

$a' = +0^s.512$ (circle E.); $a'' = +0^s.565$ (circle W.); $c = 0^s.169$ (+ with circle E.).
Chronometer No. 1254, at $2^h\ 0^m$ chron. time, $3^h\ 58^m\ 43^s.94 \pm 0^s.025$ slow, gaining $0^s.040$ per hour.
$5^h\ 27^m.3$ $3^h\ 58^m\ 43^s.80$

Transits of stars observed at Lisbon, Portugal, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Jan. 27	ν Aurigæ	W.	11	1 11 55.39	—0.71	+0.09	— 0.01	+ 1.64	0.00	56.40	5 43 4.51	+4 31 8.11	0.19
	θ Aurigæ	.	11	1 20 17.26	—0.69	+0.09	+ 0.05	+ 1.59	0.00	18.30	5 51 26.65	8.35	0.05
	δ Ursæ Min., S. P.	.	11	1 40 1.23	+5.36	—0.68	22.55	—21.44	0.00	7.02	18 11 15.95	[8.93]	. .
	ζ Geminorum . . .	.	11	2 25 45.01	—0.56	+0.07	+ 0.54	+ 1.36	0.00	46.42	6 56 54.76	8.34	0.04
	63 Aurigæ	.	11	2 32 9.39	—0.71	+0.09	— 0.03	+ 1.64	+0.01	10.39	7 3 18.66	8.27	0.03
	δ Draconis, S. P. .	.	11	2 41 18.58	+0.40	—0.05	+ 4.10	— 3.31	0.01	19.73	19 12 27.81	[8.08]	. .
	ι Geminorum . . .	W.	11	2 47 1.85	—0.61	+0.08	+ 0.34	+ 1.44	+0.01	3.11	7 18 11.55	+4 31 8.44	0.14
	α Geminorum . . .	E.	11	2 55 43.81	+0.64	+0.19	+ 0.22	— 1.54	+0.01	43.33	7 26 51.60	+4 31 8.27	0.03
	α Canis Minoris . .	.	11	3 1 48.54	0.46	0.14	0.89	1.31	0.01	48.73	7 32 57.14	8.41	0.11
	κ Geminorum . . .	.	11	3 5 59.67	0.58	0.18	0.43	1.43	0.01	59.44	7 37 7.60	8.16	0.14
	φ Geminorum . . .	.	9	3 14 56.45	+0.60	+0.18	0.37	— 1.46	0.01	56.15	7 46 4.52	8.37	0.07
	ε Draconis, S. P. .	.	9	3 17 14.58	—0.51	—0.15	4.48	+ 3.80	0.01	22.21	19 48 30.27	[8.06]	. .
	6 Cancri	.	11	3 24 56.12	+0.61	+0.18	0.34	— 1.48	0.01	55.78	7 56 4.11	8.33	0.03
	β Cancri	.	10	3 38 47.86	+0.48	+0.15	0.80	— 1.32	0.01	47.98	8 9 56.23	8.25	0.05
	κ Cephei, S. P. . .	E.	3	3 41 30.49	—1.10	—0.33	+ 6.65	+ 5.94	+0.01	41.66	20 12 50.38	+4 31 [8.72]	. .

$a' = +1^s.637$ (circle W.); $a'' = +1^s.622$ (circle E.); $c = 1^s.286$ (— with circle E.).
Chronometer No. 1295, at $1^h\ 24^m.7$ chron. time, $4^h\ 31^m\ 8^s.300 \pm 0^s.020$ slow, losing $0^s.005$ per hour.

Transits of stars observed at Funchal, Madeira, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Jan. 27	66 Orionis	E.	7	1 59 49.40	+0.24	+0.33	+ 0.38	+0.21	+0.04	50.60	5 58 33.62	+3 58 43.02	0.00
	δ Ursæ Min., S. P.	.	7	2 12 32.36	—2.24	— 2.94	+11.56	—4.19	0.03	34.58	18 11 16.01	[41.43]	. .
	23 H. Camelop. . .	.	7	2 26 49.49	+1.05	+1.27	— 3.21	+1.19	0.02	49.81	6 25 32.78	[42.97]	. .
	γ Geminorum . . .	.	7	2 31 58.00	0.28	0.33	+ 0.23	0.22	0.02	59.08	6 30 42.08	43.00	0.02
	15 Monocerotis . . .	.	7	2 35 33.67	0.26	0.30	+ 0.31	0.22	0.01	34.77	6 34 17.77	43.00	0.02
	θ Geminorum . . .	.	7	2 46 3.47	+0.33	+0.40	— 0.02	+0.26	0.01	4.45	6 44 47.46	43.01	0.01
	50 Draconis, S. P. .	.	7	2 51 28.61	—0.33	—0.41	+ 2.94	—0.98	+0.01	29.84	18 50 12.09	[42.25]	. .
	ζ Geminorum . . .	E.	7	2 58 10.64	+0.29	+0.37	+ 0.17	+0.23	0.00	11.70	6 56 54.76	+3 58 43.06	0.04
	λ Geminorum . . .	W.	7	3 12 24.59	—0.28	—0.12	+ 0.13	—0.26	—0.01	24.05	7 11 7.26	+3 58 43.21	0.19
	ρ Geminorum . . .	.	7	3 22 36.21	0.32	—0.08	0.01	0.29	0.01	35.52	7 21 18.46	42.94	0.08
	α Canis Minoris . .	.	7	3 34 14.43	0.25	0.00	+ 0.21	0.25	0.02	14.12	7 32 57.10	42.98	0.04
	Groom. 1374 . .	.	7	3 47 0.91	0.76	+0.06	— 1.13	0.91	0.03	58.14	7 45 40.82	[42.68]	. .
	6 Cancri	.	7	3 57 21.70	0.31	0.03	+ 0.04	0.28	0.03	21.15	7 56 4.11	42.96	0.06
	Groom. 1408 . .	.	7	4 5 38.93	—0.83	+0.12	— 1.32	—1.03	0.04	35.83	8 4 19.11	[43.28]	. .
	κ Cephei, S. P. . .	.	7	4 14 3.93	+0.43	—0.07	+ 1.98	+0.97	0.04	7.20	20 12 50.07	[42.87]	. .
	δ Cancri	W.	7	4 39 4.95	—0.28	+0.05	+ 0.12	—0.26	—0.06	4.52	8 37 47.53	+3 58 43.01	0.01

$a' = +0^s.785$ (circle E.); $a'' = +0^s.461$ (circle W.); $c = 0^s.230$ (+ with circle E.).
Chronometer No. 1254, at $3^h\ 0^m$ chron. time, $3^h\ 58^m\ 43^s.02 \pm 0^s.016$ slow, gaining $0^s.036$ per hour.
$1^h\ 26^m.3$ $3^h\ 58^m\ 43^s.08$

Transits of stars observed at St. Vincent, Cape Verde Islands, by Lieut.-Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Feb. 28	51 Cephei	E.	11	3 14 34.82	+3.85	+1.16	+61.53	—1.76	+0.05	99.65	6 43 12.66	+3 27 [33.01]	. .
	ζ Geminorum . . .	.	11	3 29 19.35	0.58	0.18	+ 0.23	0.09	0.04	20.29	6 56 54.57	34.28	0.07
	λ Geminorum . . .	.	11	3 43 32.22	0.57	0.17	— 0.01	0.09	0.03	32.89	7 11 7.11	34.22	0.01
	δ Geminorum . . .	.	11	3 45 17.44	0.59	0.18	+ 0.31	0.09	0.03	18.46	7 12 52.49	34.03	0.18
	Piazzi vii, 67 . .	.	11	3 50 33.73	0.93	0.28	+ 6.82	0.23	0.03	41.56	7 18 15.54	[33.98]	. .
	β Canis Minoris . .	.	8	3 52 59.77	0.55	0.16	— 0.46	0.09	0.03	59.96	7 20 34.20	34.24	0.03
	α Canis Minoris . .	.	11	4 5 22.51	0.54	0.16	— 0.62	0.09	0.02	22.52	7 32 57.02	34.50	0.29
	κ Geminorum . . .	E.	11	4 9 31.96	+0.60	+0.18	+ 0.47	—0.09	+0.02	33.14	7 37 7.51	+3 27 34.37	0.16
	3 Ursæ Majoris . .	W.	10	4 33 4.31	—0.94	+0.20	+ 6.73	+0.13	+0.01	10.44	8 0 44.93	+3 27 [34.49]	. .
	β Cancri	.	9	4 42 22.94	0.55	0.12	— 0.40	0.04	0.00	22.15	8 9 56.24	34.09	0.12
	30 Monocerotis . . .	.	11	4 52 3.29	0.51	0.11	— 1.10	0.04	0.00	1.83	8 19 36.04	34.21	0.00
	η Cancri	.	11	4 58 7.75	0.58	0.13	+ 0.23	0.05	0.00	7.58	8 25 41.75	34.17	0.04
	δ Cancri	.	11	5 10 13.85	0.58	0.13	+ 0.10	0.05	0.00	13.55	8 37 47.63	34.08	0.13
	ε Hydræ	.	11	5 12 47.88	0.54	0.12	— 0.54	0.04	—0.01	46.95	8 40 21.30	34.35	0.14
	ζ Hydræ	.	11	5 21 25.79	0.54	0.12	0.56	0.04	0.01	24.84	8 48 59.09	34.25	0.04
	α^2 Cancri	.	11	5 24 17.75	0.56	0.12	— 0.25	0.05	0.01	17.10	8 51 51.28	34.18	0.03
	σ^2 Ursæ Majoris . .	.	11	5 32 4.15	0.91	0.20	+ 6.25	0.12	0.01	9.80	8 59 44.15	[34.35]	. .
	1 Draconis	W.	11	5 51 54.35	—1.64	+0.36	+19.77	+0.32	—0.02	73.14	9 19 47.80	+3 27 [34.66]	. .

$a' = -3^s.149$ (circle E.); $a'' = -3^s.086$ (circle W.); $c = 0^s.065$ (— with circle E.).
Chronometer No. 1295, at $5^h\ 0^m$ chron. time, $3^h\ 27^m\ 34^s.213 \pm 0^s.024$ slow, gaining $0^s.026$ per hour.
$7^h\ 2^m.6$ $3^h\ 27^m\ 34^s.160$

March 2	γ Geminorum . . .	W.	10	3 3 9.01	—0.57	—0.03	— 0.02	+0.23	+0.10	8.72	6 30 41.83	+3 27 33.11	0.01
	51 Cephei	.	10	3 14 35.25	3.85	0.21	+64.17	4.56	0.10	100.02	6 43 11.90	[31.88]	. .
	ζ Geminorum . . .	.	11	3 29 21.48	0.58	0.03	+ 0.24	0.24	0.09	21.44	6 56 54.54	33.10	0.02
	λ Geminorum . . .	.	11	3 43 34.29	0.57	0.03	— 0.01	0.23	0.09	34.00	7 11 7.08	33.08	0.04
	δ Geminorum . . .	.	9	3 45 19.29	0.59	0.03	+ 0.33	0.24	0.09	19.33	7 12 52.47	33.14	0.02
	Piazzi vii, 67 . .	.	11	3 50 35.20	0.93	0.05	+ 7.11	0.61	0.09	42.03	7 18 15.46	[33.43]	. .
	α Canis Minoris . .	.	11	4 5 24.77	0.54	0.03	— 0.05	0.22	0.08	23.85	7 32 57.00	33.15	0.03
	κ Geminorum . . .	W.	11	4 9 34.13	—0.60	—0.03	+ 0.49	+0.24	+0.08	34.31	7 37 7.49	+3 27 33.18	0.06
	3 Ursæ Majoris . .	E.	11	4 33 4.00	+0.94	+0.02	+ 6.90	—0.72	+0.07	11.21	8 0 44.87	+3 27 [33.66]	. .
	β Cancri	.	11	4 42 23.22	0.55	0.01	— 0.41	0.26	0.07	23.18	8 9 56.23	33.05	0.07
	30 Monocerotis . . .	.	11	4 52 3.54	0.51	0.01	— 1.11	0.26	0.06	2.75	8 19 36.03	33.28	0.16
	δ Cancri	.	10	5 10 14.06	0.58	0.01	+ 0.10	0.27	0.06	14.54	8 37 47.62	33.08	0.04
	ε Hydræ	.	11	5 12 48.32	0.54	0.01	— 0.55	0.26	0.06	48.12	8 40 21.29	33.17	0.05
	ζ Hydræ	.	11	5 21 26.17	0.54	0.01	0.58	0.26	0.05	25.93	8 48 59.08	33.15	0.03
	α^2 Cancri	.	11	5 24 18.06	0.56	0.01	— 0.26	0.27	0.05	18.15	8 51 51.27	33.12	0.00
	σ^2 Ursæ Majoris . .	.	11	5 32 4.22	0.91	0.02	+ 6.44	0.68	0.05	10.96	8 59 44.12	[33.16]	. .
	θ Hydræ	.	11	5 40 30.67	0.53	0.01	— 0.77	0.26	0.05	30.23	9 8 3.38	33.15	0.03
	1 Draconis	E.	9	5 51 55.39	+1.64	+0.03	+20.28	—1.84	+0.04	75.54	9 19 47.71	+3 27 [32.17]	. .
	β Leonis	E.	11	8 15 19.35	+0.57	+0.09	— 0.09	—0.27	—0.01	19.64	11 42 52.78	+3 27 33.14	0.02
	β Virginis	.	9	8 16 50.18	0.53	0.09	0.76	0.26	0.01	49.77	11 44 22.82	33.05	0.07
	o Virginis	.	11	8 31 29.17	0.55	0.09	— 0.40	0.26	0.02	29.13	11 59 2.18	33.05	0.07
	4 Draconis	E.	11	8 38 50.94	+1.29	+0.21	+13.23	—1.28	—0.02	64.37	12 6 37.41	+3 27 [33.04]	. .

$a' = -3^s.284$ (circle W.); $a'' = -3^s.167$ (circle E. to 1 Draconis); $a''' = -3^s.058$ (circle W. after 1 Draconis); $c = 0^s.240$ (— with circle E.).
Chronometer No. 1295, at $7^h\ 43^m.0$ chron. time, $3^h\ 27^m\ 33^s.125 \pm 0^s.011$ slow, gaining $0^s.022$ per hour.

Transits of stars observed at Funchal, Madeira, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Feb. 28	γ Geminorum . . .	E.	7	2 32 38.59	+0.28	−0.30	−0.05	+0.07	+0.06	38.65	6 30 41.82	+3 58 3.17	0.02
	ε Geminorum . . .	.	7	2 38 24.49	0.30	0.34	−0.02	0.07	0.06	24.56	6 36 27.72	3.16	0.01
	24 H. Camelop. . .	.	7	2 44 17.57	+0.88	−1.05	+0.53	+0.30	0.05	18.28	6 42 21.81	[3.53]	. .
	50 Draconis, S. P. .	.	7	2 52 11.57	−0.33	+0.40	−0.63	−0.40	0.05	10.66	18 50 13.98	[3.32]	. .
	δ Geminorum . . .	.	6	3 14 49.22	+0.29	−0.31	0.03	+0.07	0.03	49.27	7 12 52.49	3.22	0.07
	τ Draconis, S. P. .	.	7	3 19 47.21	−0.26	+0.26	−0.55	−0.35	0.02	46.33	19 17 49.52	[3.19]	. .
	ϕ Geminorum . . .	.	7	3 23 15.14	+0.32	−0.32	0.00	+0.08	0.02	15.24	7 21 18.31	3.07	0.08
	α Geminorum . . .	E.	7	3 28 48.03	+0.32	−0.30	0.00	+0.08	+0.01	48.14	7 26 51.23	+3 58 3.09	0.06
	6 Cancri	W.	7	3 58 1.36	−0.31	−0.02	−0.03	−0.11	−0.01	0.88	7 56 4.07	+3 58 3.19	0.04
	Groom. 1408 . .	.	7	4 6 15.63	0.83	+0.16	+0.95	0.43	0.02	15.46	8 4 18.52	[3.06]	. .
	β Cancri	.	7	4 11 53.52	−0.26	+0.09	−0.13	−0.10	0.02	53.10	8 9 56.24	3.14	0.01
	κ Cephei, S. P. . .	.	6	4 14 48.80	+0.43	−0.19	1.41	+0.30	0.03	47.90	20 12 51.20	[3.30]	. .
	η Cancri	.	7	4 27 38.98	−0.29	+0.12	0.07	−0.11	0.04	38.59	8 25 41.75	3.16	0.01
	73 Draconis, S. P. .	.	7	4 34 58.13	+0.30	−0.12	1.18	+0.25	0.04	57.34	20 33 0.34	[3.00]	. .
	δ Cancri	.	7	4 39 44.91	−0.28	+0.11	0.08	−0.11	0.05	44.50	8 37 47.63	3.13	0.02
	ι Cancri	W.	7	4 41 18.79	−0.32	+0.12	−0.02	−0.12	−0.05	18.40	8 39 21.56	+3 58 3.16	0.01

$a' = -0^s.167$ (circle E.); $a'' = -0^s.330$ (circle W.); $c = 0^s.084$ (+ with circle E.).
Chronometer No. 1254, at $3^h\ 45^m$ chron. time, $3^h\ 58^m\ 3^s.15 \pm 0^s.009$ slow, gaining $0^s.053$ per hour.
$7^h\ 4^m.4$ $3^h\ 58^m\ 2^s.97$

Mar. 2	23 H. Camelop. . .	E.	6	2 27 28.88	+1.05	−0.42	−0.82	+0.69	+0.08	29.46	6 25 30.22	+3 58 [0.76]	. .
	γ Geminorum . . .	.	7	2 32 40.67	0.28	0.07	+0.06	0.13	0.08	41.15	6 30 41.79	0.64	0.01
	ε Geminorum . . .	.	7	2 38 26.55	+0.30	0.02	0.03	+0.14	0.07	27.07	6 36 27.68	0.61	0.02
	50 Draconis, S. P. .	.	7	2 52 13.90	−0.33	−0.12	+0.75	−0.63	0.06	13.63	18 50 14.13	[0.50]	. .
	α Geminorum . . .	.	7	3 28 50.03	+0.32	+0.08	0.00	+0.15	0.03	50.61	7 26 51.20	0.59	0.04
	α Canis Minoris . .	.	7	3 34 55.85	0.25	0.04	+0.09	0.12	0.02	56.37	7 32 56.96	0.59	0.04
	κ Geminorum . . .	.	7	3 39 6.25	0.30	0.05	+0.03	0.14	0.02	6.79	7 37 7.48	58 0.69	0.06
	Groom. 1374 . .	E.	7	3 47 39.29	+0.76	+0.16	−0.49	+0.46	+0.01	40.19	7 45 40.05	+3 57 [59.86]	. .
	κ Cephei, S. P. . .	W.	6	4 14 50.95	+0.43	−0.54	−0.55	+0.57	−0.01	50.85	20 12 51.33	+3 58 [0.48]	. .
	η Cancri	.	7	4 27 41.24	−0.29	+0.37	0.03	−0.17	0.02	41.10	8 25 41.73	0.63	0.00
	73 Draconis, S. P. .	.	7	4 34 59.92	+0.30	−0.39	0.46	+0.47	0.03	59.81	20 33 0.45	[0.64]	. .
	δ Cancri	.	7	4 39 47.11	−0.28	+0.37	0.03	−0.17	0.04	46.96	8 37 47.62	0.66	0.03
	ι Cancri	.	7	4 41 21.05	0.32	0.41	−0.01	0.18	0.04	20.89	8 39 21.54	0.65	0.02
	57 Cancri med. . . .	.	7	4 48 50.29	0.32	0.44	0.00	0.19	0.04	50.18	8 46 50.72	0.54	0.09
	ζ Hydræ	.	7	4 50 58.59	0.25	0.34	−0.06	0.16	0.04	58.42	8 48 59.08	0.66	0.03
	σ^2 Ursæ Majoris . .	W.	7	5 1 43.41	−0.59	+0.87	+0.19	−0.42	−0.05	43.41	8 59 43.98	+3 58 [0.57]	. .

$a' = +0^s.199$ (circle E); $a'' = -0^s.128$ (circle W.); $c = 0^s.142$ (+ with circle E.).
Chronometer No. 1254, at $4^h\ 0^m$ chron. time, $3^h\ 58^m\ 0^s.63 \pm 0^s.009$ slow, gaining $0^s.053$ per hour.
$7^h\ 45^m$ $3^h\ 58^m\ 0^s.43$

Transits of stars observed at St. Vincent, Cape Verde Islands, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Mar. 4	γ Geminorum . . .	W.	11	3 3 10.22	—0.57	+0.17	— 0.02	—0.02	+0.04	9.82	6 30 41.80	+3 27 31.98	0.22
	51 Cephei	.	8	3 14 43.57	3.85	1.16	+58.13	0.31	0.03	98.73	6 43 11.02	[32.29]	. .
	ζ Geminorum . . .	.	11	3 29 22.54	0.58	0.18	+ 0.21	0.02	0.03	22.36	6 56 54.51	32.15	0.05
	λ Geminorum . . .	.	10	3 43 35.29	0.57	0.17	— 0.01	0.02	0.02	34.88	7 11 7.06	32.18	0.02
	Piazzi vii, 67 . .	.	11	3 50 37.40	0.93	0.28	+ 6.44	0.04	0.02	43.17	7 18 15.38	[32.21]	. .
	β Canis Minoris . .	.	11	3 53 2.73	0.55	0.16	— 0.44	0.02	0.02	1.90	7 20 34.15	32.25	0.05
	α Canis Minoris . .	.	11	4 5 25.63	0.54	0.16	— 0.59	0.02	0.02	24.66	7 32 56.98	32.32	0.12
	κ Geminorum . . .	W.	11	4 9 35.10	—0.60	+0.18	+ 0.44	—0.02	+0.02	35.12	7 37 7.46	+3 27 32.34	0.14
	3 Ursæ Majoris . .	E.	11	4 33 5.23	+0.94	+0.34	+ 6.14	—0.07	+0.01	12.59	8 0 44.80	+3 27 [32.21]	. .
	β Cancri	.	11	4 42 23.57	0.55	0.20	— 0.37	0.03	+0.01	23.93	8 9 56.21	32.28	0.08
	30 Monocerotis . . .	.	11	4 52 4.15	0.51	0.19	— 0.98	0.02	0.00	3.85	8 19 36.01	32.16	0.04
	η Cancri	E.	11	4 58 8.60	+0.58	+0.21	+ 0.21	—0.03	0.00	9.57	8 25 41.72	+3 27 32.15	0.05
	δ Cancri	E.	11	5 10 14.77	+0.58	—0.07	+ 0.08	—0.03	0.00	15.33	8 37 47.60	+3 27 32.27	0.07
	ϵ Hydræ	.	11	5 12 49.03	0.54	0.07	— 0.44	0.02	0.00	49.04	8 40 21.27	32.23	0.03
	ζ Hydræ	.	11	5 21 26.87	0.54	0.07	0.46	0.02	—0.01	26.85	8 48 59.07	32.22	0.02
	a^2 Cancri	.	11	5 24 18.91	0.56	0.07	— 0.20	0.03	0.01	19.16	8 51 51.26	32.10	0.10
	σ^2 Ursæ Majoris . .	.	11	5 32 6.01	0.91	0.12	+ 5.08	0.06	0.01	11.81	8 59 44.08	[32.27]	. .
	1 Draconis	E.	11	5 51 58.39	+1.64	—0.21	+15.98	—0.18	—0.02	75.60	9 19 47.60	+3 27 [32.00]	. .

$a' = -2^s.975$ (circle W.); $a'' = -2^s.815$ (circle E. to η Cancri); $a''' = -2^s.492$ (circle E. after η Cancri); $c = 0^s.005$ (— with circle E.).
Chronometer No. 1295, at $5^h\ 0^m.0$ chron. time, $3^h\ 27^m\ 32^s.202 \pm 0^s.018$ slow, gaining $0^s.018$ per hour.
$6^h\ 46^m.5$ $3^h\ 27^m\ 32^s.170$

Mar. 5	Piazzi vii, 67 . .	E.	9	3 50 34.45	+0.93	+0.19	+ 7.54	+0.42	+0.03	43.56	7 18 15.33	+3 27 [31.77]	. .
	3 Ursæ Majoris . .	.	10	4 33 3.71	0.94	0.19	+ 7.59	0.42	0.02	12.87	8 0 44.77	[31.90]	. .
	β Cancri	.	11	4 42 24.03	0.55	0.11	— 0.45	0.16	0.02	24.42	8 9 56.19	31.77	0.01
	30 Monocerotis . . .	.	11	4 52 4.55	0.51	0.10	— 1.22	0.15	0.02	4.11	8 19 36.00	31.89	0.11
	η Cancri	.	3	4 58 8.90	0.58	0.12	+ 0.26	0.16	0.02	10.04	8 25 41.71	31.67	0.11
	δ Cancri	.	11	5 10 14.89	0.58	0.12	+ 0.11	0.16	0.01	15.87	8 37 47.59	31.72	0.06
	ϵ Hydræ	.	11	5 12 49.22	0.54	0.11	— 0.61	0.15	0.01	49.42	8 40 21.26	31.84	0.06
	ζ Hydræ	.	11	5 21 27.05	0.54	0.11	0.64	0.15	0.01	27.22	8 48 59.06	31.84	0.06
	a^2 Cancri	E.	11	5 24 19.02	+0.56	+0.11	— 0.29	+0.15	+0.01	19.56	8 51 51.25	+3 27 31.69	0.09
	σ^2 Ursæ Majoris . .	W.	11	5 32 5.99	—0.91	—0.03	+ 7.50	—0.51	+0.01	12.05	8 59 44.05	+3 27 [32.00]	. .
	θ Hydræ	.	11	5 40 33.19	0.53	0.02	— 0.90	0.19	+0.01	31.56	9 8 3.36	31.80	0.02
	1 Draconis	.	7	5 51 55.24	1.64	0.06	+23.62	1.36	0.00	75.80	9 19 47.53	[31.73]	. .
	d Ursæ Majoris . .	.	8	5 56 7.63	0.97	0.04	+ 8.82	0.57	0.00	14.87	9 23 46.52	[31.65]	. .
	o Leonis	.	11	6 7 10.26	0.55	0.02	— 0.42	0.20	0.00	9.07	9 34 40.82	31.75	0.03
	ϵ Leonis	.	11	6 11 26.85	0.60	0.02	+ 0.52	0.21	0.00	26.54	9 38 58.29	31.75	0.03
	μ Leonis	.	11	6 18 20.77	0.60	0.02	+ 0.69	0.22	0.00	20.62	9 45 52.25	31.63	0.15
	π Leonis	.	11	6 26 17.99	0.55	0.02	— 0.54	0.19	—0.01	16.68	9 53 48.48	31.80	0.02
	η Leonis	.	11	6 33 12.37	0.57	0.02	+ 0.03	0.20	0.01	11.60	10 0 43.47	31.87	0.09
	α Leonis	W.	9	6 34 24.28	—0.56	—0.02	— 0.28	—0.20	—0.01	23.21	10 1 55.06	+3 27 31.85	0.07

$a' = -3^s.483$ (circle E.); $a'' = -3^s.688$ (circle W.); $c = 0^s.173$ (+ with circle E.).
Chronometer No. 1295, at $6^h\ 0^m.0$ chron. time, $3^h\ 27^m\ 31^s.780 \pm 0^s.014$ slow, gaining $0^s.016$ per hour.
$7^h\ 29^m.5$ $3^h\ 27^m\ 31^s.756$

Transits of stars observed at Funchal, Madeira, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Mar. 4	24 H. Camelop. . .	E.	2	2 44 21.44	+0.88	+0.66	−0.46	+0.50	+0.06	23.08	6 42 21.51	+3 57 [58.43]	. .
	λ Geminorum . . .	.	5	3 13 8.29	0.28	0.15	+0.04	0.12	0.04	8.92	7 11 7.08	58.16	0.01
	Piazzi vii, 67 . .	.	6	3 20 16.03	0.61	0.29	−0.22	0.31	0.03	17.05	7 18 15.39	[58.34]	. .
	φ Geminorum . . .	.	7	3 23 19.51	0.32	0.15	0.00	0.13	0.03	20.14	7 21 18.25	58.11	0.04
	α Canis Minoris . .	.	7	3 34 58.19	0.25	0.12	+0.06	0.11	0.02	58.75	7 32 56.93	58.18	0.03
	β Geminorum . . .	.	7	3 39 54.70	0.31	0.16	+0.01	0.12	0.02	55.32	7 37 53.46	58.14	0.01
	Groom. 1374 . .	.	7	3 47 40.81	0.76	0.42	−0.33	0.41	+0.01	42.08	7 45 39.95	[57.87]	. .
	6 Cancri	E.	7	3 58 5.24	+0.31	+0.20	+0.01	+0.12	0.00	5.88	7 56 4.02	+3 57 58.14	0.01
	β Cancri	W.	7	4 11 58.10	−0.26	+0.38	−0.03	−0.15	−0.01	58.03	8 9 56.20	+3 57 58.17	0.02
	κ Cephei, S. P. . .	.	6	4 14 53.44	+0.43	−0.65	0.27	+0.51	0.01	53.45	20 12 51.48	[58.03]	. .
	30 Monocerotis . .	.	7	4 21 37.95	−0.22	+0.35	0.04	−0.15	0.02	37.87	8 19 35.98	58.11	0.04
	η Cancri	.	7	4 27 43.58	−0.29	+0.47	0.01	−0.16	0.02	43.57	8 25 41.71	58.14	0.01
	73 Draconis, S. P. .	.	6	4 35 2.46	+0.30	−0.52	−0.23	+0.42	0.03	2.40	20 33 0.55	[58.15]	. .
	ι Cancri	.	7	4 41 23.29	−0.32	+0.56	0.00	−0.17	0.03	23.33	8 39 21.52	58.19	0.04
	α^2 Cancri	.	7	4 53 53.07	0.26	0.48	−0.02	0.15	0.04	53.08	8 51 51.25	58.17	0.02
	σ^2 Ursæ Majoris . .	W.	7	5 1 45.75	−0.59	+1.09	+0.10	−0.39	−0.05	45.91	8 59 43.93	+3 57 [58.02]	. .

$a' = +0.^s134$ (circle E.); $a'' = -0^s.065$ (circle W.); $c = 0^s.129$ (+ with circle E.).
Chronometer No. 1254, at $4^h\ 0^m$ chron. time, $3^h\ 57^m\ 58^s.15 \pm 0^s.006$ slow, gaining $0^s.048$ per hour.
$6^h\ 48^m.5$ $3^h\ 57^m\ 58^s.02$

Mar. 5	35 Draconis, S. P. .	E.	7	1 56 57.10	−0.41	−0.51	−0.42	−0.57	+0.06	55.25	17 54 52.14	+3 57 [56 89]	. .
	22 H. Camelop. . .	.	7	2 7 28.96	+0.62	+0.86	+0.17	+0.26	0.05	30.92	6 5 27.62	[56.70	. .
	η Geminorum . . .	.	6	2 9 34.62	0.29	0.45	−0.02	0.10	0.05	35.49	6 7 32.67	57.18	0.12
	μ Geminorum . . .	.	7	2 17 38.69	0.29	0.45	−0.02	0.10	0.04	39.55	6 15 36.74	57.19	0.13
	23 H. Camelop. . .	.	7	2 27 29.46	1.05	1.58	+0.41	0.51	0.03	33.04	6 25 29.91	[56.87]	. .
	γ Geminorum . . .	.	7	2 32 43.84	0.28	0.42	−0.03	0.10	0.03	44.64	6 30 41.76	57.12	0.06
	ε Geminorum . . .	.	7	2 38 29.80	0.30	0.45	0.01	0.10	+0.03	30.67	6 36 27.63	56.96	0.10
	λ Geminorum . . .	E.	7	3 13 9.27	+0.28	+0.53	−0.03	+0.10	0.00	10.15	7 11 7.03	+3 57 56.88	0.18
	α Canis Minoris . .	W.	7	3 35 0.02	−0.25	+0.03	+0.11	−0.13	−0.01	59.77	7 32 56.92	+3 57 57.15	0.09
	β Geminorum . . .	.	7	3 39 56.86	0.31	0.03	+0.02	0.15	0.02	56.43	7 37 53.44	57.01	0.05
	π Geminorum . . ,	.	7	3 41 44.46	0.33	0.04	−0.01	0.15	0.02	43.99	7 39 40.98	56.99	0.07
	Groom. 1374 . .	.	7	3 47 44.80	0.76	0.08	−0.56	0.47	0.02	43.07	7 45 39.90	[56.83]	. .
	6 Cancri	.	7	3 58 7.42	0.31	0.03	+0.02	0.15	0.03	6.98	7 56 4.01	57.03	0.03
	Groom. 1408 . .	.	7	4 6 22.90	0.83	0.09	−0.66	0.53	0.04	20.93	8 4 18.26	[57.33]	. .
	β Cancri . . . , . .	.	7	4 11 59.39	0.26	0.03	+0.09	0.13	0.04	59.08	8 9 56.19	57.11	0.05
	ο Ursæ Majoris . .	W.	7	4 22 15.20	−0.50	+0.10	−0.23	−0.27	−0.05	14.25	8 20 11.45	+3 57 [57.20]	. .

$a' = -0^s.100$ (circle E.); $a'' = +0^s.230$ (circle W.); $c = 0^s.110$ (+ with circle E.).
Chronometer No. 1254, at $3^h\ 15^m$ chron. time, $3^h\ 57^m\ 57^s.06 \pm 0^s.022$ slow, gaining $0^s.043$ per hour.
$7^h\ 31^m.5$ $3^h\ 57^m\ 56^s.88$

Transits of stars observed at St. Vincent, Cape Verde Islands, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1519.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Apr. 17	1 Draconis	E.	8	5 47 14.10	+1.64	−0.03	−3.80	−1.65	−0.11	10.15	9 19 43.28	+3 32 [33.13]	. .
	o Leonis	.	11	6 2 7.56	0.55	0.01	+0.07	0.24	0.10	7.83	9 34 40.46	32.63	0.05
	π Leonis	.	11	6 21 15.25	0.55	0.01	0.09	0.24	0.07	15.57	9 53 48.19	32.62	0.04
	α Leonis	.	11	6 29 21.85	0.56	0.01	+0.05	0.24	0.07	22.14	10 1 54.78	32.64	0.06
	30 H. Camelop. . .	.	11	6 43 48.59	1.86	0.03	−4.57	1.96	0.05	43.84	10 16 16 27	[32.43]	. .
	9 Draconis	.	8	6 52 18.19	1.18	0.02	−2.16	0.99	0.04	16.16	10 24 48.48	[32.32]	. .
	ρ Leonis	.	11	6 53 52.74	0.55	0.01	+0.07	0.24	0.04	53.07	10 26 25.62	32.55	0.03
	l Leonis	E.	11	7 10 20.34	+0.56	−0.01	+0.06	−0.24	−0.02	20.69	10 42 53.13	+3 32 32.44	0.14
	3 Draconis	W.	11	8 3 13.95	−0.91	−0.20	−1.12	+0.50	+0.03	12.25	11 35 44.77	+3 32 [32.52]	. .
	β Leonis	.	10	8 10 20.82	0.57	0.12	+0.02	0.20	0.04	20.39	11 42 52.91	32.52	0.06
	β Virginis	.	11	8 11 50.80	0.53	0.12	0.14	0.19	0.04	50.52	11 44 22.98	32.46	0.12
	o Virginis	.	11	8 26 30.14	0.55	0.12	+0.07	0.20	0.06	29.80	11 59 2.40	32.60	0.02
	4 Draconis	.	11	8 34 7.73	1.30	0.28	−2.41	0.95	0.07	4.76	12 6 37.31	[32.55]	. .
	η Virginis	.	11	8 41 10.13	0.52	0.11	+0.16	0.19	0.07	9.92	12 13 42.54	32.62	0.04
	ε Draconis	.	11	8 55 51.42	0.97	0.21	−1.34	0 58	0.09	49.57	12 28 22.25	[32.68]	. .
	γ Virginis	W.	11	9 2 58.91	−0.52	−0.11	+0.17	+0.19	+0.10	58.74	12 35 31.43	+3 32 32.69	0.11

$a' = +0^s.593$ (circle E.); $a'' = +0^s.558$ circle W.); $c = 0^s.213$ (— with circle E.).
Chronometer No. 1519, at $7^h\ 31^m.3$ chron. time, $3^h\ 32^m\ 32^s.577 \pm 0^s.017$ slow, losing $0^s.064$ per hour.

Apr. 18	*δ* Cancri	E.	11	5 5 12.26	+0.58	−0.03	−0.02	+0.26	−0.06	12.99	8 37 47.04	+3 32 34.05	0.03
	e Hydræ	.	11	5 7 45.85	0.54	0.03	+0.13	0.24	0.06	46.67	8 40 20.74	34.07	0.01
	ζ Hydræ	.	11	5 16 23.53	0.54	0.03	0.14	0.24	0.05	24.37	8 48 58.56	34.19	0.11
	a^2 Cancri	.	11	5 19 15.93	0.56	0.03	+0.06	0.25	0.04	16.73	8 51 50.75	34.02	0.06
	a^2 Ursæ Majoris . .	.	11	5 27 8.70	0.91	0.05	−1.53	0.64	0.04	8.63	8 59 42.47	[33.84]	. .
	θ Hydræ	.	11	5 35 28.06	0.53	0.03	+0.18	0.24	0.03	28.95	9 8 2.91	33.96	0.12
	1 Draconis	E.	11	5 47 10.23	+1.64	−0.09	−4.81	+1.71	−0.01	8.67	9 19 43.15	+3 32 [34.48]	. .
	o Leonis	W.	11	6 2 7.25	−0.55	−0.12	+0.06	−0.29	0.00	6.35	9 34 40.45	+3 32 34.10	0.02
	π Leonis	.	11	6 21 14.92	0.55	0.12	+0.07	0.28	+0.02	14.06	9 53 48.17	34.11	0.03
	η Leonis	.	11	6 28 10.13	0.57	0.13	0.00	0.30	0.03	9.16	10 0 43.17	34.01	0.07
	30 H. Camelop. . .	.	11	6 43 50.91	1.86	0.42	−3.95	2.37	0.05	42.36	10 16 16.13	[33.77]	. .
	9 Draconis	.	7	6 52 18.58	1.18	0.27	−1.87	1.19	0.06	14.13	10 24 48.42	[34.29]	. .
	ρ Leonis	W.	10	6 53 32.25	−0.55	−0.12	+0.06	−0.29	+0.06	51.41	10 26 25.61	+3 32 34 20	0.12

$a' = +0^s.751$ (circle E.); $a'' = +0^s.512$ (circle W.); $c = 0^s.262$ (+ with circle E.).
Chronometer No. 1519, at $6^h\ 0^m.0$ chron. time, $3^h\ 32^m\ 34^s\ 079 \pm 0^s.018$ slow, losing $0^s.066$ per hour.
$7^h\ 35^m.7$ $3^h\ 32^m\ 34^s.184$

Transits of stars observed at Pernambuco, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1878.				h. m. s.	s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
Apr. 17	θ Hydræ	E.	11	6 22 30.78	+0.53	—0.09	— 0.17	+0.88	—0.02	31.91	9 8 2.92	+2 45 31.01	0.20
	β Argus	.	9	6 26 15.47	0.72	0.43	+ 2.23	2.48	0.02	20.45	9 11 51.88	[31.43]	. .
	α Hydræ	.	11	6 36 5.19	0.54	—0.36	0.00	0.89	0.02	6.24	9 21 37.38	31.14	0.07
	π Leonis	.	11	7 8 15.97	0.52	0.00	— 0.26	0.89	0.01	17.11	9 53 48.18	31.07	0.14
	μ^1 Chameleon. . . .	.	8	7 18 10.52	1.05	+0.24	+ 5.95	6.04	—0.01	23.79	10 3 55.67	[31.68]	. .
	α Ursæ Majoris . .	.	11	8 10 43.75	0.39	—0.16	— 1.84	1.90	0.00	44.04	10 56 15.77	[31.73]	. .
	β Crateris	.	11	8 20 9.06	0.56	0.24	+ 0.24	0.95	0.00	10.57	11 5 41.84	31.27	0.06
	δ Crateris	E.	10	8 27 44.37	+0.55	—0.26	+ 0.10	+0.91	+0.01	45.68	11 13 16.90	+2 45 31.22	0.01
	τ Leonis	W.	11	8 36 12.84	- 0.53	—0.49	+ 0.11	—0.92	+0.01	11.02	11 21 42.37	+2 45 31.35	0.14
	ε Chameleon. . . .	.	11	9 8 15.74	0.88	0.41	— 2.45	4.27	0.02	7.75	11 53 40.09	[32.34]	. .
	ε Corvi	.	7	9 18 24.08	0.56	0.07	0.15	0.99	0.02	22.33	12 3 53.64	31.31	0.10
	β Chameleon. . . .	.	10	9 25 55.06	0.91	0.24	2.71	4.68	0.02	46.54	12 11 17.71	[31.17]	. .
	α^1 Crucis.	.	11	9 34 25.40	0.68	0.28	— 0.99	1.99	0.02	21.48	12 19 52.04	[30.56]	. .
	f Virginis.	.	10	9 45 3.32	0.54	0.16	+ 0.03	0.92	0.02	1.75	12 30 33.06	31.31	0.10
	γ^1 Virginis	W.	11	9 50 1.73	—0.54	—0.16	+ 0.07	—0.92	+0.02	0.20	12 35 31.43	+2 45 31.23	0.02

$a' = -0^s.903$ (circle E.); $a'' = +0^s.565$ (circle W.); $c = 0^s.900$ (+ with circle E.).
Chronometer No. 1254, at $8^h\ 0^m$ chron. time, $2^h\ 45^m\ 31^s.21 \pm 0^s.026$ slow, losing $0^s.013$ per hour.
$7^h\ 38^m.8$ $2^h\ 45^m\ 31^s.21$

Apr. 18	Groom. 1586 . .	W.	8	7 1 58.37	—0.28	—0.03	+ 6.37	—3.12	—0.04	1.27	9 47 32.15	+2 45 [30.88]	. .
	Lacaille 4744 . .	.	11	8 34 0.95	0.76	0.58	— 5.20	2.81	0.02	51.58	11 19 22.73	[31.15]	. .
	v Leonis	.	11	8 45 14.46	0.54	0.24	+ 0.25	0.89	0.01	13.03	11 30 44.63	31.60	0.03
	3 Draconis	.	11	8 50 12.27	0.35	0.18	3.67	2.31	0.01	13.09	11 35 44.73	[31.64]	. .
	β Virginis	.	11	8 58 52.75	0.53	0.28	0.34	0.89	0.01	51.38	11 44 22.97	31.59	0.02
	π Virginis	.	11	9 9 9 52	0.52	0.29	0.49	0.90	0.01	8.29	11 54 39.88	31.59	0.02
	o Virginis	.	11	9 13 31.85	0.52	0.30	+ 0.56	0.90	0.01	30.68	11 59 2.36	31.68	0.11
	ε Corvi	W.	11	9 18 24.28	—0.56	—0.33	— 0.47	—0.96	—0.01	21.95	12 3 53.64	+2 45 31.69	0.12
	κ Draconis	E.	9	9 42 32.15	+0.32	—0.11	+15.77	+2.53	0.00	50.66	12 28 22.06	+2 45 [31.40]	. .
	γ^1 Viginis	.	8	9 49 57.91	0.54	0.15	0.68	0.85	0.00	59.83	12 35 31.43	31.60	0.03
	8 Draconis	.	6	10 4 55.22	0.36	0.06	12.78	2.09	0.00	10.39	12 50 42.16	[31.77]	. .
	ε Virginis	.	7	10 10 34.48	0.51	0.14	1.85	0.86	+0.01	37.57	12 56 9.02	31.45	0.12
	θ Virginis	.	9	10 18 7.83	0.54	0.22	+ 0.30	0.85	0.01	9.31	13 3 40.79	31.48	0.09
	61 Virginis	.	11	10 26 32.28	0.56	0.03	— 0.94	0.89	0.01	32.77	13 12 4.25	31.48	0.09
	α Virginis	E.	11	10 33 16.28	+0.55	—0.16	— 0.24	+0.86	+0.01	17.30	13 18 48.83	+2 45 31.53	0.04

$a' = +1^s.833$ (circle W.); $a'' = +5^s.383$ (circle E); $c = 0^s.867$ (+ with circle E.).
Chronometer No. 1254, at $9^h\ 45^m$ chron. time, $2^h\ 45^m\ 31^s.57 \pm 0^s.018$ slow, losing $0^s.014$ per hour.
$7^h\ 43^m.3$ $2^h\ 45^m\ 31^s.54$

Transits of stars observed at St. Vincent, Cape Verde Islands, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1519.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Apr. 20	δ Cancri	W.	II	5 5 10.65	−0.58	+0.08	−0.01	−0.22	−0.15	9.77	8 37 47.01	+3 32 37.24	0.07
	ε Hydræ	.	II	5 7 44.14	0.54	0.08	+0.07	0.21	0.15	43.39	8 40 20.71	37.32	0.01
	ζ Hydræ	.	II	5 16 21.97	0.54	0.08	0.07	0.21	0.14	21.23	8 48 58.53	37.30	0.01
	α^2 Cancri	.	II	5 19 14.23	0.56	0.08	+0.03	0.21	0.14	13.43	8 51 50.72	37.29	0.02
	σ^8 Ursæ Majoris . .	.	8	5 27 7.25	0.91	0.13	−0.81	0.55	0.13	4.98	8 59 42.38	[37.40]	. .
	κ Cancri	.	II	5 28 33.67	0.56	0.08	+0.04	0.21	0.13	32.89	9 1 10.27	37.38	0.07
	θ Hydræ	.	II	5 35 26.21	0.53	0.08	+0.10	0.21	0.12	25.53	9 8 2.89	37.36	0.05
	1 Draconis . . , .	W.	II	5 47 11.20	−1.64	+0.24	−2.56	−1.47	−0.10	5.67	9 19 42.89	+3 32 [37.22]	. .
	o Leonis	E.	II	6 2 2.51	+0.55	−0.06	+0.07	+0.17	−0.09	3.15	9 34 40.43	+3 32 37.28	0.03
	π Leonis	.	II	6 21 10.22	0.55	0.06	+0.09	0.17	0.07	10.90	9 53 48.15	37.25	0.06
	η Leonis	.	II	6 28 5.26	0.57	0.06	0.00	0.18	0.06	5.89	10 0 43.15	37.26	0.05
	α Leonis	.	8	6 29 16.77	0.56	0.06	+0.05	0.17	0.06	17.43	10 1 54.74	37.31	0.00
	30 H. Camelop. . .	.	II	6 43 40.55	1.86	0.20	−4.82	1.42	0.05	38.76	10 16 15.86	[37.10]	. .
	9 Draconis	.	8	6 52 11.42	1.18	0.13	−2.28	0.71	0.04	10.86	10 24 48.28	[37.42]	. .
	ρ Leonis	E.	II	6 53 47.58	+0.55	−0.06	+0.08	+0.17	−0.04	48.28	10 26 25.59	+3 32 37.31	0.00
	β Leonis	E.	II	8 10 14.73	+0.57	0.00	+0.02	+0.17	+0.04	15.53	11 42 52.90	+3 32 37.37	0.06
	β Virginis	.	II	8 11 44.82	0.53	0.00	0.15	0.17	0.04	45.71	11 44 22.97	37.26	0.05
	o Virginis	.	II	8 26 24.10	0.55	0.00	+0.08	0.17	0.06	24.96	11 59 2.39	37.43	0.12
	4 Draconis	.	II	8 33 60.24	1.30	0.00	−2.64	0.83	0.07	59.80	12 6 37.16	[37.36]	. .
	η Virginis	E.	II	8 41 4.24	+0.52	0.00	+0.18	+0.17	+0.08	5.19	12 13 42.54	+3 32 37.35	0.04

$a' = +0^s.400$ (circle W.); $a'' = +0^s.625$ (circle E. to ρ Leonis); $a''' = +0^s.610$ (circle E. after ρ Leonis); $c = 0^s.188$ (+ with circle E.).
Chronometer No. 1519, at $7^h\ 28^m.2$ chron. time, $3^h\ 32^m\ 37^s.314 \pm 0^s.009$ slow, losing $0^s.063$ per hour.

Apr. 23	δ Cancri	W.	II	5 5 6.11	−0.58	+0.06	−0.01	−0.06	−0.06	5.46	8 37 46.96	+3 32 41.50	0.01
	ε Hydræ	.	II	5 7 39.73	0.54	0.06	+0.06	0.05	0.06	39.20	8 40 20.67	41.47	0.04
	ζ Hydræ	.	II	5 16 17.37	0.54	0.06	0.06	0.05	0.05	16.85	8 48 58.49	41.64	0.13
	α^2 Cancri	.	5	5 19 9.80	0.56	0.06	+0.03	0.05	0.05	9.23	8 51 50.68	41.45	0.06
	σ^8 Ursæ Majoris . .	W.	7	5 27 2.45	−0.91	+0.10	−0.71	−0.14	−0.04	0.75	8 59 42.24	+3 32 [41.49]	. .
	30 H. Camelop. . .	E.	II	6 43 35.91	+1.86	−0.50	−3.44	+0.11	+0.03	33.97	10 16 15.40	+3 32 [41.43]	. .
	9 Draconis	.	8	6 52 7.16	1.18	0.32	−1.63	0.06	0.04	6.49	10 24 48.07	[41.58]	. .
	ρ Leonis	.	II	6 53 43.48	0.55	0.15	+0.05	0.01	0.04	43.98	10 26 25.56	41.58	0.07
	l Leonis . . . , .	.	II	7 10 11.01	0.56	0.15	0.04	0.01	0.05	11.52	10 42 53.07	41.55	0.04
	χ Leonis	.	II	7 26 3.77	0.55	0.15	0.07	0.01	0.07	4.32	10 58 45.85	41.53	0.02
	θ Leonis	.	II	7 35 10.89	0.57	0.16	0.01	0.01	0.08	11.40	11 7 52.85	41.45	0.06
	σ Leonis	.	II	7 42 11.12	0.54	0.15	0.08	0.01	0.08	11.68	11 14 53.20	41.52	0.01
	ι Leonis	.	II	7 44 54.46	0.56	0.15	0.04	0.01	0.09	55.01	11 17 36.43	41.42	0.09
	τ Leonis	E.	II	7 49 0.19	+0.53	−0.14	+0.10	+0.01	+0.09	0.78	11 21 42.32	+3 32 41.54	0.03

$a' = +0^s.348$ (circle W.); $a'' = +0^s.446$ (circle E.); $c = 0^s.033$ (+ with circle E.).
Chronometer No. 1519, at $6^h\ 10^m.7$ chron. time, $3^h\ 32^m\ 41^s.514 \pm 0^s.013$ slow, losing $0^s.055$ per hour.

Transits of stars observed at Pernambuco, Brazil, by Lieut. Commander C. H. Davis, U. S. N, to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Apr. 20	σ^2 Ursæ Majoris . .	W.	9	6 14 10.08	—0.35	—0.06	+2.22	—1.83	—0.04	10.02	8 59 42.23	+2 45 [32.21]	. .
	o Leonis	.	8	6 49 9.23	0.52	0.09	0.28	0.71	0.03	8.16	9 34 40.42	32.26	0.14
	Groom. 1586 . .	.	11	7 1 59.39	0.28	0.05	3.03	2.45	0.02	59.60	9 47 32.03	[32.43]	. .
	π Leonis	.	11	7 8 16.83	0.52	0.09	0.25	0.70	0.02	15.75	9 53 48.14	32.39	0.01
	α Leonis	.	11	7 16 23.26	0.52	0.12	0.32	0.71	0.02	22.21	10 1 54.70	32.49	0.09
	α Ursæ Majoris . .	.	11	8 10 43.49	0.44	0.21	1.78	1.50	—0.01	43.11	10 56 15.69	[32.58]	. .
	χ Leonis	.	11	8 13 14.84	0.52	0.29	0.24	0.70	0.00	13.57	10 58 45.95	32.38	0.02
	τ Leonis	W.	11	8 36 11.35	—0.53	—0.31	+0.18	—0.70	0.00	9.99	11 21 42.36	+2 45 32.37	0.03
	υ Leonis	E.	11	8 45 11.09	+0.54	—0.24	+0.10	+0.65	0.00	12.14	11 30 44.62	+2 45 32.48	0.08
	3 Draconis	.	11	8 50 8.62	0.35	0.12	1.48	1.71	0.00	12.04	11 35 44.68	[32.64]	. .
	β Leonis	.	11	8 57 19.10	0.51	0.10	0.30	0.68	+0.01	20.50	11 42 52.87	32.37	0.03
	o Virginis	.	11	9 13 28.61	0.52	0.12	0.22	0.66	0.01	29.90	11 59 2.35	32.45	0.05
	η Virginis	.	11	9 28 8.92	0.54	—0.12	+0.10	0.65	0.02	10.11	12 13 42.48	32.37	0.03
	α^1 Crucis	.	11	9 34 18.86	0.68	+0.08	—1.30	1.41	0.02	19.75	12 19 52.02	[32.27]	. .
	κ Draconis	.	11	9 42 45.65	0.32	—0.11	+2.16	1.96	0.02	50.00	12 28 22.02	[32.02]	. .
	γ^1 Virginis	E.	11	9 49 57.88	+0.54	—0.19	+0.09	+0.65	+0.02	58.99	12 35 31.43	+2 45 32.44	0.04

$a' = +0^s.873$ (circle W.); $a'' = +0^s.738$ (circle E.); $c = 0^s.676$ (+ with circle E.).
Chronometer No. 1254, at $8^h\ 30^m$ chron. time, $2^h\ 45^m\ 32^s.40 \pm 0^s.014$ slow, losing $0^s.016$ per hour.
$7^h\ 35^m.8$ $2^h\ 45^m\ 32^s.39$

Apr. 23	ϵ Hydræ	E.	11	5 54 46.11	+0.52	—0.09	—0.14	+0.82	—0.03	47.19	8 40 20.64	+2 45 33.45	0.09
	o Leonis	.	11	6 49 5.83	0.52	0.12	0.17	0.82	0.02	6.86	9 34 40.38	33.52	0.02
	Groom. 1586 . .	.	11	7 1 57.16	0.28	0.06	1.85	2.85	0.02	58.36	9 47 31.84	[33.48]	. .
	π Leonis	.	11	7 8 13.52	0.52	0.12	—0.16	0.82	0.01	14.57	9 53 48.11	33.54	0.00
	υ^2 Hydræ	.	11	7 13 38.10	0.55	0.13	+0.04	0.83	0.01	39.38	9 59 12.98	33.60	0.06
	30 H. Ursæ Maj. .	.	11	7 29 49.18	0.36	0.04	—1.27	2.01	—0.01	50.23	10 15 23.59	[33.36]	. .
	9 Draconis	.	11	7 39 12.86	0.22	—0.01	—2.25	3.44	0.00	14.26	10 24 48.04	[33.78]	. .
	ν Hydræ	E.	11	7 58 3.48	+0.56	0.00	+0.07	+0.84	0.00	4.95	10 43 38.46	+2 45 33.51	0.03
	α Ursæ Majoris . .	W.	11	8 10 47.00	—0.39	—0.11	—3.10	—1.84	0.00	41.56	10 56 15.61	+2 45 [34.05]	. .
	β Crateris	.	11	8 20 9.58	0.56	0.16	+0.40	0.92	0.00	8.34	11 5 41.79	33.45	0.09
	Groom. 1771 . .	.	11	8 30 12.63	0.36	0.12	—3.45	2.02	+0.01	6.69	11 15 40.03	[33.34]	. .
	τ Leonis	.	11	8 36 10.54	0.53	0.17	0.31	0.86	0.01	8.68	11 21 42.34	33.66	0.12
	υ Leonis	.	11	8 45 12.80	0.54	0.18	0.21	0.85	0.01	11.03	11 30 44.61	33.58	0.04
	3 Draconis	.	11	8 50 17.28	0.35	0.12	3.05	2.22	0.01	11.55	11 35 44.59	[33.04]	. .
	β Virginis	.	11	8 58 51.25	0.53	0.17	0.28	0.85	0.02	49.44	11 44 22.95	33.51	0.03
	π Virginis	W.	11	9 9 8.24	—0.52	—0.15	—0.41	—0.86	+0.02	6.32	11 54 39.87	+2 45 33.55	0.01

$a' = -0^s.533$ (circle E.); $a'' = -1^s.523$ (circle W.); $c = 0^s.833$ (+ with circle E.).
Chronometer No. 1254, at $8^h\ 0^m$ chron. time, $2^h\ 45^m\ 33^s.54 \pm 0^s.014$ slow, losing $0^s.015$ per hour.
$6^h\ 18^m.3$ $2^h\ 45^m\ 33^s.51$

Transits of stars observed at Pernambuco, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Sept. 7	f Draconis	W.	II	14 54 51.28	—0.34	+0.10	+0.65	—0.40	—0.01	51.28	17 32 27.24	+2 37 [35.96]	. .
	ω Draconis	.	II	15 0 3.68	0.34	0.10	0.67	0.41	0.01	3.69	17 37 40.04	[36.35]	. .
	γ Ophiuchi	.	II	15 4 16.98	0.53	0.15	+0.05	0.15	—0.01	16.49	17 41 52.74	36.25	0.07
	ν Ophiuchi	.	II	15 14 49.78	0.55	0.16	—0.01	0.15	0.00	49.23	17 52 25.57	36.34	0.02
	67 Ophiuchi	.	II	15 17 2.64	0.53	0.15	+0.05	0.15	0.00	2.16	17 54 38.45	36.29	0.03
	p^1 Ophiuchi	.	II	15 21 47.84	0.53	0.15	0.05	0.15	0.00	47.36	17 59 23.65	36.29	0.03
	72 Ophiuchi	.	II	15 24 4.00	0.52	0.15	0.08	0.15	0.00	3.56	18 1 40.02	36.46	0.14
	36 Draconis	W.	II	15 35 36.90	—0.38	+0.11	+0.54	—0.34	0.00	36.83	18 13 13.21	+2 37 [36.38]	. .
	Groom. 2640 . .	E.	II	15 58 13.14	+0.37	+0.21	+1.34	+0.26	0.00	15.32	18 35 51.42	+2 37 [36.10]	. .
	o Draconis	.	II	16 11 48.64	0.41	0.27	1.05	0.21	0.00	50.58	18 49 26.73	[36.15]	. .
	v Draconis	.	II	16 18 14.03	0.31	0.22	1.77	0.33	0.00	16.66	18 55 53.45	[36.79]	. .
	λ Aquilæ	.	II	16 22 16.22	0.54	0.39	0.03	0.11	0.00	17.29	18 59 53.52	36.23	0.09
	η Aquilæ	.	II	17 8 44.90	0.53	0.47	0.09	0.11	0.00	46.10	19 46 22.44	36.34	0.02
	τ Aquilæ	.	II	17 20 40.06	0.52	0.43	+0.15	0.11	+0.01	41.28	19 58 17.57	36.29	0.03
	β^2 Capricorni	.	II	17 36 39.86	0.56	0.40	—0.07	0.11	0.01	40.87	20 14 17.12	36.25	0.07
	π Capricorni	E.	II	17 42 50.77	+0.56	+0.39	—0.11	+0.11	+0.01	51.73	20 20 28.17	+2 37 36.44	0.12

$a' = +0^s.247$ (circle W.); $a'' = +0^s.581$ (circle E.); $c = 0^s.128$ (+ with circle E.).
Chronometer No. 1254, at $16^h\ 15^m$ chron. time, $2^h\ 37^m\ 36^s.32 \pm 0^s.017$ slow, losing $0^s.005$ per hour.
$16^h\ 49^m.4$ $2^h\ 37^m\ 36^s.32$

Sept. 8	f Draconis	E.	II	14 54 52.24	+0.34	+0.14	—2.52	+0.58	0.00	50.78	17 32 27.18	+2 37 [36.40]	. .
	ω Draconis	.	II	15 0 4.70	0.34	0.15	2.60	0.60	0.00	3.19	17 37 39.98	[36.79]	. .
	γ Ophiuchi	.	II	15 4 15.43	0.53	0.26	0.18	0.22	0.00	16.26	17 41 52.73	36.47	0.00
	ψ^1 Draconis (pr.) .	.	II	15 6 30.85	0.30	0.15	—3.11	0.71	0.00	28.90	17 44 5.20	[36.30]	. .
	ν Ophiuchi	.	II	15 14 47.90	0.55	0.30	+0.03	0.22	0.00	49.00	17 52 25.56	36.56	0.09
	67 Ophiuchi	.	II	15 17 1.09	0.53	0.29	—0.18	0.22	0.00	1.95	17 54 38.43	36.48	0.01
	p^1 Ophiuchi	.	II	15 21 46.42	0.53	0.30	0.18	0.22	0.00	47.29	17 59 23.64	36.35	0.12
	72 Ophiuchi	E.	II	15 24 2.81	+0.52	+0.30	—0.30	+0.22	0.00	3.55	18 1 40.00	+2 37 36.45	0.02
	36 Draconis	W.	II	15 35 40.94	—0.38	+0.11	—3.50	—0.60	0.00	36.57	18 13 13.16	+2 37 [36.59]	. .
	η Serpentis	.	II	15 37 30.86	0.54	0.16	—0.17	0.26	0.00	30.05	18 15 6.45	36.40	0.07
	1 Aquilæ	.	II	15 51 5.28	0.55	0.09	+0.01	0.26	0.00	4.57	18 28 41.07	36.50	0.03
	Groom. 2460 . .	.	II	15 58 19.64	0.37	0.11	—3.66	0.62	0.00	15.10	18 35 51.38	[36.28]	. .
	o Draconis	.	II	16 11 53.86	0.41	0.07	2.88	0.51	0.00	50.13	18 49 26.69	[36.56]	. .
	λ Aquilæ	W.	II	16 22 17.73	—0.54	+0.12	—0.08	—0.26	0.00	16.97	18 59 53.50	+2 37 36.53	0.06

$a' = -0^s.963$ (circle E.); $a'' = -1^s.591$ (circle W.); $c = 0^s.238$ (+ with circle E.).
Chronometer No. 1254, at $15^h\ 30^m$ chron. time, $2^h\ 37^m\ 36^s.47 \pm 0^s.016$ slow, losing $0^s.005$ per hour.
$17^h\ 14^m.4$ $2^h\ 37^m\ 36^s.48$

Transits of stars observed at Bahia, Brazil, by Lieut. J. A. Norris, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1879.				h. m. s.	s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
Sept. 7	ε Pavonis	E.	6	17 12 0.31	+0.95	+0.14	−0.96	−0.76	0.00	59.68	19 46 44.62	+2 34 [44.94]	. .
	δ Pavonis	.	11	17 22 14.40	0.82	0.10	−0.64	0.55	0.00	14.13	19 56 58.39	[44.26]	. .
	ϑ Aquilæ	.	11	17 30 22.55	0.54	0.06	+0.06	0.22	0.00	22.99	20 5 7.72	44.73	0.04
	α^2 Capricorni	.	11	17 36 39.71	0.56	0.05	0.00	0.22	0.00	40.10	20 11 24.74	44.64	0.05
	β^2 Capricorni	.	11	17 39 32.06	0.57	0.05	−0.01	0.23	0.00	32.44	20 14 17.12	44.68	0.01
	π Capricorni	.	11	17 45 43.13	0.58	+0.02	−0.03	0.23	0.00	43.47	20 20 28.19	44.72	0.03
	ϑ Cephei	.	11	17 52 50.76	0.30	−0.01	+0.67	0.47	0.00	51.25	20 27 35.83	[44.58]	. .
	κ Delphini	E.	11	17 58 33.92	+0.52	+0.03	+0.12	−0.22	0.00	34.37	20 23 19.04	+2 34 44.67	0.02
	η Cephei	W.	11	18 8 7.30	−0.31	−0.03	+0.71	+0.37	0.00	8.04	20 42 52.74	+2 34 [44.70]	. .
	μ Aquarii	.	11	18 11 27.77	0.56	0.07	+0.02	0.18	0.00	27.34	20 46 11.91	44.57	0.12
	α Octantis	.	11	18 15 30.32	1.09	0.15	−1.48	0.82	0.00	28.42	20 50 12.94	[44.52]	. .
	α Equulei	.	11	18 35 6.32	0.53	0.10	+0.11	0.18	0.00	5.98	21 9 50.62	44.64	0.05
	β Cephei	.	11	18 52 23.34	0.20	0.04	1.03	0.52	0.00	24.65	21 27 9.19	[44.54]	. .
	ξ Aquarii	.	11	18 56 38.76	0.55	0.13	0.03	0.18	0.00	38.29	21 31 23.00	44.71	0.02
	ε Pegasi	.	11	19 3 34.34	0.52	0.12	+0.14	0.18	0.00	34.02	21 38 18.76	44.74	0.05
	μ Capricorni	W.	11	19 12 2.25	−0.57	−0.13	−0.01	+0.18	0.00	1.72	21 46 46.49	+2 34 44.77	0.08

$a' = +0^s.318$ (circle E.); $a'' = +0^s.355$ (circle W.); $c = 0^s.198$ (— with circle E.).
Chronometer No. 1295, at $18^h\ 0^m$ chron. time, $2^h\ 34^m\ 44^s.69 \pm 0^s.013$ slow, losing $0^s.002$ per hour.
$16^h\ 38^m$ $2^h\ 34^m\ 44^s.69$

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
Sept. 8	α Trianguli Aust. .	W.	11	14 1 14.40	−0.86	−0.26	−0.52	+0.40	−0.01	13.15	16 35 57.90	+2 34 [44.75]	. .
	α^1 Herculis	.	11	14 34 26.24	0.50	0.05	+0.11	0.15	−0.01	25.94	17 9 10.69	44.75	0.03
	δ Aræ	.	11	14 45 32.29	0.76	0.06	−0.34	0.29	0.00	31.32	17 20 16.33	[44.91]	. .
	ω Draconis	.	11	15 2 54.23	0.22	0.02	+0.62	0.40	0.00	55.01	17 37 39.99	[44.98]	. .
	γ Ophiuchi	.	11	15 7 8.46	0.53	0.07	0.06	0.14	0.00	8.06	17 41 52.74	44.68	0.10
	ν Ophiuchi	.	11	15 17 41.25	0.56	0.11	0.01	0.14	0.00	40.73	17 52 25.57	44.84	0.06
	67 Ophiuchi	.	11	15 19 54.07	0.53	−0.08	0.06	0.14	0.00	53.66	17 54 38.44	44.78	0.00
	72 Ophiuchi	W.	11	15 26 55.57	−0.52	+0.02	+0.09	+0.14	0.00	55.30	18 1 40.02	+2 34 44.72	0.06
	36 Draconis	E.	11	15 38 28.51	+0.28	+0.02	+0.20	−0.42	0.00	28.59	18 13 13.20	+2 34 [44.61]	. .
	ϕ Draconis	.	11	15 47 45.63	0.17	0.02	0.28	0.57	0.00	45.53	18 22 30.19	[44.66]	. .
	ι Aquilæ	.	11	15 53 55.79	0.55	0.08	+0.01	0.18	0.00	56.25	18 28 41.08	44.83	0.05
	κ Pavonis	.	11	16 9 50.84	0.83	0.11	−0.19	0.48	0.00	51.11	18 44 35.64	[44.53]	. .
	ϑ Serpentis	.	11	16 15 30.54	0.53	0.06	+0.03	0.18	0.00	30.98	18 50 15.74	44.76	0.02
	d' Sagittarii	.	11	16 35 52.20	0.58	0.12	−0.01	0.19	+0.01	52.71	19 10 37.56	44.85	0.07
	ω Aquilæ	.	11	16 37 26.43	0.51	0.11	+0.04	0.18	0.01	26.92	19 12 11.73	44.81	0.03
	δ Aquilæ	E.	11	16 44 42.23	+0.53	+0.14	+0.02	−0.18	+0.01	42.75	19 19 27.55	+2 34 44.80	0.02

$a' = +0^s.226$ (circle W.); $a'' = +0^s.089$ (circle E.); $c = 0^s.163$ (— with circle E.).
Chronometer No. 1295, at $15^h\ 30^m$ chron. time, $2^h\ 34^m\ 44^s.78 \pm 0^s.012$ slow, losing $0^s.006$ per hour.
$17^h\ 3^m$ $2^h\ 34^m\ 44^s.79$

Transits of stars observed at Pernambuco, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Sept. 10	*f* Draconis	W.	11	14 54 50.13	—0.34	—0.06	+1.59	—0.71	—0.01	50.60	17 32 27.06	+2 37 [36.46]	. .
	ω Draconis	.	11	15 0 2.44	0.34	0.07	1.63	0.73	—0.01	2.92	17 37 39.85	[36.93]	. .
	γ Ophiuchi	.	11	15 4 16.91	0.53	0.12	+0.11	0.27	0.00	16.10	17 41 52.69	36.59	0.05
	ν Ophiuchi	.	11	15 14 49.82	0.55	0.13	—0.02	0.27	0.00	48.85	17 52 25.52	36.67	0.03
	67 Ophiuchi	.	11	15 17 2.50	0.53	0.12	+0.12	0.27	0.00	1.70	17 54 38.40	36.70	0.06
	p^1 Ophiuchi	.	11	15 21 47.84	0.53	0.12	0.11	0.27	0.00	47.03	17 59 23.60	36.57	0.07
	72 Ophiuchi	W.	11	15 24 4.00	—0.52	—0.12	+0.19	—0.27	0.00	3.28	18 1 39.97	+2 37 36.69	0.05
	υ Draconis	E.	4	16 18 12.41	+0.31	+0.01	+3.03	+0.69	0.00	16.45	18 55 53.25	+2 37 [36.80]	. .
	λ Aquilæ	.	11	16 22 15.94	0.54	+0.02	0.05	0.22	0.00	16.77	18 59 53.48	36.71	0.07
	δ Draconis	.	8	16 34 52.93	0.35	0.00	2.52	0.58	0.00	56.38	19 12 32.92	[36.54]	. .
	θ Aquilæ	.	11	17 27 30.10	0.54	+0.06	+0.12	0.22	0.00	31.04	20 5 7.67	36.63	0.01
	β^2 Capricorni	.	6	17 36 39.68	0.56	0.07	—0.13	0.23	0.00	40.41	20 14 17.08	36.67	0.03
	β Delphini	.	8	17 54 18.46	0.52	0.09	+0.39	0.23	+0.01	19.70	20 31 56.24	36.54	0.10
	α Delphini	.	11	17 56 26.98	0.51	0.09	0.41	0.23	0.01	28.23	20 34 4.88	36.65	0.01
	η Cephei	E.	11	18 5 13.05	+0.40	+0.05	+1.95	+0.47	+0.01	15.92	20 42 52.62	+2 37 [36.70]	. .

$a' = +0^s.606$ (circle W.); $a'' = +0^s.997$ (circle E.); $c = 0^s.244$ (+ with circle E.).
Chronometer No. 1254, at $16^h\ 20^m$ chron. time, $2^h\ 37^m\ 36^s.64 \pm 0^s.012$ slow, losing $0^s.004$ per hour.
$16^h\ 57^m.4$ $2^h\ 37^m\ 36^s.64$

Sept. 11	ω Draconis	E.	11	15 0 5.44	+0.34	+0.03	—3.36	+0.68	0.00	3.13	17 37 39.79	+2 37 [36.66]	. .
	ψ^1 Draconis, (pr.) .	.	5	15 6 31.13	0.30	0.05	—4.02	0.81	0.00	28.27	17 44 4.99	[36.72]	. .
	ν Ophiuchi	.	11	15 14 47.77	0.55	0.07	+0.04	0.25	0.00	48.68	17 52 25.51	36.83	0.11
	67 Ophiuchi	.	11	15 17 1.14	0.53	0.06	—0.24	0.25	0.00	1.74	17 54 38.38	36.64	0.08
	p^1 Ophiuchi	.	11	15 21 46.37	0.53	0.05	0.23	0.25	0.00	46.97	17 59 23.58	36.61	0.11
	72 Ophiuchi	.	11	15 24 2.78	0.52	0.04	0.38	0.25	0.00	3.21	18 1 39.95	36.74	0.02
	36 Draconis	.	8	15 35 38.02	0.38	0.03	2.74	0.57	0.00	36.26	18 13 13.02	[36.76]	. .
	η Serpentis	E.	11	15 37 28.99	+0.54	+0.05	—0.11	+0.25	0.00	29.72	18 15 6.40	+2 37 36.68	0.04
	φ Draconis	W.	11	15 45 1.42	—0.31	—0.19	—7.12	—0.90	0.00	52.90	18 22 29.94	+2 37 [37.04]	. .
	1 Aquilæ	.	11	15 51 5.59	0.55	0.35	+0.01	0.29	0.00	4.41	18 28 41.02	36.61	0.11
	Groom. 2640 . .	.	11	15 58 21.32	0.37	0.25	—5.35	0.69	0.00	14.66	18 35 51.22	[36.56]	. .
	o Draconis (seq.) .	.	10	16 11 55.59	0.41	0.35	4.19	0.56	0.00	50.08	18 49 26.58	[36.50]	. .
	λ Aquilæ	.	11	16 22 18.05	0.54	0.42	0.12	0.29	0.00	16.68	18 59 53.46	36.78	0.06
	ω Aquilæ	.	11	16 34 36.90	0.52	0.40	0.79	0.29	0.00	34.90	19 12 11.67	36.77	0.05
	δ Aquilæ	.	11	16 41 40.79	0.52	0.40	0.80	0.29	0.00	38.78	19 19 15.63	36.85	0.13
	κ Aquilæ	W.	11	16 52 51.51	—0.54	—0.44	—0.03	—0.29	0.00	50.21	19 30 26.85	+2 37 36.64	0.08

$a' = -1^s.245$ (circle E.); $a'' = -2^s.325$ (circle W.); $c = 0^s.267$ (+ with circle E.).
Chronometer No. 1254, at $15^h\ 56^m$ chron. time, $2^h\ 37^m\ 36^s.72 \pm 0^s.019$ slow, losing $0^s.003$ per hour.
$18^h\ 35^m.4$ $2^h\ 37^m\ 36^s.72$

Transits of stars observed at Bahia, Brazil, by Lieut. J. A. Norris, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1879.				h. m. s.	s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
Sept. 10	ζ Pavonis	E.	8	15 54 17.11	+0.91	+0.05	−0.57	−0.47	−0.01	17.02	18 29 1.56	+2 34 [44.54]	. .
	λ Pavonis	.	9	16 6 21.68	0.77	−0.01	−0.34	0.32	−0.01	21.77	18 41 6.96	[45.19]	. .
	η Aquilæ	.	11	17 11 36.72	0.54	+0.11	+0.05	0.15	0.00	37.27	19 46 22.43	45.16	0.06
	ε Draconis	.	8	17 13 51.20	0.20	0.03	0.61	0.44	0.00	51.60	19 48 36.26	[44.66]	. .
	τ Aquilæ	.	11	17 23 31.88	0.52	+0.01	0.07	0.15	0.00	32.33	19 58 17.56	45.23	0.01
	ϑ Aquilæ	.	11	17 30 21.98	0.54	−0.05	+0.04	0.15	0.00	22.36	20 5 7.69	45.33	0.11
	a^2 Capricorni	.	11	17 36 38.97	0.56	+0.02	0.00	0.15	0.00	39.40	20 11 24.71	45.31	0.09
	β^2 Capricorni	E.	11	17 39 31.40	+0.57	+0.05	−0.01	−0.16	0.00	31.85	20 14 17.09	+2 34 45.24	0.02
	μ^1 Octantis	W.	11	17 52 31.80	−1.06	−0.08	−1.08	+0.47	0.00	30.05	20 27 14.96	+2 34 [44.91]	. .
	η Cephei	.	11	18 8 6.99	0.31	0.01	+0.56	0.23	0.00	7.46	20 42 52.66	[45.20]	. .
	μ Aquarii	.	11	18 11 27.17	0.56	0.02	0.02	0.11	0.00	26.72	20 46 11.89	45.17	0.05
	α Equulei	.	11	18 35 5.80	0.53	0.09	0.09	0.11	0.00	4.38	21 9 50.60	45.22	0.00
	α Cephei	.	11	18 40 59.43	0.30	0.07	0.58	0.23	0.00	59.87	21 15 44.89	[45.02]	. .
	β Aquarii	.	11	18 50 30.90	0.55	−0.02	0.03	0.11	0.00	30.47	21 25 15.68	45.21	0.01
	ξ Aquarii	.	11	18 56 38.22	0.55	+0.05	0.02	0.11	0.00	37.85	21 31 22.99	45.14	0.08
	ε Pegasi	W.	11	19 3 33.83	−0.52	−0.01	+0.11	+0.11	+0.01	33.53	21 38 18.75	+2 34 45.22	0.00

$a' = +0^s.210$ (circle E.); $a'' = +0^s.279$ (circle W.); $c = 0^s.130$ (— with circle E.).
Chronometer No. 1295, at $18^h\ 0^m$ chron. time, $2^h\ 34^m\ 45^s.22 \pm 0^s.013$ slow, losing $0^s.006$ per hour.
$17^h\ 9^m$ $2^h\ 34^m\ 45^s.21$

Sept. 11	α Trianguli Aust.	W.	11	14 1 12.90	−0.86	−0.22	+0.22	+0.33	0.00	12.37	16 35 57.76	+2 34 [45.39]	. .
	δ Aræ	.	11	14 45 31.66	0.76	0.17	+0.15	0.24	0.00	31.12	17 20 16.22	[45.10]	. .
	α Ophiuchi	.	11	14 54 37.30	0.51	0.08	−0.04	0.12	0.00	36.79	17 29 21.99	45.20	0.10
	β Ophiuchi	.	10	15 2 48.14	0.53	0.05	0.03	0.12	0.00	47.65	17 37 32.86	45.21	0.09
	η Serpentis	.	11	15 40 21.83	0.54	0.26	0.02	0.12	0.00	21.13	18 15 6.36	45.23	0.07
	ι Aquilæ	.	11	15 53 56.31	0.55	0.20	0.01	0.12	0.00	55.67	18 28 41.03	45.36	0.06
	ϑ Serpentis	W.	11	16 15 31.05	−0.53	−0.20	−0.03	+0.12	0.00	30.41	18 50 15.69	+2 34 45.28	0.02
	λ Aquilæ	E.	11	16 25 7.55	+0.55	+0.11	0.00	−0.16	0.00	8.05	18 59 53.48	+2 34 45.43	0.13
	τ Pavonis	.	10	16 28 51.38	0.86	0.17	+0.05	0.45	0.00	52.01	19 3 36.84	[44.83]	. .
	δ Draconis	.	11	16 37 48.38	0.24	0.04	−0.06	0.41	0.00	48.19	19 12 32.88	[44.69]	. .
	δ Aquilæ	.	8	16 44 41.77	0.53	0.12	0.01	0.16	0.00	42.25	19 19 27.51	45.26	0.04
	β Cephei	.	10	18 52 23.96	0.20	0.07	0.07	0.46	0.00	23.70	21 27 9.08	[45.38]	. .
	ε Pegasi	.	11	19 3 32.74	0.52	0.24	−0.01	0.16	0 00	33.33	21 38 18.74	45.41	0.11
	μ Capricorni	.	11	19 12 0.47	0.57	0.31	0.00	0.16	0.00	1.19	21 46 46.48	45.29	0.01
	ϑ Pegasi	E.	11	19 29 24.12	+0.52	+0.21	−0.01	−0.16	0.00	24.68	22 4 10.01	+2 34 45.33	0.03

$a' = -0^s.096$ (circle W.); $a'' = -0^s.023$ (circle E.); $c = 0^s.138$ (— with circle E.).
Chronometer No. 1295, at $16^h\ 30^m$ chron. time, $2^h\ 34^m\ 45^s.30 \pm 0^s.017$ slow, losing $0^s.001$ per hour.
$18^h\ 24^m$ $2^h\ 34^m\ 45^s.30$

Transits of stars observed at Rio de Janeiro, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Aug. 12	δ^1 Apodis	W.	II	13 58 0.82	−1.52	+0.04	−1.39	+1.45	+0.03	59.43	16 2 28.64	+2 4 [29.21]	. .
	ε Ophiuchi	.	II	14 7 29.07	0.51	−0.03	+0.11	0.29	0.02	28.95	16 11 58.49	29.54	0.09
	α Scorpii	.	II	14 17 34.11	0.60	+0.02	- 0.02	0.32	0.02	33.85	16 22 3.29	29.44	0.01
	β Apodis	.	II	14 21 31.96	1.43	0.11	−1.25	1.32	0.02	30.73	16 26 0.28	[29.55]	. .
	ζ Ophiuchi	.	II	14 26 3.97	0.54	0.07	+0.08	0.30	0.02	3.90	16 30 33.36	29.46	0.01
	α Trianguli Aust. .	.	II	14 31 30.41	1.04	+0.18	−0.68	0.81	0.02	29.70	16 35 59.16	[29.46]	. .
	κ Ophiuchi	.	II	14 47 30.32	0.46	−0.03	+0.18	0.30	+0.01	30.32	16 51 59.67	29.35	0.10
	η Ophiuchi	W.	II	14 59 1.08	−0.56	−0.03	+0.04	+0.30	0.00	0.83	17 3 30.21	+2 4 29.38	0.07
	θ Ophiuchi	E.	II	15 10 9.23	+0.60	0.00	0.00	−0.36	0.00	9.47	17 14 38.92	+2 4 29.45	0.00
	44 Ophiuchi	.	II	15 14 33.27	+0.59	+0.07	0.00	−0.36	0.00	33.57	17 19 3.07	29.50	0.05
	γ Mensæ, S. P. . .	.	II	15 32 5.38	−0.37	−0.08	+0.23	+1.24	−0.01	6.39	5 36 35.31	[28.92]	. .
	Lacaille 2138, S. P.	.	II	15 42 9.14	−0.76	−0.22	0.33	+1.77	0.02	10.24	5 46 39.55	[29.31]	. .
	ν Ophiuchi	.	II	15 47 56.14	+0.53	+0.18	0.01	−0.34	0.02	56.50	17 52 25.91	29.41	0.04
	67 Ophiuchi	.	II	15 50 9.02	0.49	0.17	0.02	0.33	0.02	9.35	17 54 38.78	29.43	0.02
	η Serpentis	.	II	16 10 36.93	0.51	0.18	+0.02	0.33	0.03	37.28	18 15 6.78	29.50	0.05
	ζ Pavonis	E.	II	16 24 33.99	+1.13	+0.45	−0.13	−1.04	−0.03	34.37	18 29 3.24	+2 [28.87]	. .

$a' = +0^s.340$ (circle W.); $a'' = +0^s.055$ (circle E.); $c = 0^s.311$ (— with circle E.).
Chronometer No. 1254, at $15^h\ 5^m$ chron. time, $2^h\ 4^m\ 29^s.45 \pm 0^s.012$ slow, gaining $0^s.026$ per hour.
$16^h\ 46^m.8$ $2^h\ 4^m\ 29^s.41$

Aug. 13	ν^2 Scorpii	E.	II	14 0 32.49	+0.57	−0.18	−0.13	−0.31	+0.02	32.46	16 5 1.46	+2 4 29.00	0.05
	2 Ophiuchi	.	II	14 7 30.08	0.51	0.17	−0.62	0.30	0.01	29.51	16 11 58.47	28.96	0.01
	γ Ap dis	.	8	14 10 29.60	1.54	0.53	+8.11	1.50	0.01	37.23	16 15 6.17	[28.94]	. .
	α Scorpii	.	II	14 17 34.18	0.60	0.19	0.12	0.33	0.01	34.39	16 22 3.28	28.89	0.06
	β Apodis	.	II	14 21 23.86	1.43	0.44	+7.14	1.34	0.01	30.66	16 26 0.20	[29.54]	. .
	ζ Ophiuchi	.	II	14 26 4.78	0.54	0.16	−0.43	0.30	0.01	4.44	16 30 33.34	28.90	0.05
	α Trianguli Aust. .	.	II	14 31 26.72	1.04	0.30	+3.85	0.82	+0.01	30.50	16 35 59.13	[28.63]	. .
	κ Ophiuchi	E.	II	14 47 31.69	+0.46	−0.13	−1.05	−0.30	0.00	30.67	16 51 59.66	+2 4 28.99	0.04
	η Ophiuchi	W.	8	14 59 2.10	−0.56	−0.29	−0.20	+0.27	0.00	1.32	17 3 30.19	+2 4 28.87	0.08
	ι Apodis	.	3	15 4 13.63	1.08	0.58	+3.19	0.75	0.00	15.91	17 8 44.80	[28.89]	. .
	44 Ophiuchi	.	II	15 14 34.69	0.59	0.34	+0.03	0.28	0.00	34 07	17 19 3.06	28.99	0.04
	ν Ophiuchi	.	II	15 47 57.83	0.53	0.38	−0.34	0.26	−0.01	56.83	17 52 25.90	29.07	0.12
	67 Ophiuchi	.	II	15 50 10.99	0.49	0.35	0.65	0.26	0.01	9.75	17 54 38.76	29.01	0.06
	72 Ophiuchi	.	II	15 57 12.88	0.46	0.34	0.81	0.26	0.01	11.52	18 1 40.35	28.83	0.12
	η Serpentis	.	II	16 10 39.03	0.51	0.41	−0.51	0.26	0.02	37.84	18 15 6.76	28.92	0.03
	ζ Pavonis	W.	10	16 24 31.85	−1.13	−0.78	+3.52	+0.81	−0.02	34.25	18 29 3.21	+2 4 [28.96]	. .

$a' = -1^s.936$ (circle E.); $a'' = -1^s.488$ (circle W.); $c = 0^s.276$ (— with circle E.).
Chronometer No. 1254, at $15^h\ 4^m$ chron. time, $2^h\ 4^m\ 28^s.95 \pm 0^s.015$ slow, gaining $0^s.015$ per hour.
$16^h\ 48^m.8$ $2^h\ 4^m\ 28^s.92$

Transits of stars observed at Bahia, Brazil, by Lieut. J. A. Norris, U. S. N., to determine the correction of sidereal chronometer Negus 1295

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	v.
1879.				h. m. s.	s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
Aug. 12	β^1 Scorpii	E.	11	13 23 49.78	+0.58	—0.02	+0.02	—0.13	—0.02	50.21	15 58 27.68	+2 34 37.47	0.06
	ϑ Draconis	.	8	13 25 1.70	0.33	—0.01	—0.26	0.23	0.02	1.51	15 59 39.16	[37.65]	. .
	δ Ophiuchi	.	11	13 33 25.86	0.54	0.00	0.02	0.12	0.02	26.24	16 8 3.61	37.37	0.04
	ϵ Ophiuchi	.	11	13 37 20.66	0.55	+0.01	0.02	0.12	0.02	21.06	16 11 58.50	37.44	0.03
	A Draconis	.	11	13 53 38.58	0.21	0.02	—0.39	0.33	0.01	38.08	16 28 15.03	[36.95]	. .
	α Trianguli Aust. .	.	11	14 1 20.99	0.86	0.09	+0.32	0.33	0.01	21.92	16 35 59.18	[37.26]	. .
	49 Herculis	.	11	14 11 59.75	0.50	0.01	—0.07	0.12	—0.01	0.06	16 46 37.48	37.42	0.01
	60 Herculis	E.	11	14 25 11.47	+0.51	+0.01	—0.06	—0.12	0.00	11.81	16 59 49.33	+2 34 37.52	0.11
	ζ Draconis	W.	11	14 33 51.22	—0.26	—0.03	—0.48	+0.19	0.00	50.64	17 8 28.17	+2 34 [37.53]	. .
	44 Ophiuchi	.	11	14 44 26.18	0.59	0.12	+0.04	0.09	0.00	25.60	17 19 3.08	37.48	0.07
	α Ophiuchi	.	11	14 54 45.79	0.51	0.14	—0.09	0.08	+0.01	45.14	17 29 22.45	37.31	0.10
	ω Draconis	.	11	15 3 4.77	0.22	0.06	0.56	0.22	0.01	4.16	17 37 41.59	[37.43]	. .
	γ Ophiuchi	.	11	15 7 16.44	0.53	0.14	0.06	0.08	0.01	15.80	17 41 53.11	37.31	0.10
	ν Ophiuchi	.	11	15 17 49.13	0.54	0 14	0.01	0.08	0.02	48.54	17 52 25.93	37.39	0.02
	67 Ophiuchi	.	11	15 20 2.09	0.53	0.14	0.06	0.08	0.02	1.46	17 54 38.81	37.35	0.06
	36 Draconis	W.	11	15 38 37.73	—0.28	—0.09	—0.46	+0.18	+0.02	37.10	18 13 14.40	+2 34 [37.30]	. .

$a' = -0^s.141$ (circle E.); $a'' = -0^s.205$ (circle W.); $c = 0^s.098$ (— with circle E.).
Chronometer No. 1295, at $14^h\ 30^m$ chron. time, $2^h\ 34^m\ 37^s.41 \pm 0^s.015$ slow, losing $0^s.020$ per hour.
$16^h\ 34^m.7$ $2^h\ 34^m\ 37^s.45$

Aug. 13	κ Libræ	W.	11	13 0 24.59	—0.58	—0.08	+0.04	+0.17	—0.02	24.12	15 35 1.92	+2 34 37.80	0.02
	α Serpentis	.	11	13 3 44 27	0.52	0.07	—0.13	0.16	0.02	43.69	15 38 21.46	37.77	0.05
	β Serpentis	.	11	13 6 1.82	0.50	0.06	0.19	0.16	0.02	1.21	15 40 39.04	37.83	0.01
	12 H. Draconis . . .	.	11	13 10 14.00	0.29	0.03	—0.81	0.35	0.01	13.21	15 44 51.14	[37.93]	. .
	49 Libræ	.	11	13 18 58.34	0.57	0.09	+0.02	0.16	0.01	57.85	15 53 35.67	37.82	0.00
	β^1 Scorpii	.	11	13 23 50.33	0.58	0.11	0.05	0.17	0.01	49.85	15 58 27.67	37.82	0.00
	δ^1 Apodis	.	11	13 27 49.44	1.14	0.23	1.72	0.78	—0.01	50.56	16 2 28.55	[37.99]	. .
	γ Apodis	W.	11	13 40 27.11	—1.15	—0.32	+1.76	+0.80	0.00	28.20	16 15 6.17	+2 34 [37.97]	. .
	A Draconis	E.	11	13 53 38.15	+0.21	+0.02	—1.08	—0.55	0.00	36.75	16 28 14.97	+2 34 [38.22]	. .
	α Trianguli Aust. .	.	11	14 1 20.09	0.86	0.09	+0.89	0.55	0.00	21.38	16 35 59.13	[37.75]	. .
	49 Herculis	.	11	14 11 59.53	0.50	0.06	—0.19	0.20	0.00	59.70	16 47 37.46	37.76	0.06
	η Ophiuchi	.	11	14 28 51.91	0.57	+0.02	+0.02	0.21	+0.01	52.32	17 3 30.20	37.88	0.06
	ζ Draconis	.	11	14 33 51.74	0.26	—0.01	—0.91	0.48	0.01	50.61	17 8 28.12	[37.51]	. .
	44 Ophiuchi	.	11	14 44 24.60	0.59	+0.04	+0.08	0.22	0.01	25.10	17 19 3.07	37.97	0.15
	α Ophiuchi	.	11	14 54 44.40	0.51	0.09	—0.17	0.20	0.02	44.65	17 29 22.44	37.79	0.03
	β Ophiuchi	E.	11	15 2 55.10	+0.53	+0.14	—0.12	—0.20	+0.02	55.47	17 37 33.28	+2 34 37.81	0.01

$a' = -0^s.380$ (circle W.); $a'' = -0^s.390$ (circle E.); $c = 0^s.178$ (— with circle E.).
Chronometer No. 1295, at $14^h\ 0^m$ chron. time, $2^h\ 34^m\ 37^s.82 \pm 0^s.013$ slow, losing $0^s.018$ per hour.
$16^h\ 31^m.7$ $2^h\ 34^m\ 37^s.87$

Transits of stars observed at Rio de Janeiro, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Aug. 15	δ^1 Apodis	W.	11	13 58 3.66	−1.52	+0.18	−4.07	+1.06	0.00	59.31	16 2 28.37	+2 4 [29.06]	. .
	δ Ophiuchi	.	11	14 3 34.74	0.51	0.06	+0.33	0.21	0.00	34.83	16 8 3.58	28.75	0.07
	ε Ophiuchi	.	11	14 7 29.81	0.51	0.06	+0.32	0.21	0.00	29.89	16 11 58.44	28.55	0 13
	γ Apodis	.	11	14 10 42.09	1.54	0.18	−4.17	1.09	0.00	37.65	16 15 6.00	[28.35]	. .
	α Scorpii	.	11	14 17 34.99	0.60	0.05	−0.06	0.24	0.00	34.62	16 22 3.25	28.63	0.05
	λ Ophiuchi	.	11	14 20 23.07	0.49	0.03	+0.42	0.21	0.00	23.24	16 24 52.00	28.76	0.08
	ζ Ophiuchi	.	11	14 26 4.68	0.54	+0.02	+0.22	0.22	0.00	4.60	16 30 33.32	28.72	0.04
	α Trianguli Aust. .	W.	11	14 31 32.77	−1.04	0.00	−1.98	+0.59	0.00	30.34	16 35 59.05	+2 4 [28.71]	. .
	κ Ophiuchi	E.	11	14 47 30.20	+0.46	+0.13	+0.32	−0.26	0.00	30.85	16 51 59.63	+2 4 28.78	0.10
	η Ophiuchi	.	11	14 59 1.06	0.56	0.10	+0.08	0.26	0.00	1.54	17 3 30.17	28.63	0.05
	ι Apodis	.	11	15 4 16.91	1.08	0.21	−1.26	0.74	0.00	16.20	17 8 44.72	[28.52]	. .
	θ Ophiuchi	.	11	15 10 9.77	0.60	0.13	0.02	0.28	0.00	10.20	17 14 38.88	28.68	0.00
	44 Ophiuchi	.	11	15 14 33.97	+0.59	+0.14	−0.01	−0.28	0.00	34.41	17 19 3.04	28.63	0.05
	γ Mensæ, S. P. . .	.	11	15 32 4.27	−0.37	−0.10	+2.47	+0.91	0.00	7.18	5 36 35.54	[28.36]	. .
	ι Mensæ, S. P. . .	.	11	15 38 21.54	−0.57	−0.16	2.99	+1.11	0.00	24.91	5 42 53.64	[28.73]	. .
	ν Ophiuchi	E.	11	15 47 56.66	+0.53	+0.15	+0.14	−0.26	0.00	57.22	17 52 25.88	+2 4 28.66	0.02

$a' = +0^s.994$ (circle W.); $a'' = +0^s.589$ (circle E.); $c = 0^s.234$ (— with circle E.).
Chronometer No. 1254, at $14^h\ 43^m$ chron. time, $2^h\ 4^m\ 28^s.68 \pm 0^s.016$ slow, gaining $0^s.005$ per hour.
$16^h\ 41^m.3$ $2^h\ 4^m\ 28^s.67$

Aug. 18	δ^1 Apodis	E.	11	13 58 3.94	+1.52	+0.61	−4.60	−1.49	+0.02	0.00	16 2 28.11	+2 4 [28.11]	. .
	δ Ophiuchi	.	11	14 3 34.61	0.51	0.20	+0.38	0.30	0.01	35.41	16 8 3.53	28.12	0.11
	ε Ophiuchi	.	11	14 7 29.60	0.52	0.21	+0.36	0.30	0.01	30.40	16 11 58.39	27.99	0.02
	γ Apodis	.	10	14 10 42.43	1.54	0.62	−4.71	1.52	0.01	38.37	16 15 5.72	[27.35]	. .
	α Scorpii	.	11	14 17 34.78	0.60	0.21	0.07	0.33	0.01	35.20	16 22 3.20	28.00	0.01
	β Apodis	.	11	14 21 34.93	1.43	0.48	−4.14	1.36	0.01	31.35	16 25 59.81	[28.46]	. .
	ζ Ophiuchi	.	11	14 26 4.61	0.54	0.17	+0.25	0.30	+0.01	5.28	16 30 33.27	27.99	0.02
	κ Ophiuchi	E.	11	14 47 30.66	+0.46	+0.21	+0.61	−0.30	0.00	31.64	16 51 59.58	+2 4 27.94	0.07
	η Ophiuchi	W.	11	14 59 2.17	−0.56	0.00	+0.20	+0.27	0.00	2.08	17 3 30.13	+2 4 28.05	0.04
	ι Apodis	.	11	15 4 20.07	1.08	0.00	−3.31	0.76	0.00	16.44	17 8 44.58	[28.14]	. .
	θ Ophiuchi	.	11	15 10 11.17	0.60	0.00	0.06	0.29	0.00	10.80	17 14 38.84	28.04	0.03
	44 Ophiuchi	.	11	15 14 35.30	−0.59	0.00	−0.03	+0.28	0.00	34.96	17 19 2.98	28.02	0.01
	γ Mensæ, S. P. . .	.	11	15 32 1.72	+0.37	+0.09	+6.48	−1.11	−0.01	7.54	5 36 35.77	[28.23]	. .
	ι Mensæ, S. P. . .	.	11	15 38 19.12	+0.57	+0.20	7.83	−1.35	0.01	26.36	5 42 53.92	[27.56]	. .
	ν Ophiuchi	.	11	15 47 57.96	−0.53	−0.13	0.36	+0.26	0.01	57.91	17 52 25.84	27.93	0.08
	67 Ophiuchi	W.	11	15 50 10.39	−0.49	−0.12	+0.67	+0.26	−0.01	10.70	17 54 38.71	+2 4 28.01	0.00

$a' = +1^s.124$ (circle E.); $a'' = +1^s.544$ (circle W.); $c = 0^s.280$ (— with circle E.).
Chronometer No. 1254, at $14^h\ 52^m$ chron. time, $2^h\ 4^m\ 28^s.01 \pm 0^s.012$ slow, gaining $0^s.014$ per hour.
$17^h\ 1^m.8$ $2^h\ 4^m\ 27^s.98$

Transits of stars observed at Bahia, Brazil, by Lieut. J. A. Norris, U. S. N., to determine the correction of sidereal chronometer Negus 1295

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	ν.
1879.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
Aug. 15	β^2 Capricorni	W.	11	17 39 38.96	—0.57	+0.19	+0.02	+0.12	—0.02	38.70	20 14 17.22	+2 34 38.52	0.11
	π Capricorni	.	11	17 45 50.02	0.58	0.19	0.06	0.13	0.02	49.80	20 20 28.29	38.49	0.14
	ρ Capricorni	.	11	17 47 23.74	0.58	0.18	+0.05	0.12	0.02	23.49	20 22 2.05	38.56	0.07
	ε Delphini	.	11	17 52 51.76	0.51	0.16	—0.22	0.12	0.02	51.29	20 27 29.88	38.59	0.04
	κ Delphini	.	10	17 58 41.02	0.52	0.16	0.21	0.12	0.02	40.55	20 33 19.12	38.57	0.06
	η Cephei	.	11	18 8 15.41	0.31	0.10	1.10	0.25	0.02	14.33	20 42 53.14	[38.81]	. .
	Groom. 3415. . .	.	8	18 34 9.54	0.33	0.14	—1.03	0.23	—0.01	8.54	21 8 47.03	[38.49]	. .
	ε Octantis	W.	10	19 31 55.76	—1.32	+0.05	+3.25	+0.76	0.00	58.50	22 6 37.28	+2 34 [38.78]	. .
	β Octantis	E.	6	19 59 10.59	+1.42	+0.11	+2.02	—1.14	+0.01	13.01	22 33 51.58	+2 34 [38.57]	. .
	λ Aquarii	.	8	20 11 43.13	0.55	0.10	—0.03	0.16	0.01	43.60	22 46 22.25	38.65	0.02
	δ Aquarii	.	11	20 13 38.68	0.57	0.11	+0.02	0.17	0.01	39.22	22 48 17.96	38.74	0.11
	4 Cassiop.	.	8	20 44 54.14	0.31	0.01	—0.61	0.32	0.02	53.55	23 19 32.48	[38.93]	. .
	41 H. Cephei	.	10	21 7 35.94	0.24	0.01	0.76	0.41	0.02	35.04	23 42 12.71	[37.67]	. .
	ω Piscium	.	11	21 18 30.75	0.52	0.07	0.10	0.16	0.02	31.10	23 53 9.86	38.76	0.13
	30 Piscium	.	11	21 21 10.08	0.55	0.06	—0.03	0.16	0.02	10.52	23 55 49.18	38.66	0.03
	2 Ceti	E.	11	21 22 57.20	+0.56	+0.05	+0.03	—0.17	+0.03	57.70	23 57 36.46	+2 34 38.76	0.13

$a' = -0^s.546$ (circle W.); $a'' = -0^s.301$ (circle E.); $c = 0^s.139$ (— with circle E.).
Chronometer No. 1295, at $19^h\ 30^m$ chron. time, $2^h\ 34^m\ 38^s.63 \pm 0^s.021$ slow, losing $0^s.014$ per hour.
$16^h\ 29^m.2$ $2^h\ 34^m\ 38^s.59$

Aug. 18	δ Ophiuchi	W.	11	13 33 24.38	—0.54	0.00	—0.08	+0.20	—0.02	23.94	16 8 3.53	+2 34 39.59	0.02
	ε Ophiuchi	.	11	13 37 19.29	0.55	0.00	—0.07	0.20	0.02	18.85	16 11 58.42	39.57	0.04
	γ Apodis	.	9	13 40 25.85	1.15	0.00	+2.07	1.02	0.02	27.77	16 15 5.72	[37.95]	. .
	β Apodis	.	9	13 51 17.72	1.08	+0.03	+1.83	0.91	0.02	19.39	16 25 59.81	[40.42]	. .
	ζ Ophiuchi	.	11	13 55 54.13	0.56	0.02	—0.02	0.20	0.02	53.75	16 30 33.30	39.55	0.06
	α Trianguli Aust. .	.	11	14 1 18.34	0.86	0.04	+1.02	0.56	0.02	19.08	16 35 58.92	[39.84]	. .
	49 Herculis	.	11	14 11 58.34	0.50	0.09	—0.22	0.21	—0.01	57.91	16 46 37.39	39.48	0.13
	44 Ophiuchi	W.	11	14 44 23.66	—0.59	+0.10	+0.09	+0.22	0.00	23.48	17 19 3.01	+2 34 39.53	0.08
	α Ophiuchi	E.	11	14 54 42.58	+0.51	+0.08	—0.25	—0.25	0.00	42.67	17 29 22.38	+2 34 39.71	0.10
	β Ophiuchi	.	11	15 2 53.22	0.53	0.12	—0.17	0.24	0.00	53.46	17 37 33.22	39.76	0.15
	μ^1 Sagittarii	.	8	15 31 55.58	0.58	0.26	+0.08	0.26	+0.01	56.25	18 6 35.93	39.68	0.07
	τ Pavonis	.	11	16 28 56.77	0.86	0.20	1.31	0.68	0.02	58.48	19 3 37.72	[39.24]	. .
	d Sagittarii	.	11	16 35 57.60	0.58	+0.13	+0.06	0.26	0.02	58.13	19 10 37.71	39.58	0.03
	α^2 Capricorni	.	11	17 36 44.80	0.56	—0.01	0.00	0.25	0.04	45.14	20 11 24.82	39.68	0.07
	μ^1 Octantis	.	11	17 52 34.98	1.06	+0.06	+2.17	1.04	0.04	37.27	20 27 15.73	[38.46]	. .
	η Cephei	E.	11	18 8 14.81	+0.31	+0 05	—1.13	—0.50	+0.05	13.59	20 42 53.11	+2 34 [39.52]	. .

$a' = -0^s.447$ (circle W.); $a'' = -0^s.561$ (circle E.); $c = 0^s.221$ (— with circle E.).
Chronometer No. 1295, at $15^h\ 0^m$ chron. time, $2^h\ 34^m\ 39^s.61 \pm 0^s.019$ slow, losing $0^s.016$ per hour.
$16^h\ 49^m.7$ $2^h\ 34^m\ 39^s.64$

Transits of stars observed at Rio de Janeiro, Brazil, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
July 1	61 Virginis	W.	11	10 58 27.90	−0.58	−0.23	+0.05	+0.50	+0.03	27.67	13 12 3.99	+2 13 36.32	0.02
	κ Octantis	.	5	11 7 62.49	3.04	1.21	−5.22	5.65	0.02	58.69	13 21 36.68	[37.99]	. .
	ζ Virginis	.	11	11 14 54.84	0.51	0.20	+0.19	0.48	0.02	54.82	13 28 31.24	36.42	0.08
	m Virginis	.	11	11 21 39.11	0 54	0.22	+0.13	0.48	0.02	38.98	13 35 15.39	36.41	0.07
	θ Apodis	.	11	11 39 59.91	1.38	0.55	−1.67	2.00	+0.01	58.32	13 53 34.54	[36.22]	. .
	η Apodis	.	11	11 49 34.44	1.78	0.71	−2.52	2.87	0.00	32.30	14 3 8.04	[35.74]	. .
	ι Virginis	.	11	11 56 3.82	0.53	0.21	+0.15	0.48	0.00	3.71	14 9 40.02	36.31	0.03
	φ Vi ginis	W.	11	12 8 21.79	−0.51	−0.20	+0.18	+0.48	0.00	21.74	14 21 57.96	+2 13 36.22	0.12
	α Apodis	E.	11	12 19 21.96	+1.56	−0.23	−2.45	−2.59	−0.01	18.24	14 32 54.48	+2 13 [36.24]	. .
	μ Virginis	.	11	12 23 4.49	0.53	0.08	+0.18	0.52	0.01	4.59	14 36 40.93	36.34	0.00
	α^2 Libræ	.	11	12 30 34.66	0.57	0.08	0.08	0.54	0.01	34.68	14 44 10.98	36.30	0.04
	ξ^2 Libræ	.	10	12 36 35.85	0.55	0.08	+0.12	0.52	0.02	35.90	14 50 12.14	36.24	0.10
	20 Libræ	.	11	12 43 22.88	0.60	0.09	−0.02	0.57	0.02	22.78	14 56 59.23	36.45	0.11
	ρ Octantis	.	11	13 2 12.82	2.56	0.37	−4.99	4.97	0.03	5.02	15 15 41.47	[36.45]	. .
	κ Libræ	.	11	13 21 22.15	0.58	0.08	+0.04	0.55	0.03	22.11	15 34 58.63	36.52	0.18
	λ Libræ	.	11	13 32 42.58	0.58	0.08	0.04	0.55	0.04	42.53	15 46 18.78	36.25	0.09
	δ Scorpii	E.	11	13 39 34.78	+0.59	−0.09	+0.01	−0.56	−0.04	34.69	15 53 10.97	+2 13 36.28	0.06

$a' = +0^s.497$ (circle W.); $a'' = +0^s.590$ (circle E.); $c = 0^s.496$ (— with circle E.).
Chronometer No. 1295, at $12^h\ 0^m$ chron. time, $2^h\ 13^m\ 36^s.338 \pm 0^s.018$ slow, gaining $0^s.025$ per hour.
$14^h\ 28^m.9$ $2^h\ 13^m\ 36^s.276$

July 5	ζ Virginis	E.	10	11 14 56.99	+0.51	0.00	+0.18	−0.45	+0.02	57.25	13 28 31.21	+2 13 33.96	0.01
	Lac. 505, S. P. .	.	10	11 19 13.92	−0.60	0.00	2.39	+2.39	0.02	18.12	1 32 52.39	[34.27]	. .
	B. A. C. 554, S. P.	.	5	11 27 41.61	−0.68	0.00	+2.53	+2.54	0.01	46.01	1 41 19.12	[33.11]	. .
	θ Apodis	.	11	11 40 2.56	+1.38	0.00	−1.55	−1.89	+0.01	0.51	13 53 34.28	[33.77]	. .
	ι Virginis	.	11	11 56 5.66	0.53	0.00	+0.14	0.45	0.00	5.88	14 9 39.99	34.11	0.14
	λ Virginis	.	11	11 58 59.43	0.56	0.00	0.08	0.46	0.00	59.61	14 12 32.53	33.92	0.05
	φ Virginis	E.	11	12 8 23.80	+0.51	0.00	+0.17	−0.45	0.00	24.03	14 21 57.93	+2 13 33.90	0.07
	α Apodis	W.	11	12 19 22.13	−1.56	+0.26	−2.06	+2.08	−0.01	20.84	14 32 54.19	+2 13 [33.35]	. .
	μ Virginis	.	11	12 23 6.87	0.53	0.08	+0.15	0.41	0.01	6.97	14 36 40.90	33.93	0.04
	α^2 Libræ	.	11	12 30 36.90	0.57	0.09	0.06	0.43	0.01	36.90	14 44 10.95	34.05	0.08
	ξ^2 Libræ	.	11	12 36 38.13	0.55	0.09	+0.10	0.42	0.02	38.17	14 50 12.11	33.94	0.03
	20 Libræ	.	11	12 43 25.32	0.60	0.10	−0.02	0.45	0.02	25.23	14 56 59.20	33.97	0.00
	β Libræ	.	11	12 56 55.74	0.54	0.09	+0.12	0.42	0.02	55.81	15 10 29.81	34.00	0.03
	ρ Octantis	W.	10	13 2 8.43	−2.56	+0.42	−4.20	+3.98	−0.03	6.04	15 15 40.97	+2 13 [34.93]	. .

$a' = +0^s.461$ (circle E.); $a'' = +0^s.497$ (circle W.); $c = 0^s.432$ (— with circle E.).
Chronometer No. 1295, at $12^h\ 0^m$ chron. time, $2^h\ 13^m\ 33^s.975 \pm 0^s.015$ slow, gaining $0^s.025$ per hour.
$14^h\ 34^m.9$ $2^h\ 13^m\ 33^s.910$

Transits of stars observed at Montevideo, Uruguay, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
July 1	δ Corvi	W.	11	11 3 24.36	−0.53	+0.09	− 1.14	− 1.85	+0.11	21.04	12 23 35.43	+1 20 14.39	0.19
	β Corvi	.	11	11 7 49.78	0.57	0.19	0.77	1.93	0.10	46.80	12 28 0.98	14.18	0.02
	f Virginis . , . . .	.	11	11 10 22.28	0.47	0.08	1.66	1.79	0.10	18.54	12 30 32.63	14.09	0.11
	γ^1 Virginis	.	11	11 15 20.76	0.45	0.08	1.87	1.78	0.10	16.84	12 35 31.02	14.18	0.02
	α Virginis	.	11	11 58 38.01	0.50	0.11	− 1.40	1.81	0.04	34.45	13 18 48.60	14.15	0.05
	κ Octantis	.	7	12 1 17.13	−4.10	+1.00	+30.51	−21.18	0.04	23.40	13 21 36.59	[13.19]	. .
	Lacaille 505, S. P.	.	11	12 12 43.74	+1.16	−0.30	−16.17	+ 9.45	0.02	37.90	1 32 51.96	[14.06]	. .
	B. A. C. 554, S. P.	W.	10	12 21 10.67	+1.27	−0.37	−17.08	+10.02	+0.01	4.52	1 41 18.44	+1 20 [13.92]	. .
	θ Centauri	E.	11	12 39 16.28	+0.67	+0.29	+ 0.08	+ 2.14	−0.01	19.45	13 59 33.54	+1 20 14.09	0.11
	η Apodis	.	11	12 42 22.51	+2.28	+0.93	+18.15	+10.54	0.02	54.39	14 3 7.99	[13.60]	. .
	Lacaille 709, S. P.	.	11	12 50 35.08	−0.92	−0.35	−17.64	− 7.89	0.03	8.25	2 10 21.74	[13.49]	. .
	ι Apodis . . . , .	.	11	13 12 14.57	+1.97	+0.60	+14.65	+ 8.80	0.05	40.54	14 32 54.46	[13.92]	. .
	α^2 Libræ	.	11	13 23 55.81	0.53	0.14	− 1.45	1.82	0.07	56.78	14 44 10.99	14.21	0.01
	ξ^2 Libræ	.	11	13 29 57.31	0.50	0.12	1.75	1.78	0.08	57.88	14 50 12.11	14.23	0.03
	20 Libræ	.	11	13 36 43.29	0.59	0.12	0.81	1.93	0.09	45.03	14 56 59.23	14.20	0.00
	β Libræ	E.	11	13 50 15.14	+0.49	+0.09	− 1.87	+ 1.77	−0.10	15.52	15 10 29.82	+1 20 14.30	0.10

$a' = -3^s.342$ (circle W.); $a'' = -4^s.222$ (circle E.); $c = 1^s.767$ (+ with circle E.).
Chronometer No. 1254, at $12^h\ 30^m$ chron. time, $1^h\ 20^m\ 14^s.20 \pm 0^s.019$ slow, gaining $0^s.076$ per hour.
$14^h\ 23^m.1$ $1^h\ 20^m\ 14^s.06$

July 5	B. A. C. 554, S. P.	W.	11	12 21 10.19	+1.27	+0.22	−12.67	+10.16	+0.04	9.21	1 41 18.90	+1 20 [9.69]	. .
	θ Apodis	.	11	12 33 27.36	−1.70	−0.29	+ 6.87	− 7.58	0.04	24.70	13 53 34.25	[9.55]	. .
	θ Centauri	.	10	12 39 27.06	0.66	−0.25	+ 0.05	2.23	0.04	24.01	13 59 33.49	9.48	0.09
	λ Virginis	.	8	12 52 27.45	0.51	0.00	− 0.96	1.85	0.03	24.16	14 12 33.48	9.32	0.07
	ϕ Virginis . . , . .	.	11	13 1 52.22	0.45	0.00	− 1.36	1.81	0.03	48.63	14 21 57.97	9.34	0.05
	α Apodis	.	17	13 12 47.31	1.97	−0.05	+ 8.61	9.09	0.02	44.83	14 32 54.18	[9.35]	. .
	α^2 Libræ	.	11	13 24 4.82	0.53	0.02	− 0.85	1.88	0.02	1.56	14 44 10.96	9.40	0.01
	ξ^2 Libræ	W.	11	13 30 6.11	−0.50	−0.03	− 1.03	− 1.84	+0.02	2.73	14 50 12.08	+1 20 9.35	0.04
	ζ Ophiuchi	E.	11	15 10 19.52	+0.50	−0.03	− 1.08	+ 1.77	−0.02	20.66	16 30 30.13	+1 20 9.47	0.08
	α Trianguli Aust. .	.	11	15 15 34.25	1.24	0.36	+ 3.96	4.82	0.02	43.89	16 35 53.38	[9.49]	. .
	ϵ Scorpii	.	11	15 22 8.16	+0.65	−0.20	− 0.05	+ 2.11	0.02	10.65	16 42 20.08	9.43	0.04
	Lacaille 1752, S.P.	.	11	15 38 44.42	−0.72	+0.24	9.38	− 6.79	0.03	27.74	4 58 36.89	[9.15]	. .
	η Ophiuchi	.	11	15 43 16.03	+0.53	−0.18	− 0.88	+ 1.81	0.03	17.28	17 3 26.73	9.45	0.06
	ι Apodis	.	11	15 48 18.86	1.29	0.45	+ 4.31	5.10	0.03	29.08	17 8 38.47	[9.39]	. .
	θ Ophiuchi	.	11	15 54 24.00	0.58	−0.20	− 0.49	1.92	0.04	25.77	17 14 35.16	9.39	0.00
	d Ophiuchi	E.	11	15 59 26.44	+0.62	+0.04	− 0.26	+ 2.01	−0.04	28.81	17 19 38.12	+1 20 [9.31]	0.08

$a' = -2^s.480$ (circle W.); $a'' = -2^s.565$ (circle E.); $c = 1^s.791$ (+ with circle E.).
Chronometer No. 1254, at $14^h\ 15^m$ chron. time, $1^h\ 20^m\ 9^s.39 \pm 0^s.013$ slow, gaining $0^s.022$ per hour.
$14^h\ 35^m.6$ $1^h\ 20^m\ 9^s.39$

Transits of stars observed at Rio de Janeiro, Brazil, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
July 6	61 Virginis	W.	11	10 58 30.72	—0.58	—0.14	+0.05	+0.51	+0.03	30.59	13 12 3.93	+2 13 33.34	0.00
	κ Octantis	.	11	11 8 5.21	3.04	0.72	—5.16	5.81	0.02	2.12	13 21 35.59	[33.47]	. .
	ζ Virginis	.	11	11 14 57.90	0.51	0.12	+0.19	0.49	0.02	57.97	13 28 31.20	33.23	0.11
	m Virginis	.	11	11 21 41.84	0.54	0.13	+0.13	0.49	0.02	41.81	13 35 15.34	33.53	0.19
	θ Apodis	.	11	11 40 2.28	1.38	0.33	—1.65	2.05	+0.01	0.98	13 53 34.21	[33.23]	. .
	κ Virginis	.	11	11 52 52.95	0.54	0.13	+0.11	0.49	0.00	52.88	14 6 26.17	33.29	0.05
/	ι Virginis	.	11	11 56 6.73	0.53	0.12	0.15	0.49	0.00	6.72	14 9 39.98	33.26	0.08
	λ Virginis	W.	11	11 59 0.19	—0.56	—0.13	+0.09	+0.50	0.00	0.09	14 12 33.52	+2 13 33.43	0.09
	α Apodis	E.	11	12 19 23.73	+1.56	+0.40	—2.18	—2.66	—0.01	20.84	14 32 54.12	+2 13 [33.28]	. .
	μ Virginis	.	11	12 23 7.26	0.53	0.13	+0.16	0.53	0.01	7.54	14 36 40.89	33.35	0.01
	a^2 Libræ	.	11	12 30 37.41	0.57	0.14	0.07	0.55	0.01	37.63	14 44 10.95	33.32	0.02
	ξ^2 Libræ	.	11	12 36 38.51	0.55	0.14	+0.11	0.54	0.02	38.75	14 50 12.10	33.35	0.01
	20 Libræ	.	11	12 43 25.60	0.60	0.15	—0.02	0.58	0.02	25.73	14 56 59.19	33.46	0.12
	β Libræ	.	11	12 56 56.29	0.54	0.14	+0.13	0.53	0.02	56.55	15 10 29.79	33.24	0.10
	ρ Octantis	E.	10	13 2 13.91	+2.56	+0.65	—4.44	—5.10	—0.03	7.55	15 15 40.85	+2 13 [33.30]	. .

$a' = +0^s.492$ (circle W.); $a'' = +0^s.526$ (circle E.); $c = 0^s.509$ (— with circle E.).
Chronometer No. 1295, at 12^h 0^m chron. time, 2^h 13^m $33^s.345 \pm 0^s.019$ slow, gaining $0^s.025$ per hour.
14^h $13^m.9$ 2^h 13^m $33^s.289$

Transits of stars observed at Montevideo, Uruguay, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
July 6	ι Octantis	E.	11	11 22 12.74	+3.64	+0.21	—3.14	—3.02	+0.02	10.45	12 42 20.39	+1 20 [9.94]	. .
	θ Virginis	.	8	11 43 31.10	0.47	0.00	+0.20	0.29	0.02	31.50	13 3 40.43	8.93	0.06
	61 Virginis	.	11	11 51 54.53	0.54	+0.01	0.12	0.31	0.02	54.91	13 12 3.93	9.02	0.03
	α Virginis	.	11	11 58 39.21	0.50	—0.01	+0.17	0.30	0.02	39.59	13 18 48.55	8.96	0.03
	θ Centauri	.	11	12 39 24.07	0.66	+0 06	—0.01	0.36	0.01	24.43	13 59 33.48	9.05	0.06
	η Apodis	.	9	12 42 59.90	2.28	0.21	—1.71	1.75	0.01	58.94	14 3 7.52	[8.58]	. .
	λ Virginis	.	11	12 52 24.06	0.51	0.06	+0.15	0.30	+0.01	24.49	14 12 33.48	8.99	0.00
	α Apodis	E.	11	13 12 45.62	+1.97	+0.69	—1.38	—1.46	0.00	45.44	14 32 54.11	+1 20 [8.67]	. .
	c^2 Libræ	W.	11	13 24 2.17	—0.53	—0.09	+0.30	+0.27	0.00	2.12	14 44 10.96	+1 20 8.84	0.15
	20 Libræ	.	11	13 36 50.58	—0.59	—0.17	0.17	+0.28	0.00	50.27	14 56 59.19	8.92	0.07
	Lacaille 1105, S.P.	.	11	13 51 25.62	+1.22	+0.37	+4 37	—1.41	0.00	30.17	3 11 39.59	[9.42]	. .
	α Trianguli Aust. .	.	11	15 15 46.78	—1.24	—0.50	—1.35	+0.71	—0.02	44.38	16 35 53.37	[8.99]	. .
	ε Scorpii	.	11	15 22 11.61	—0.65	—0.27	+0.02	+0.31	0.02	11.00	16 42 20.08	9.08	0.09
	Lacaille 1752, S.P.	.	11	15 38 24.93	+0.72	+0.32	3.21	—1.00	0.02	28.16	4 58 36.93	[8.77]	. .
	η Ophiuchi	.	11	15 43 17.95	—0.53	—0.24	0.30	+0.27	0.02	17.73	17 3 26.74	9.01	0.02
	θ Ophiuchi	W.	11	15 54 26.52	—0.59	—0.27	+0.17	+0.28	—0.02	26.09	17 14 35.17	+1 20 9.08	0.09

$a' = +0^s.397$ (circle E.); $a'' = +0^s.877$ (circle W.); $c = 0^s.274$ (— with circle E.).
Chronometer No. 1254, at 13^h 30^m chron. time, 1^h 20^m $8^s.99 \pm 0^s.016$ slow, gaining $0^s.010$ per hour.
14^h $14^m.6$ 1^h 20^m $8^s.98$

Transits of stars observed at Montevideo, Uruguay, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Aximuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
July 26	δ Octantis	W.	6	12 46 11.96	−3.05	+1.00	−5.44	+ 9.42	−0.10	13.79	14 7 38.88	+1 21 [25.09]	. .
	α Apodis	.	11	13 11 27.42	2.00	0.66	−3.03	5.68	0.07	28.66	14 32 52.51	[23.85]	. .
	α^2 Libræ	.	11	13 22 45.65	0.54	0.18	+0.31	1.18	0.05	46.73	14 44 10.75	24.02	0.12
	ξ^2 Libræ	.	11	13 28 46.62	0.51	0.17	0.36	1.15	0.04	47.75	14 50 11.92	24.17	0.03
	20 Libræ	.	11	13 35 33.92	0.59	+0.19	0.17	1.24	0.03	34.90	14 56 59.00	24.10	0.04
	β Libræ	.	11	13 49 4.39	0.50	0.00	+0.38	1.14	0.02	5.39	15 10 29.62	24.23	0.09
	ρ Octantis	.	7	13 54 13.28	3.47	0.00	−6.39	10.91	−0.01	14.32	15 15 37.76	[23.44]	. .
	ζ Libræ	W.	8	14 0 0.85	−0.54	0.00	+0.29	+ 1.18	0.00	1.78	15 21 25.81	+1 21 24.03	0.11
	β Mensæ, S. P. . .	E.	10	14 12 50.98	−1.13	+0.06	+4.63	+ 5.98	+0.02	60.54	3 34 24.49	+1 21 [23.95]	. .
	λ Libræ	.	11	14 24 54.87	+0.57	−0.03	0.28	− 1.23	0.04	54.50	15 46 18.60	24.10	0.04
	δ Scorpii	.	11	14 31 47.03	0.58	0.03	0.24	1.26	0.05	46.61	15 53 10.81	24.20	0.06
	β^1 Scorpii	.	11	14 37 0.38	0.56	0.03	+0.28	1.23	0.05	0.01	15 58 24.17	24.16	0.02
	δ^1 Apodis	.	11	14 40 63.52	1.98	0.11	−3.36	5.79	0.06	56.30	16 2 20.26	[23.96]	. .
	δ Ophiuchi	.	11	14 46 36.49	0.47	0.02	+0.51	1.16	0.07	36.36	16 8 0.48	24.12	0.02
	ϵ Ophiuchi	E.	11	14 50 31.30	+0.47	−0.03	+0.50	− 1.16	+0.07	31.15	16 11 55.37	+1 21 24.22	0.08

$a' = +0^s.876$ (circle W.); $a'' = +0^s.985$ (circle E.); $c = 1^s.148$ (— with circle E.).
Chronometer No. 1295, at 14^h 0^m chron. time, $1^h\ 21^m\ 24^s.135 \pm 0^s.016$ slow, losing $0^s.086$ per hour.
$15^h\ 23^m.1$ $1^h\ 21^m\ 24^s.254$

July 27	θ Centauri	W.	11	12 38 6.70	−0.68	−0.02	−0.03	+1.31	−0.10	7.18	13 59 33.19	+1 21 26.01	0.00
	η Apodis	.	10	12 41 40.97	2.32	0.08	−5.76	6.43	0.10	39.14	14 3 5.37	[26.23]	. .
	ι Virginis	.	8	12 48 12.52	0.48	0.02	+0.66	1.07	0.09	13.66	14 9 39.76	26.10	0.09
	λ Virginis	.	11	12 51 6.17	−0 52	−0.02	0.52	+1.11	0.09	7.17	14 12 33.29	26.12	0.11
	μ Hydri, S. P. . .	.	11	13 12 49.68	+1.27	+0.05	6.79	−5.94	0.06	51.79	2 34 18.02	[26.23]	. .
	α^2 Libræ	.	11	13 22 43.88	−0.54	−0.02	0.47	+1.11	0.05	44.85	14 44 10.74	25.89	0.12
	ξ^2 Libræ	.	11	13 28 44.93	0.51	0.02	0.55	1.09	0.04	46.00	14 50 11.90	25.90	0.11
	20 Libræ	W.	11	13 35 32.31	−0.59	−0.02	+0.25	+1.17	−0.03	33.09	14 56 58.99	+1 21 25.90	0.11
	ι Hydri, S. P. . . .	E.	11	13 57 23.54	−1.01	−0.37	+6.23	+5.22	0.00	33.61	3 18 59.79	+1 21 [26.18]	. .
	β Mensæ, S. P. . .	.	11	14 12 47 80	−1.13	−0.41	6.69	+5.65	+0.02	58.62	3 34 24.58	[25.96]	. .
	λ Libræ	.	11	14 24 52.56	+0.57	+0.21	0.40	−1.17	0.03	52.60	15 46 18.59	25.99	0.02
	δ Scorpii	.	11	14 31 44.68	0.58	0.21	0.34	1.19	0.04	44.66	15 53 10.80	26.14	0.13
	β^1 Scorpii	.	11	14 36 58.12	0.56	0.20	+0.40	1.17	0.05	58.16	15 58 24.16	26.00	0.01
	δ^1 Apodis	.	11	14 40 61.70	1.98	0.72	−4.85	5.48	0.05	54.12	16 2 20.19	[26.07]	. .
	δ Ophiuchi	.	10	14 46 34.11	0.47	0.17	+0.74	1.10	0.06	34.45	16 8 0.47	26.02	0.01
	ϵ Ophiuchi	E.	11	14 50 29.02	+0.47	+0.17	+0.72	−1.10	+0.06	29.34	16 11 55.36	+1 21 26.02	0.01

$a' = +1^s.342$ (circle W.); $a'' = +1^s.423$ (circle E.); $c = 1^s.085$ (— with circle F.).
Chronometer No. 1295, at 14^h 0^m chron. time, $1^h\ 21^m\ 26^s.008 \pm 0^s.018$ slow, losing $0^s.074$ per hour.
$15^h\ 34^m.1$ $1^h\ 21^m\ 26^s.124$

Transits of stars observed at Buenos Ayres, Argentine Republic, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
July 26	ξ^2 Libræ	E.	11	13 39 12.59	+0.50	−0.10	+ 0.36	+0.38	+0.11	13.84	14 50 11.88	+1 10 58.04	0.00
	20 Libræ	.	11	13 45 59.78	0.59	0.15	0.17	0.41	0.11	0.91	14 56 59.00	58.09	0.05
	β Libræ	.	8	13 59 30.36	+0.49	−0.09	0.39	+0.38	0.09	31.62	15 10 29.61	57.99	0.05
	ι Hydri, S. P. . . .	.	11	14 8 0.55	−0.98	0.00	3.87	−1.77	0.08	1.75	3 18 59.66	[57.91]	. .
	β Mensæ, S. P. . .	.	11	14 23 25.58	−1.10	0.00	4.16	−1.92	0.07	26.79	3 34 24.50	[57.71]	. .
	λ Libræ	.	11	14 35 19.42	+0.56	−0.08	0.24	+0.40	0.06	20.60	15 46 18.62	58.02	0.02
	β^1 Scorpii	.	11	14 47 24.99	0.55	0.14	+ 0.24	0.40	0.04	26.08	15 58 24.14	58.06	0.02
	δ^1 Apodis	E.	12	14 51 22.75	+1.94	−0.56	− 3.04	+1.86	+0.04	22.99	16 2 20.26	+1 10 [57.27]	. .
	α Scorpii	W.	11	15 11 1.90	−0.60	−0.22	+ 0.89	−0.45	+0.02	1.54	16 21 59.57	+1 10 58.03	0.01
	γ Mensæ, S. P. . .	.	11	16 25 14.16	+0.82	+0.31	21.69	+1.74	−0.03	38.69	5 36 36.91	[58 22]	. .
	Lacaille 2138, S.P.	.	11	16 35 11.48	+1.40	+0.46	30.08	+2.49	0.07	45.84	5 46 43.43	[57.59]	. .
	ν Ophiuchi	.	11	16 41 23.26	−0.50	−0.14	2.32	−0.41	0.08	24.45	17 52 22.49	58.04	0.00
	γ^2 Sagittarii	.	11	16 47 5.53	0.62	0.17	0.46	0.47	0.08	4.65	17 58 2.80	58.15	0.11
	μ^1 Sagittarii	.	11	16 55 34.10	0.56	0.13	1.37	0.44	0.09	34.25	18 6 32.26	58.01	0.03
	ϵ Sagittarii	.	11	17 5 12.57	0.66	0.12	+ 0.02	0.49	0.10	11.22	18 16 9.20	57.98	0.06
	ζ Pavonis	W.	13	17 18 11.58	−1.36	−0.16	−10.34	−1.29	−0.11	58.32	18 28 56.14	+1 10 [57.82]	. .

$a' = +0^s.883$ (circle E.); $a'' = +5^s.456$ (circle W.); $c = 0^s.391$ (+ with circle E.).
Chronometer No. 1254, at $15^h\ 30^m$ chron. time, $1^h\ 10^m\ 58^s.04 \pm 0^s.011$ slow, gaining $0^s.061$ per hour.
$15^h\ 24^m.4$ $1^h\ 10^m\ 58^s.04$

July 27	α^2 Libræ	E.	11	13 33 12.54	+0.53	−0.03	+ 0.58	+0.39	+0.07	14.08	14 44 10.75	+1 10 56.67	0.01
	ξ^2 Libræ	.	11	13 39 13.74	0.50	−0.09	0.70	0.38	0.06	15.29	14 50 11.87	56.58	0.10
	20 Libræ	.	11	13 46 0.98	+0.59	+0.07	0.32	+0.41	0.06	2.43	14 56 58.99	56.56	0.12
	ι Hydri, S. P. . . .	.	11	14 7 58.64	−0.98	0.00	7.48	−1.78	0.05	3.41	3 18 59.76	[56.35]	. .
	β Mensæ, S. P. . .	.	11	14 23 23.36	1.10	+0.06	8.03	1.93	0.02	28.44	3 34 24.60	[56.16]	. .
	γ Hydri, S. P. . . .	.	11	14 38 5.52	−0.67	−0.05	6.07	−1.42	+0.01	9.46	3 49 6.67	[57.21]	. .
	δ Scorpii	.	11	14 42 12.53	+0.57	+0.07	0.39	+0.41	0.00	13.97	15 53 10.78	56.81	0.13
	β^1 Scorpii	E.	11	14 47 25.90	+0.55	+0.09	+ 0.47	+0.40	0.00	27.41	15 58 24.13	+1 10 56.72	0.04
	ν^2 Scorpii	W.	11	14 54 0.70	−0.55	−0.14	+ 1.61	−0.44	−0.01	1.17	16 4 57.92	+1 10 56.75	0.07
	δ Ophiuchi	.	11	14 57 1.81	0.46	0.11	+ 3.01	0.41	0.01	3.83	16 8 0.44	56.61	0.07
	γ Apodis	.	11	15 4 25.85	1.97	0.34	−20.45	2.08	0.02	0.99	16 14 57.58	[56.59]	. .
	α Scorpii	.	11	15 11 3.07	0.60	0.11	+ 0.95	0.46	0.02	2.83	16 21 59.56	56.73	0.05
	β Apodis	.	11	15 15 16.51	1.80	0.36	−17.84	1.86	0.03	54.62	16 25 52.00	[57.38]	. .
	η Ophiuchi	.	11	15 52 29.20	0.53	0.14	+ 1.97	0.43	0.07	30.00	17 3 26.69	56.69	0.01
	ι Apodis	.	11	15 57 53.93	1.29	0.34	− 9.04	1.20	0.07	41.99	17 8 38.11	[56.12]	. .
	θ Ophiuchi	W.	11	16 3 38.61	−0.59	−0.14	+ 1.08	−0.45	−0.08	38.43	17 14 35.13	+1 10 56.70	0.02

$a' = +1^s.707$ (circle E.); $a'' = +5^s.802$ (circle W.); $c = 0^s.393$ (+ with circle E.).
Chronometer No. 1254, at $14^h\ 45^m$ chron. time, $1^h\ 10^m\ 56^s.68 \pm 0^s.017$ slow, gaining $0^s.059$ per hour.
$15^h\ 35^m.4$ $1^h\ 10^m\ 56^s.63$

Transits of stars observed at Montevideo, Uruguay, by Lieut. Commander F. M. Green, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
July 28	θ Centauri	E.	11	12 38 6.05	+0.68	+0.02	−0.03	−1.31	−0.05	5.36	13 59 33.18	+1 21 27.82	0.20
	η Apodis	.	10	12 41 48.35	2.32	0.08	−6.72	6.43	0.05	37.55	14 3 5.26	[27.71]	. .
	ι Virginis	.	11	12 48 12.10	0.48	0.02	+0.77	1.07	0.04	12.26	14 9 39.75	27.49	0.13
	λ Virginis	.	11	12 51 5.73	+0.52	+0.02	0.61	−1.11	0.04	5.73	14 12 33.28	27.55	0.07
	μ Hydri, S. P. . . .	.	11	13 12 37.47	−1.27	−0.05	7.93	+5.94	0.02	50.00	2 34 18.14	[28.14]	. .
	α^2 Libræ	.	11	13 22 43.11	+0.54	+0.02	0.55	−1.11	−0.01	43.10	14 44 10.73	27.63	0.01
	ξ^2 Libræ	.	11	13 28 44.22	0.51	0.02	0.64	1.09	0.00	44.30	14 50 11.89	27.59	0.03
	20 Libræ	E.	11	13 35 31.57	+0.59	+0.02	+0.30	−1.17	+0.01	31.32	14 56 58.97	+1 21 27.65	0.03
	β Libræ	W.	11	13 49 0.80	−0.50	+0.05	+0.56	+1.04	+0.02	1.97	15 10 29.60	+1 21 27.63	0.01
	ι Hydri, S. P. . . .	.	11	13 57 30.57	+1.01	−0.11	5.58	−4.90	0.03	32.18	3 18 59.89	[27.71]	. .
	κ Libræ	W.	11	14 13 29.78	−0.56	+0.06	+0.36	+1.10	+0.04	30.78	15 34 58.42	+1 21 27.64	0.02

$a' = +1^s.567$ (circle E.); $a'' = +1^s.273$ (circle W.); $c = 1^s.051$ (— with circle E.).
Chronometer No. 1295, at $13^h\ 30^m$ chron. time, $1^h\ 21^m\ 27^s.625 \pm 0^s.023$ slow, losing $0^s.061$ per hour.
$14^h\ 39^m.1$ $1^h\ 21^m\ 27^s.695$

Transits of stars observed at Buenos Ayres, Argentine Republic, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1878.				*h. m. s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*s.*
July 28	θ Centauri	E.	11	12 48 36.73	+0.66	+0.12	−0.02	+0.29	+0.08	37.86	13 59 33.18	+1 10 55.32	0.01
	η Apodis	.	9	12 52 8.87	2.27	0.18	−2.80	1.41	0.08	10.01	14 3 5.21	[55.20]	. .
	κ Virginis	.	8	12 55 29.51	0.50	+0.01	+0.28	0.24	0.08	30.62	14 6 25.95	55.33	0.02
	λ Virginis	.	11	13 1 36.84	0.51	−0.06	+0.25	0.24	0.07	37.85	14 12 33.24	55.39	0.08
	α Apodis	.	11	13 21 56.90	1.96	0.11	−2.26	1.18	0.05	57.72	14 32 52.32	[54.60]	. .
	α^2 Libræ	.	11	13 33 14.40	0.53	0.03	+0.22	0.24	0.04	15.40	14 44 10.74	55.34	0.03
	20 Libræ	.	11	13 46 2.81	+0.59	−0.03	0.12	+0.26	0.03	3.78	14 56 58.97	55.19	0.12
	θ Hydri, S. P. . . .	E.	11	13 51 5.40	−0.52	+0.03	+2.04	−0.77	+0.02	6.20	3 2 1.02	+1 10 [54.82]	. .
	β Libræ	W.	11	13 59 34.46	−0.49	−0.16	+0.83	−0.27	+0.01	34.38	15 10 29.59	+1 10 55.21	0.10
	ι Hydri, S. P. . . .	.	11	14 7 53.22	+0.98	+0.25	8.33	+1.27	+0.01	4.06	3 18 59.86	[55.80]	. .
	β Mensæ, S. P. . .	.	11	14 23 18.01	+1.10	+0.19	8.95	+1.37	−0.01	29.61	3 34 24.70	[55.09]	. .
	α Scorpii	.	11	15 11 4.97	−0.60	−0.16	0.31	−0.30	0.05	4.17	16 21 59.55	55.38	0.07
	ζ Ophiuchi	.	11	15 19 34.90	0.50	0.12	+0.79	0.27	0.06	34.74	16 30 30.02	55.28	0.03
	α Trianguli Aust. .	.	11	15 25 2.67	1.24	0.30	−2.96	0.74	0.07	57.36	16 35 52.87	[55.51]	. .
	ϵ Scorpii	.	11	15 31 25.82	0.65	0.15	+0.02	0.32	0.07	24.65	16 42 19.97	55.32	0.01
	η Ophiuchi	W.	11	15 52 31.86	−0.53	−0.25	+0.64	−0.28	−0.09	31.35	17 3 26.69	+1 10 55.34	0.03

$a' = +0^s.647$ (circle E.); $a'' = +1^s.901$ (circle W.); $c = 0^s.251$ (+ with circle E.).
Chronometer No. 1254, at $14^h\ 15^m$ chron. time, $1^h\ 10^m\ 55^s.31 \pm 0^s.014$ slow, gaining $0^s.057$ per hour.
$14^h\ 40^m.4$ $1^h\ 10^m\ 55^s.29$

Transits of stars observed at Pernambuco, Brazil, by Lieut. Commander C. H. Davis, U. S. N., to determine the correction of sidereal chronometer Negus 1254.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Sept. 28	λ Aquilæ	W.	11	16 22 12.52	—0.54	—0.22	—0.02	—0.24	—0.01	11.49	18 59 53.20	+2 37 41.68	0.03
	δ Draconis	.	11	16 34 52.01	0.35	0.16	0.73	0.62	0.01	50.14	19 12 31.87	[41.73]	. .
	τ Draconis	.	6	16 40 12.02	0.28	0.16	0.98	0.82	0.01	9.77	19 17 51.71	[41.94]	. .
	κ Aquilæ	.	11	16 52 46.12	0.54	0.34	0.00	0.24	—0.01	44.99	19 30 26.59	41.60	0.11
	γ Aquilæ	.	11	17 2 53.26	0.52	0.36	0.09	0.24	0.00	52.05	19 40 33.82	41.77	0.06
	ε Draconis	.	11	17 10 55.87	0.33	0.24	0.82	0.69	0.00	53.79	19 48 35.26	[41.47]	. .
	τ Aquilæ	.	11	17 20 36.74	0.53	0.41	0.08	0.24	0.00	35.48	19 58 17.28	41.80	0.09
	θ Aquilæ	W.	11	17 27 26.97	—0.54	—0.43	—0.03	—0.24	0.00	25.73	20 5 7.44	+2 37 41.71	0.00
	α^1 Capricorni	E.	11	17 33 18.31	+0.55	—0.32	—0.02	+0.20	0.00	18.72	20 11 0.51	+2 37 41.79	0.08
	π Capricorni	.	11	17 42 45.77	0.56	0.32	0.04	0.21	0.00	46.18	20 20 27.92	41.74	0.03
	ϕ Capricorni	.	11	17 44 19.56	0.56	0.32	—0.03	0.21	0.00	19.98	20 22 1.71	41.73	0.02
	θ Cephei	.	11	17 49 52.22	0.39	0.22	+0.38	0.42	0.00	53.19	20 27 35.05	[41.86]	. .
	73 Draconis	.	11	17 55 23.12	0.26	0.15	0.69	0.73	+0.01	24.66	20 33 6.23	[41.57]	. .
	3 Aquarii	.	11	18 3 43.09	0.54	0.31	0.01	0.20	0.01	43.54	20 41 25.14	41.60	0.11
	η Cephei	.	9	18 5 9.37	0.40	0.23	0.36	0.41	0.01	10.32	20 42 52.05	[41.73]	. .
	α Equulei	E.	11	18 32 8.24	+0.53	—0.30	+0.04	+0.20	+0.01	8.72	21 9 50.42	+2 37 41.70	0.01

$a' = -0^s.288$ (circle W.); $a'' = +0^s.186$ (circle E.); $c = 0^s.216$ (+ with circle E.).
Chronometer No. 1254, at $17^h\ 28^m$ chron. time, $2^h\ 37^m\ 41^s.71 \pm 0^s.015$ slow, losing $0^s.013$ per hour.
$19^h\ 15^m.4$ $2^h\ 37^m\ 41^s.73$

Sept. 30	o Draconis (seq.) .	W.	11	16 11 46.09	—0.41	—0.15	—1.53	—0.38	—0.01	46.61	18 49 25.79	+2 37 [42.18]	. .
	v Draconis	.	8	16 18 13.18	0.31	0.14	2.58	0.60	0.01	9.54	18 55 51.85	[42.31]	. .
	λ Aquilæ	.	11	16 22 12.10	0.54	0.26	0.05	0.20	0.01	11.04	18 59 53.17	42.13	0.02
	ω Aquilæ	.	9	16 34 30.50	0.52	0.28	0.29	0.20	0.01	29.20	19 12 11.35	42.15	0.04
	δ Aquilæ	.	11	16 41 46.28	0.53	0.30	0.16	0.20	—0.01	45.08	19 19 27.25	42.17	0.06
	γ Aquilæ	.	11	17 2 53.03	0.52	0.36	0.27	0.20	0.00	51.68	19 40 33.78	42.10	0.01
	β Aquilæ	W.	11	17 11 44.78	—0.53	—0.39	—0.21	—0.20	0.00	43.45	19 49 25.67	+2 37 42.22	0.11
	τ Aquilæ	E.	7	17 20 34.88	+0.52	—0.30	—0.10	+0.15	0.00	35.15	19 58 17.25	+2 37 42.10	0.01
	α^2 Capricorni	.	8	17 33 42.01	0.55	0.29	+0.03	0.16	0.00	42.46	20 11 24.44	41.98	0.13
	β^2 Capricorni	.	7	17 36 34.29	0.56	0.29	0.05	0.16	+0.01	34.78	20 14 16.86	42.08	0.03
	ϕ Capricorni	.	8	17 44 19.17	0.56	0.31	+0.07	0.16	0.01	19.66	20 22 1.68	42.02	0.09
	θ Cephei	.	4	17 49 52.92	0.39	0.22	—0.75	0.33	0.01	52.68	20 27 34.97	[42.29]	. .
	Groom. 3415 . .	E.	10	18 31 4.59	+0.41	—0.28	—0.67	+0.30	+0.02	4.37	21 8 46.16	+2 37 [41.79]	. .

$a' = -0^s.850$ (circle W.); $a'' = -0^s.368$ (circle E.); $c = 0^s.174$ (+ with circle E.).
Chronometer No. 1254, at $17^h\ 11^m$ chron. time, $2^h\ 37^m\ 42^s.11 \pm 0^s.017$ slow, losing $0^s.013$ per hour.
$19^h\ 16^m.4$ $2^h\ 37^m\ 42^s.13$

Transits of stars observed at Para, Brazil, by Lieut. F. A. Norris, U. S. N., to determine the correction of sidereal chronometer Negus 1295.

Date.	Name of Star.	Circle.	No. of threads.	Transit over mean of threads.	Flexure.	Level.	Azimuth.	Aberration and collimation.	Rate.	Seconds of corr. transit.	R. A.	Chron. correction.	*v.*
1879.				*h.* *m.* *s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*s.*
Sept. 28	*o* Cephei	W.	10	21 18 47.62	—0.52	+0.12	+0.78	—1.18	—0.01	46.81	23 13 44.87	+1 54 [58.06]	. .
	γ Cephei	.	11	21 39 33.29	0.49	0.07	1.39	2.01	0.01	32.24	23 34 30.28	[58.04]	. .
	41 H. Cephei	.	10	21 47 16.49	0.52	0.06	0.76	1.17	0.01	15.61	23 42 13.42	[57.81]	. .
	ω Piscium	.	11	21 58 13.33	0.55	0.04	0.04	0.46	—0.01	12.39	23 53 10.37	57.98	0.10
	γ Pegasi	.	11	22 12 7.80	0.55	0.06	+0.09	0.47	0.00	6.93	0 7 4.84	57.91	0.03
	ι Ceti	.	11	22 18 23.41	0.55	0.07	—0.05	0.46	0.00	22.42	0 13 20.29	57.87	0.01
	12 Ceti	.	11	22 28 59.40	0.55	0.11	—0.02	0.45	0.00	58.49	0 23 56.34	57.85	0.03
	ε Andromedæ . . .	W.	11	22 37 17.39	—0.54	+0.13	+0.18	—0.52	0.00	16.64	0 32 14.45	+1 54 57.81	0.07
	γ Cassiopeæ	E.	11	22 54 30.94	+0.53	+0.21	+0.22	+0.82	0.00	32.72	0 49 30.76	+1 54 [58.04]	. .
	ε Piscium	.	11	23 1 45.33	0.55	0.24	0.02	0.41	+0.01	46.56	0 56 44.35	57.79	0.09
	τ Piscium	.	11	23 10 5.52	0.54	0.25	0.07	0.47	0.01	6.86	1 5 4.78	57.92	0.04
	υ Piscium	.	11	23 17 54.60	0.54	0.29	0.06	0.46	0.01	55.96	1 12 53.79	57.83	0.05
	ψ Cassiopeæ	.	11	23 22 31.16	0.52	0.30	0.30	1.07	0.01	33.36	1 17 31.08	[57.72]	. .
	ν Piscium	.	11	23 40 13.65	0.55	0.03	0.01	0.41	0.01	14.66	1 35 12.58	57.92	0.04
	o Piscium	.	11	23 44 5.85	0.55	0.07	0.02	0.42	0.01	6.92	1 39 4.87	57.95	0.07
	ε Cassiopeæ	E.	11	23 50 49.34	+0.52	+0.12	+0.24	+0.91	+0.01	51.14	1 45 48.91	+1 54 [57.77]	. .

$a' = +0^s.319$ (circle W.); $a'' = +0^s.122$ (circle E.); $c = 0^s.432$ (+ with circle E.).
Chronometer No. 1295, at $22^h\ 30^m$ chron. time, $1^h\ 54^m\ 57^s.88 \pm 0^s.013$ slow, losing $0^s.011$ per hour.
$19^h\ 3^m.6$ $1^h\ 54^m\ 57^s.84$

Sept. 30	*θ* Serpentis (pr.) .	E.	11	16 55 16.18	+0.55	+0.08	—0.19	+0.38	—0.02	16.98	18 50 15.38	+1 54 58.40	0.03
	υ Draconis	.	11	17 0 57.54	0.51	0.07	—5.87	1.17	0.02	53.40	18 55 51.90	[58.50]	. .
	λ Aquilæ	.	11	17 4 53.63	0.55	0.07	+0.13	0.38	0.02	54.74	18 59 53.18	58.44	0.07
	ω Aquilæ	.	11	17 17 12.47	0.55	0.04	—0.45	0.39	0.01	12.99	19 12 11.37	58.38	0.01
	τ Draconis	.	11	17 22 58.03	0.50	0.02	—6.60	1.31	0.01	53.25	19 17 51.68	[58.43]	. .
	κ Aquilæ	.	11	17 35 27.12	0.55	0.02	+0.20	0.38	0.01	28.26	19 30 26.59	58.33	0.04
	γ Aquilæ	.	11	17 45 34.85	0.55	0.02	—0.41	0.39	0.01	35.39	19 40 33.77	58.38	0.01
	α Aquilæ	.	11	17 49 57.29	0.55	+0.01	0.35	0.38	0.01	57.87	19 44 56.20	58.33	0.04
	ε Draconis	E.	7	17 53 40.82	+0.51	—0.01	—5.50	+1.11	—0.01	36.92	19 48 35.08	+1 54 [58.16]	. .
	κ Cephei	W.	8	18 18 8.74	—0.49	+0.03	—8.77	—1.93	0.00	57.58	20 12 55.81	+1 54 [58.23]	. .
	α Cephei	.	11	19 20 51.54	0.52	—0.14	3.74	0.90	+0.01	46.25	21 15 44.34	[58.09]	. .
	α Aquarii	.	11	20 4 41.29	0.55	0.28	0.02	0.42	0.02	40.04	21 59 38.36	58.32	0.05
	θ Pegasi	.	11	20 9 13.05	0.55	0.29	0.24	0.42	0.02	11.57	22 4 9.94	58.37	0.00
	24 Cephei	.	11	20 12 42.11	0.51	0.28	—5.98	1.35	0.02	34.01	22 7 32.81	[58.80]	. .
	θ Aquarii	.	11	20 15 33.96	0.55	0.31	+0.24	0.43	0.02	32.93	22 10 31.29	58.36	0.01
	γ Aquarii	.	11	20 20 31.51	0.55	0.32	+0.02	0.42	0.02	30.26	22 15 28.68	58.42	0.05
	π Aquarii	.	11	20 24 13.08	0.55	0.32	—0.08	0.42	0.02	11.73	22 19 10.12	58.39	0.02
	η Aquarii	W.	11	20 34 15.56	—0.55	—0.34	—0.02	—0.42	+0.02	14.25	22 29 12.60	+1 54 58.35	0.02

$a' = -1^s.987$ (circle E.); $a'' = -1^s.957$ (circle W.); $c = 0^s.401$ (+ with circle E.).
Chronometer No. 1295, at $18^h\ 30^m$ chron. time, $1^h\ 54^m\ 58^s.37 \pm 0^s.007$ slow, losing $0^s.011$ per hour.
$19^h\ 4^m.6$ $1^h\ 54^m\ 58^s.38$

Deduced difference of longitude between the Greenwich and Lisbon stations.

	June 25.		June 26.		June 27.		June 29.	
Middle time of exchange of cable signals to which clock corrections and comparisons are reduced.		*h. m.*		*h. m.*		*h. m.*		*h. m.*
	Greenwich . .	19 47		19 22		20 27		19 17
	Lisbon	14 40		14 25		15 30		14 20
		s.		*s.*		*s.*		*s.*
Greenwich clock slow at 16^h		6.81		6.57		6.48		6.09
Greenwich clock daily rate		—0.26		—0.17		—0.20		—0.20
Greenwich clock slow at epoch of reduction . .		6.78		6.55		6.44		6.06
Porthcurnow clock slow of Greenwich clock—	*h. m. s.*							
by signals from Greenwich to Porthcurnow .	5 8 22.29		5 8 23.39		5 8 24.36		5 8 26.28	
by signals from Porthcurnow to Greenwich .	5 8 22.41		5 8 23.48		5 8 24.52		5 8 26.36	
	5 8 22.35		5 8 23.44		5 8 24.44		5 8 26.32	
Correction to reduce this comparison to the middle time of exchange of cable signals . .	+ 0.16	5 8 22.51	+ 0.12	5 8 23.56	+ 0.12	5 8 24.56	+ 0.09	5 8 26.41
Porthcurnow clock slow of Greenwich time . .		5 8 29.29		5 8 30.11		5 8 31.00		5 8 32.47
Carcavellos clock slow of Porthcurnow clock—	*m. s.*							
by signals from Porthcurnow to Carcavellos	1 56.19		1 54.60		1 52.99		1 50.14	
by signals from Carcavellos to Porthcurnow	1 57.00	1 56.60	1 55.52	1 55.06	1 53.97	1 53.48	1 50.98	1 50.56
Carcavellos clock slow of Greenwich time . . .		5 10 25.89		5 10 25.17		5 10 24.48		5 10 23.03
Carcavellos clock slow of Lisbon clock—								
before exchange of cable signals	12 53.78		12 53.93		12 53.99		12 54.02	
after exchange of cable signals	12 53.80	12 53.79	12 54.04	12 53.98	12 54.05	12 54.02	12 53.98	12 54.00
Lisbon clock slow of Greenwich time		4 57 32.10		4 57 31.19		4 57 30.46		4 57 29.03
Lisbon clock slow of Lisbon time at epoch of reduction		4 20 47.39		4 20 46.62		4 20 45.77		4 20 44.43
Resulting difference of longitude between Greenwich and Lisbon stations		36 44.71		36 44.57		36 44.69		36 44.60
Wave and armature time Greenwich–Porthcurnow		*s.* 0.06		0.045		0.08		0.04
Wave and armature time Porthcurnow–Carcavellos		0.40		0.41		0.49		0.42

Mean difference of longitude between Greenwich and Lisbon from four nights' work. } $36^m\ 44.^s64$

Differences of longitude deduced from exchange of signals.

Eastern and western stations.	Date.	Observer.	Position of observer.	Number of time stars observed.	Chronometer correction at epoch of comparison. ΔT_e } +slow. ΔT_w } −fast.	Excess of correction of eastern chronometer. $\Delta T_e - \Delta T_w$.	Number of chronometer comparisons.	Comparison of chronometers by east and west signals. T' and T''.	Difference of longitude by east and west signals. λ' and λ''.	Difference of longitude for each night. $\frac{1}{2}(\lambda' + \lambda'')$.	Wave and armature time. $\frac{1}{2}(\lambda' - \lambda'')$.
					h. *m.* *s.*	*m.* *s.*		*m.* *s.*	*m.* *s.*	*m.* *s.*	*s.*
Lisbon	Jan. 21, 1878	G.	E.	12	+4 31 6.30	32 17.36	69	1 25.84	30 51.52	30 51.10	0.42
Funchal	Jan. 21, 1878	D.	W.	10	+3 58 48.92		63	1 26.68	30 50.68		
Lisbon	Jan. 22, 1878	G.	E.	13	+4 31 6.91	32 19.05	63	1 27.55	30 51.50	30 51.11	0.39
Funchal	Jan. 22, 1878	D.	W.	9	+3 58 47.86		64	1 28.33	30 50 72		
Lisbon	Jan. 23, 1878	G.	E.	14	+4 31 7.29	32 20.25	64	1 29.03	30 51.22	30 50.86	0.36
Funchal	Jan. 23, 1878	D.	W.	9	+3 58 47.04		65	1 29.75	30 50.50		
Lisbon	Jan. 26, 1878	G.	E.	15	+4 31 8.22	32 24.42	64	1 33.20	30 51.22	30 50.86	0.37
Funchal	Jan. 26, 1878	D.	W.	10	+3 58 43.80		78	1 33.93	30 50.49		
Lisbon	Jan. 27, 1878	G.	E.	11	+4 31 8.30	32 25.22	62	1 33.99	30 51.23	30 50.81	0.42
Funchal	Jan. 27, 1878	D.	W.	10	+3 58 43.08		60	1 34.84	30 50.38		
										30 50.95	0.39
Funchal	Feb. 28, 1878	D.	E.	10	+3 58 2.97	30 28.81	65	1 53.67	32 22.48	32 21.94	0.53
Porto Grande . . .	Feb. 28, 1878	G.	W.	13	+3 27 34.16		65	1 52.60	32 21.41		
Funchal	Mar. 2, 1878	D.	E.	10	+3 58 0.43	30 27.30	63	1 55.30	32 22.60	32 22.03	0.57
Porto Grande . . .	Mar. 2, 1878	G.	W.	16	+3 27 33.13		65	1 54.15	32 21.45		
Funchal	Mar. 4, 1878	D.	E.	10	+3 57 58.02	30 25.85	62	1 56.60	32 22.45	32 21.91	0.54
Porto Grande . . .	Mar. 4, 1878	G.	W.	13	+3 27 32.17		62	1 55.52	32 21.37		
Funchal	Mar. 5, 1878	D.	E.	10	+3 57 56.88	30 25.12	64	1 57.46	32 22.58	32 22.04	0.54
Porto Grande . . .	Mar. 5, 1878	G.	W.	14	+3 27 31.76		58	1 56.37	32 21.49		
										32 21.98	0.54
Porto Grande . . .	Apr. 17, 1878	G.	E.	10	+3 32 32.58	47 1.37	64	7 29.46	39 31.91	39 31.38	0.53
Pernambuco	Apr. 17, 1878	D.	W.	9	+2 45 31.21		64	7 30.53	39 30.84		
Porto Grande . . .	Apr. 18, 1878	G.	E.	9	+3 32 34.18	47 2.64	64	7 30.53	39 32.11	39 31.58	0.53
Pernambuco	Apr. 18, 1878	D.	W.	10	+2 45 31.54		65	7 31.59	39 31.05		
Porto Grande . . .	Apr. 20, 1878	G.	E.	15	+3 32 37.31	47 4.92	62	7 33.05	39 31.87	39 31.34	0.53
Pernambuco	Apr. 20, 1878	D.	W.	10	+2 45 32.39		65	7 34.12	39 30.80		
Porto Grande . . .	Apr. 23, 1878	G.	E.	11	+3 32 41.51	47 8.00	61	7 36.20	39 31.80	39 31.26	0.54
Pernambuco	Apr. 23, 1878	D.	W.	10	+2 45 33.51		64	7 37.28	39 30.72		
										39 31.39	0.53
Pernambuco	Sept. 7, 1879	D.	E.	10	+2 37 36.32	2 51.63	65	11 48.27	14 39.90	14 39.52	0.38
Bahia	Sept. 7, 1879	N.	W.	10	+2 34 44.69		64	11 47.51	14 39.14		
Pernambuco	Sept. 8, 1879	D.	E.	8	+2 37 36.48	2 51.69	65	11 48.22	14 39.91	14 39.54	0.37
Bahia	Sept. 8, 1879	N.	W.	11	+2 34 44.79		65	11 47.48	14 39.17		
Pernambuco	Sept. 10, 1879	D.	E.	10	+2 37 36.64	2 51.43	65	11 48.31	14 39.74	14 39.38	0.36
Bahia	Sept. 10, 1879	N.	W.	10	+2 34 45.21		64	11 47.60	14 39.03		
Pernambuco	Sept. 11, 1879	D.	E.	10	+2 37 36.72	2 51.42	63	11 48.35	14 39.77	14 39.35	0.42
Bahia	Sept. 11, 1879	N.	W.	10	+2 34 45.30		64	11 47.51	14 38.93		
										14 39.45	0.38

Differences of longitude deduced from exchange of signals—Continued.

Eastern and western stations.	Date.	Observer.	Position of observer.	Number of time stars observed.	Chronometer correction at epoch of comparison. ΔT_e / ΔT_w } +slow. —fast.	Excess of correction of eastern chronometer. $\Delta T_e - \Delta T_w$.	Number of chronometer comparisons.	Comparison of chronometers by east and west signals. T' and T''.	Difference of longitude by east and west signals. λ' and λ''.	Difference of longitude for each night. $\frac{1}{2}(\lambda' + \lambda'')$.	Wave and armature time. $\frac{1}{2}(\lambda' - \lambda'')$.
					h. *m.* *s.*	*m.* *s.*		*m.* *s.*	*m.* *s.*	*m.* *s.*	*s.*
Bahia	Aug. 12, 1879	N.	E.	10	+2 34 37.45	30 8.04	64	11 34.63	18 33.41	18 32.96	0.45
Rio de Janeiro	Aug. 12, 1879	D.	W.	10	+2 4 29.41		65	11 35.53	18 32.51		
Bahia	Aug. 13, 1879	N.	E.	10	+2 34 37.87	30 8.95	64	11 35.48	18 33.47	18 33.05	0.42
Rio de Janeiro	Aug. 13, 1879	D.	W.	11	+2 4 28.92		65	11 36.32	18 32.63		
Bahia	Aug. 15, 1879	N.	E.	10	+2 34 38.59	30 9.92	65	11 36.58	18 33.34	18 32.88	0.46
Rio de Janeiro	Aug. 15, 1879	D.	W.	10	+2 4 28.67		65	11 37.51	18 32.41		
Bahia	Aug. 18, 1879	N.	E.	10	+2 34 39.64	30 11.67	65	11 38.11	18 33.56	18 33.11	0.45
Rio de Janeiro	Aug. 18, 1879	D.	W.	10	+2 4 27.98			11 39.01	18 32.66		
										18 33.00	0.45
Rio de Janeiro	July 1, 1878	G.	E.	12	+2 13 36.28	53 22.22	61	1 14.04	52 8.18	52 7.47	0.71
Montevideo	July 1, 1878	D.	W.	10	+1 20 14.06		57	1 15.46	52 6.76		
Rio de Janeiro	July 5, 1878	G.	E.	9	+2 13 33 91	53 24.52	65	1 16.15	52 8.37	52 7.62	0.75
Montevideo	July 5, 1878	D.	W.	10	+1 20 9.39		65	1 17.66	52 6.86		
Rio de Janeiro	July 6, 1878	G.	E.	11	+2 13 33.29	53 24.31	65	1 16.04	52 8.27	52 7.51	0.76
Montevideo	July 6, 1878	D.	W.	10	+2 20 8.98		63	1 17.56	52 6.75		
										52 7.53	*0.74
Montevideo	July 26, 1878	G.	E.	10	+1 21 24 25	10 26.21	64	1 45.68	8 40.53	8 40.23	0.30
Buenos Ayres	July 26, 1878	D.	W.	10	+1 10 58.04		65	1 46.28	8 39.93		
Montevideo	July 27, 1878	G.	E.	11	+1 21 26.12	10 29.49	63	1 48.87	8 40.62	8 40.33	0.28
Buenos Ayres	July 27, 1878	D.	W.	10	+1 10 56.63		65	1 49.44	8 40.05		
Montevideo	July 28, 1878	G.	E.	7	+1 21 27.70	10 32.41	65	1 52.00	8 40.41	8 40.18	0.23
Buenos Ayres	July 28, 1878	D.	W.	10	+1 10 55.29		64	1 52.46	8 39.95		
										8 40.25	0.27
Pernambuco	Sept. 28, 1879	D.	E.	10	+2 37 41.73	42 43.89	61	11 47.80	54 31.69	54 31.21	0.48
Para	Sept. 28, 1879	N.	W.	10	+1 54 57.84		61	11 46.84	54 30.73		
Pernambuco	Sept. 30, 1879	D.	E.	9	+2 37 42.13	42 43.75	65	11 47.88	54 31.61	54 31.21	0.40
Para	Sept. 30, 1879	N.	W.	12	+1 54 58.38		65	11 47.06	54 30.81		
										54 31.21	0.44

*The remarkably large wave and armature time in this case is attributed by the superintendent of the line, J. R. France, esq., to the small diameter of the copper conductor.

Latitude of observing station, Funchal, Madeira, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Feb. 17	1748	+ 9 24 22.56	+32 32 36.67	+ 5 28.10	—2.40	+0.11	+32 38 2.48	1.97
	1854	55 40 50.78						
	1766	9 13 23.72	32 27 7.25	+10 57.85	1.60	+0.22	32 38 3.72	0.73
	1854	55 40 50.78						
	1989	16 9 28.12	32 45 15.06	— 7 3.45	9.60	—0.14	32 38 1.87	2.58
	2044	49 21 1.99						
	2201	48 55 8.94	32 34 58.56	+ 3 7.27	1.60	+0.06	32 38 4.29	0.16
	2285	16 14 48.19						
	2201	48 55 8.94	32 38 31.49	— 0 24.32	—1.33	—0.01	32 38 5.83	1.38
	2362	16 21 54.04						
	2416	36 59 24.84	32 30 54.27	+ 7 8.26	+3.60	+0.12	32 38 6.25	1.80
	2442	28 2 23.70						
	2416	36 59 24.84	32 34 43.68	+ 3 19.91	+3.87	+0.06	32 38 7.52	3.07
	2469	28 10 2.51						
	2609	47 52 48.86	32 45 38.54	— 7 33.96	—1.60	—0.15	32 38 2.83	1.62
	2659	17 38 28.23						
	2714	21 56 4.28	32 45 22.95	— 7 18.43	0.93	—0.15	32 38 3.44	1.01
	2793	43 34 41.62						
	2815	28 17 37.52	32 33 54.80	+ 4 11.02	—0.53	+0.07	32 38 5.36	0.91
	2871	36 50 12.07						
	2909	46 15 38.93	32 25 49.98	12 14.93	+1.07	0.24	32 38 6.22	1.77
	2953	18 36 1.02						
	3109	30 8 33.06	32 31 27.28	6 37.73	—0.67	0.11	32 38 4.45	0.00
	3178	34 54 21.50						
Feb. 20	1748	9 24 22.47	32 32 36.54	5 24.46	+1.20	0.11	32 38 2.31	2.14
	1854	55 40 50.61						
	1766	9 13 23.62	32 27 7.12	10 53.52	+2.00	0.22	32 38 2.86	1.59
	1854	55 40 50.61						
	1876	20 15 12.81	32 35 42.36	2 28.87	—8.27	0.05	32 38 3.01	1.44
	1895	44 56 11.92						
	1895	44 56 11.92	32 18 52.57	+19 19.35	8.27	+0.38	32 38 4.03	0.42
	1934	19 41 33.22						
	1989	16 9 28.09	32 45 15.18	— 7 11.70	—0.27	—0.14	32 38 3.07	1.38
	2044	49 21 2.28						
	2009	16 10 49.66	32 45 55.97	— 7 53.88	0.00	—0.16	32 38 1.93	2.52
	2044	49 21 2.28						
	2201	48 55 9.28	32 34 58.73	+ 3 11.74	—4.94	+0.06	32 38 5.59	1.14
	2285	16 14 48.18						
	2201	48 55 9.28	+32 31 18.94	+ 6 51.36	—4.94	+0.14	+32 38 5.50	1.05
	2330	+16 7 28.60						

Latitude of observing station, Funchal, Madeira, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Feb. 20	2201	+48 55 9.28	+32 38 31.66	— 0 21.92	—4.94	—0.01	+32 38 4.79	0.34
	2362	16 21 54.03						
Feb. 21	1748	9 24 22.44	32 32 36.56	+ 5 28.79	3.07	+0.11	32 38 2.39	2.06
	1854	55 40 50.69						
	1766	9 13 23.59	32 27 7.14	+10 59.78	3.07	+0.22	32 38 4.07	0.38
	1854	55 40 50.69						
	1989	16 9 28.08	32 45 15.23	— 7 9.78	2.40	—0.14	32 38 2.91	1.54
	2044	49 21 2.38						
	2090	20 17 20.33	32 27 57.15	+10 7.36	—1.07	+0.20	32 38 3.64	0.81
	2182	44 38 33.97						
	2201	48 55 9.40	32 34 58.79	3 4.11	+3.60	0.06	32 38 6.56	2.11
	2285	16 14 48.18						
Feb. 22	1876	20 15 12.81	32 35 42.42	2 28.32	—4.80	0.05	32 38 5.99	1.54
	1895	44 56 12.04						
	1895	44 56 12.04	32 18 52.63	+19 14.26	+0.13	+0.38	32 38 7.40	2.95
	1934	19 41 33.22						
	1989	16 9 28.07	32 45 15.28	— 7 12.94	1.07	—0.14	32 38 3.27	1.18
	2044	49 21 2.48						
	2009	16 10 49.65	32 45 56.06	— 7 53.60	+1.33	—0.16	32 38 3.63	0.82
	2044	49 21 2.48						
	2090	20 17 20.36	32 27 57.22	+10 7.98	—2.40	+0.22	32 38 3.02	1.43
	2182	44 38 34.07						
	2416	36 59 25.28	32 30 54.62	+ 7 12.18	1.07	+0.12	32 38 5.85	1.40
	2442	28 2 23.95						
	2416	36 59 25.28	32 40 46.82	— 2 40.48	0.80	—0.04	32 38 5.50	1.05
	2467	28 22 8.37						
	2416	36 59 25.28	32 34 44.03	+ 3 23.90	0.80	+0.06	32 38 7.19	2.74
	2469	28 10 2.78						
	2815	28 17 37.85	32 33 55.24	4 11.50	1.87	0.07	32 38 4.94	0.49
	2871	36 50 12.62						
	2909	46 15 39.72	32 25 50.41	12 18.23	1.47	0.25	32 38 7.42	2.97
	2953	18 36 1.10						
	3109	30 8 33.46	+32 31 27.74	+ 6 39.48	—3.87	+0.11	+32 38 3.46	0.99
	3178	+34 54 22.03						
Mean (37 determinations)							+32 38 4.45	±0.19

Latitude of observing station, Porto Grande, St. Vincent, Cape de Verde Islands, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Mar. 20	2199	+13 21 0.14	+17 2 57.51	— 9 40.55	+3.35	—0.16	+16 53 20.15	0.41
	2305	20 44 54.88						
	2206	13 1 33.40	16 53 14.14	+ 0 2.57	3.35	0.00	16 53 20.06	0.32
	2305	20 44 54.88						
	2255	13 19 54.47	17 2 24.68	— 9 6.54	3.00	—0.15	16 53 20.99	1.25
	2305	20 44 54.88						
	2410	22 12 22.57	17 3 23.76	10 11.54	8.70	0.17	16 53 20.75	1.01
	2444	11 54 24.96						
	2460	21 41 38.30	16 58 32.43	5 19.92	+8.00	0.09	16 53 20.42	0.68
	2473	12 15 26.56						
	2522	5 32 7.00	16 55 40.65	— 2 19.36	—1.94	—0.05	16 53 19.30	0.44
	2555	28 19 14.30						
	2522	5 32 7.00	16 50 7.17	+ 3 14.18	—1.80	+0.06	16 53 19.61	0.13
	2672	28 8 7.34						
	2786	27 36 42.45	16 52 8.34	+ 1 9.35	+2.15	+0.02	16 53 19.86	0.12
	2901	6 7 34.22						
	2786	27 36 42.45	16 56 55.11	— 3 38.59	2.30	—0.06	16 53 18.76	0.98
	2978	6 17 7.77						
	2786	27 36 42.45	17 0 33.14	— 7 17.57	2.30	—0.12	16 53 17.75	1.99
	3032	6 24 23.82						
	2817	27 19 58.73	16 43 46.48	+ 9 32.92	1.85	+0.16	16 53 21.41	1.67
	2901	6 7 34.22						
	2817	27 19 58.73	16 48 33.25	4 44.98	2.00	0.08	16 53 20.31	0.57
	2978	6 17 7.77						
	2817	27 19 58.73	16 52 11.28	1 5.93	2.00	0.03	16 53 19.24	0.50
	3032	6 24 23.82						
	3111	11 9 23.90	16 50 48.28	+ 2 32.20	+0.55	+0.04	16 53 21.07	1.33
	3117	22 32 12.67						
	3246	23 30 12.64	16 58 25.47	— 5 6.36	—1.05	—0.08	16 53 17.98	1.76
	3312	10 26 38.30						
	3366	24 58 11.86	16 47 52.54	+ 5 22.29	+4.90	+0.09	16 53 19.82	0.08
	3415	8 37 33.21						
Mar. 21	2199	13 21 0.14	17 2 57.52	— 9 32.33	—5.35	—0.16	16 53 19.68	0.06
	2305	20 44 54.90						
	2206	13 1 33.40	16 53 14.15	+ 0 10.40	5.85	0.00	16 53 18.70	1.04
	2305	20 44 54.90						
	2255	13 19 54.48	17 2 24.69	— 8 59.30	—5.75	—0.15	16 53 19.49	0.25
	2305	20 44 54.90						
	2410	22 12 22.60	+17 3 23.78	—10 9.44	+6.10	—0.17	+16 53 20.27	0.53
	2444	+11 54 24.96						

Latitude of observing station, Porto Grande, St. Vincent, Cape de Verde Islands, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Mar. 21	2460	+21 41 38.33	+16 58 32.44	−5 14.85	+ 2.20	−0.09	+16 53 19.70	0.04
	2473	12 15 26.56						
	2522	5 32 7.00	16 55 40.67	−2 21.60	0.50	−0.05	16 53 19.52	0.22
	2555	28 19 14.34						
	2522	5 32 7.00	16 50 7.20	+3 10.29	0.50	+0.06	16 53 18.05	1.69
	2672	28 8 7.40						
	2786	27 36 42.51	16 52 8.36	+1 7.58	3.20	+0.02	16 53 19.16	0.58
	2901	6 7 34.20						
	2786	27 36 42.51	16 56 55.14	−3 39.11	3.30	−0.06	16 53 19.27	0.47
	2978	6 17 7.77						
	2786	27 36 42.51	17 0 33.16	−7 16.12	2.70	−0.12	16 53 19.62	0.12
	3032	6 24 23.80						
	2817	27 19 58.79	16 43 46.50	+9 29.63	4.00	+0.16	16 53 20.29	0.55
	2901	6 7 34.20						
	2817	27 19 58.79	16 48 33.28	4 42.94	4.10	0.08	16 53 20.40	0.66
	2978	6 17 7.77						
	2817	27 19 58.79	16 52 11.30	+1 5.93	3.50	+0.03	16 53 20.76	1.02
	3032	6 24 23.80						
Mar. 29	2522	5 32 7.00	16 55 40.83	−2 28.51	6.75	−0.05	16 53 19.02	0.72
	2555	28 19 14.66						
	2522	5 32 7.00	16 50 7.38	+3 3.84	+ 6.95	+0.06	16 53 18.23	1.51
	2672	28 8 7.76						
	2786	27 36 42.97	16 52 8.62	+1 15.80	− 4.10	+0.02	16 53 20.34	0.60
	2901	6 7 34.28						
	2786	27 36 42.97	16 56 55.38	−3 31.48	4.20	−0.06	16 53 19.64	0.10
	2978	6 17 7.78						
	2786	27 36 42.97	17 0 33.40	−7 9.94	4.20	−0.12	16 53 19.14	0.60
	3032	6 24 23.84						
	2817	27 19 59.28	16 43 46.78	+9 37.07	2.65	+0.16	16 53 21.36	1.62
	2901	6 7 34.28						
	2817	27 19 59.28	16 48 33.53	4 49.78	2.75	0.08	16 53 20.64	0.90
	2978	6 17 7.78						
	2817	27 19 59.28	16 52 11.56	1 11.46	− 2.75	0.03	16 53 20.30	0.56
	3032	6 24 23.84						
	3111	11 9 24.08	16 50 48.66	+2 17.78	+15.50	+0.04	16 53 21.98	2.24
	3117	22 32 13.24						
	3246	23 30 13.32	16 58 25.91	−5 15.38	8.85	−0.09	16 53 19.29	0.45
	3312	10 26 38.50						
	3366	24 58 12.64	+16 47 53.02	+5 24.86	+ 0.95	+0.09	+16 53 18.92	0.82
	3415	+ 8 37 33.40						

Latitude of observing station, Porto Grande, St. Vincent, Cape de Verde Islands, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Mar. 29	3508	+24 1 21.66	+16 58 35.70	−5 19.00	+2.65	−0.09	+16 53 19.26	0.48
	3609	9 55 49.74						
	3609	9 55 49.74	+16 52 37.57	+0 38.43	+2.35	+0.01	+16 53 18.36	1.38
	3671	+23 49 25.40						
Mean (42 determinations)							+16 53 19.74	±0.10

Latitude of observing station, Pernambuco, Brazil, from zenith telescope observations, Lieut. F. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	″	″	″	° ′ ″	″
Apr. 29	3227	+ 9 35 6.57	−8 14 16.19	+10 33.01	+ 0.88	+0.21	−8 3 42.09	1.20
	3248	−26 3 38.95						
	3312	+10 26 39.56	8 9 42.09	+ 5 55.27	3.65	+0.12	8 3 43.05	0.24
	3385	−26 46 3.74						
	3312	+10 26 39.56	7 59 38.04	− 4 9.68	4.53	−0,08	8 3 43.27	0.02
	3489	−26 25 55.64						
	3385	−26 46 3.74	8 5 15.28	+ 1 31.29	1.21	+0.03	8 3 42.75	0.54
	3457	+10 35 33.17						
	3457	+10 35 33.17	7 55 11.24	− 8 33.66	+ 2.11	−0.17	8 3 42.96	0.33
	3489	−26 25 55.64						
	3782	+ 0 39 5.87	8 11 2.61	+ 7 18.87	− 0.55	+0.12	8 3 44.17	0.88
	3883	−17 1 11.09						
	3832	+ 0 35 22.67	8 12 54.21	9 10.41	− 0.10	0.15	8 3 43.75	0.46
	3883	−17 1 11.09						
	3928	−31 11 20.38	8 6 56.26	+ 3 7.79	+ 2.62	+0.06	8 3 45.79	2.50
	3992	+14 57 27.87						
	3928	−31 11 20.38	7 58 9.34	− 5 37.76	2.62	−0.11	8 3 44.59	1.30
	3995	+15 15 1.70						
May 1	3227	+ 9 35 6.66	8 14 22.95	+10 34.42	5.38	+0.21	8 3 42.94	0.35
	3243	−26 3 52.56						
	3227	+ 9 35 6.66	8 14 16.16	10 26.73	5.38	0.21	8 3 43.84	0.55
	3248	−26 3 38.97						
	3312	+10 26 39.64	8 9 42.08	+ 5 52.80	6.04	+0.12	8 3 43.12	0.17
	3385	−26 46 3.80						
	3312	+10 26 39.64	−7 59 38.04	− 4 17.30	+11.47	−0.08	−8 3 43.95	0.66
	3349	−26 25 55.72						

Latitude of observing station, Pernambuco, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
May 1	3385	−26 46 3.80	−8 5 15.26	+ 1 33.16	−0.35	+0.03	−8 3 42.42	0.87
	3457	+10 35 33.27						
	3457	+10 35 33.27	7 55 11.22	− 8 36.93	+5.08	−0.17	8 3 43.24	0.05
	3489	−26 25 55.72						
May 4	3227	+ 9 35 6.78	8 14 22.90	+10 42.50	−1.51	+0.21	8 3 41.70	1.59
	3243	−26 3 52.57						
	3227	+ 9 35 6.78	8 14 16.10	10 34.22	−1.51	0.21	8 3 43.18	0.11
	3248	−26 3 38.98						
	3312	+10 26 39.78	8 9 42.04	+ 5 58.28	+0.75	+0.12	8 3 42.89	0.40
	3385	−26 46 3.86						
	3312	+10 26 39.78	7 59 38.02	− 4 8.07	+3.12	−0.08	8 3 43.05	0.24
	3489	−26 25 55.82						
	3385	−26 46 3.86	8 5 15.22	+ 1 32.89	−1.01	+0.03	8 3 43.31	0.02
	3457	+10 35 33.42						
	3457	+10 35 33.42	7 55 11.20	− 8 33.39	+1.36	−0.17	8 3 43.40	0.11
	3489	−26 25 55.82						
	3928	−31 11 20.86	8 6 56.24	+ 3 7.06	+4.28	+0.06	8 3 44.84	1.55
	3992	+14 57 28.39						
	3928	−31 11 20.86	7 58 9.34	− 5 38.69	4.28	−0.11	8 3 43.86	0.57
	3995	+15 15 2.18						
	4042	−25 14 6.22	7 54 51.30	− 8 50.76	+1.21	−0.15	8 3 41.00	2.29
	4072	+ 9 24 23.62						
May 5	3312	+10 26 39.82	8 9 42.03	+ 5 59.68	−1.21	+0.12	8 3 43.44	0.15
	3385	−26 46 3.88						
	3312	+10 26 39.82	7 59 38.01	− 4 2.19	1.61	−0.08	8 3 41.89	1.40
	3489	−26 25 55.84						
	3312	+10 26 39.82	8 10 16.28	+ 6 36.04	1.46	+0.13	8 3 41.57	1.72
	3638	−26 47 12.39						
	3385	−26 46 3.88	8 5 15.20	+ 1 31.42	0.60	+0.03	8 3 44.35	1.06
	3457	+10 35 33.47						
	3457	+10 35 33.47	7 55 11.18	− 8 30.51	1.01	−0.17	8 3 42.87	0.42
	3489	−26 25 55.84						
	3457	+10 35 33.47	8 5 49.46	+ 2 7.78	0.86	+0.04	8 3 42.50	0.79
	3638	−26 47 12.39						
	3693	+14 50 8.04	8 10 45.84	7 4.57	−1.21	0.14	8 3 42.34	0.95
	3774	−31 11 39.73						
	3928	−31 11 20.95	8 6 56.23	+ 3 10.06	+0.25	+0.06	8 3 45.86	2.57
	3992	+14 57 28.49						
	3928	−31 11 20.95	−7 58 9.34	− 5 36.02	+0.25	−0.11	−8 3 45.22	1.93
	3995	+15 15 2.28						

Latitude of observing station, Pernambuco, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
May 5	4042	−25 14 6.29	−7 54 51.30	− 8 55.71	+3.37	−0.15	−8 3 43.79	0.50
	4072	+ 9 24 23.70						
	4145	+ 0 0 23.62	7 58 40.28	5 5.41	1.41	0.08	8 3 44.36	1.07
	4198	−15 57 44.19						
	4145	+ 0 0 23.62	7 55 2.94	8 42.68	1.41	−0.14	8 3 44.35	1.06
	4211	−15 50 29.50						
	4230	8 47 3.11	8 3 23.85	− 0 21.59	+2.01	0.00	8 3 43.43	0.14
	4257	− 7 19 44.59						
	3227	+ 9 35 6.87	8 14 22.84	+10 41.03	0.00	+0.21	8 3 41.60	1.69
	3243	−26 3 52.56						
	3227	+ 9 35 6.87	8 14 16.04	10 33.88	0.00	0.21	8 3 41.95	1.34
	3248	−26 3 38.96						
	3693	+14 50 8.11	8 10 45.85	7 2.97	−0.30	0.14	8 3 43.04	0.25
	3774	−31 11 39.81						
	3782	+ 0 39 6.11	8 11 2.62	7 18.81	+1.11	0.12	8 3 42.58	0.71
	3883	−17 1 11.34						
	3832	+ 0 35 22.91	−8 12 54.22	+ 9 8.01	+2.26	+0.15	−8 3 43.80	0.51
	3883	−17 1 11.34						
Mean (42 determinations)							−8 3 43.29	±0.11

Latitude of observing station, Bahia, Brazil, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 22	5489	−18 10 59.03	−13 8 32.96	+8 0.11	−5.30	+0.13	−13 0 38.02	0.91
	5495	− 8 6 6.89						
	5520	+ 2 14 56.57	12 51 32.28	−8 59.63	6.85	−0.18	13 0 38.94	1.83
	5539	−27 58 1.12						
	5548	10 19 19.80	12 56 54.96	−3 42.27	2.25	−0.06	13 0 39.54	2.43
	5781	15 34 30.12						
	5637	10 34 9.37	13 4 19.74	+3 45.50	2.65	+0.06	13 0 36.83	0.28
	5781	15 34 30.12						
	5844	20 58 59.50	12 58 51.98	−1 43.24	2.05	−0.03	13 0 37.30	0.19
	5890	4 58 44.46						
	5866	21 19 42.84	−13 9 13.65	+8 37.85	−2.50	+0.14	−13 0 38.16	1.05
	5890	− 4 58 44.46						

Latitude of observing station, Bahia, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 22	5941	+12 39 3.00	−13 9 32.98	+ 8 59.86	−5.70	+0.19	−13 0 38.63	1.52
	5970	−38 58 8.96						
	5987	21 37 24.00	12 50 36.00	− 9 58.94	1.95	−0.17	13 0 37.06	0.05
	6071	− 4 3 48.00						
	6123	+ 2 31 48.39	12 58 12.10	2 15.05	8.00	0.05	13 0 35.20	1.91
	6127	−28 28 12.58						
	6155	+ 3 58 32.57	12 57 5.92	− 3 28.92	3.30	−0.07	13 0 38.21	1.10
	6209	−29 52 44.42						
	6267	17 52 18.48	13 5 55.94	+ 5 22.82	2.70	+0.09	13 0 35.73	1.38
	6325	8 19 33.40						
	6267	17 52 18.48	13 7 53.74	7 20.73	2.85	0.12	13 0 35.74	1.37
	6367	8 23 28.99						
	6388	4 52 28.88	13 11 25.46	+10 48.92	2.60	+0.18	13 0 38.96	1.85
	6432	−21 30 22.03						
	6460	+ 4 2 58.83	13 0 3.04	− 0 34.58	−0.55	−0.01	13 0 38.18	1.07
	6489	−30 3 4.90						
	6526	5 3 38.46	13 8 12.68	+ 7 35.57	0.00	+0.13	13 0 36.98	0.13
	6548	21 12 46.91						
	6704	24 58 50.74	12 56 20.72	− 4 16.59	+1.80	−0.09	13 0 35.60	1.51
	6736	0 53 50.69						
	6706	25 8 50.42	13 1 20.56	+ 0 42.87	1.80	+0.01	13 0 35.88	1.23
	6736	0 53 50.69						
Aug. 24	5303	22 16 45.12	12 49 51.80	−10 47.67	3.40	−0.22	13 0 36.29	0.82
	5414	3 22 58.49						
	5489	18 10 58.96	13 8 32.89	+ 7 50.47	3.45	+0.13	13 0 38.84	1.73
	5495	− 8 6 6.82						
	5520	+ 2 14 56.68	12 51 32.20	− 9 6.67	2.70	−0.18	13 0 36.35	0.76
	5539	−27 58 1.08						
	5548	10 19 19.70	12 56 54.89	− 3 44.08	3.25	−0.06	13 0 35.78	1.33
	5781	15 34 30.08						
	5637	10 34 9.29	13 4 19.68	+ 3 39.87	+3.15	+0.06	13 0 36.60	0.51
	5781	15 34 30.08						
	5844	20 58 59.48	12 58 51.93	− 1 43.80	−2.70	−0.03	13 0 38.46	1.35
	5890	4 58 44.38						
	5866	21 19 42.82	13 9 13.60	+ 8 37.65	−2.40	+0.14	13 0 38.21	1.10
	5890	− 4 58 44.38						
	5996	+ 4 37 11.72	12 54 10.06	− 6 34.73	+6.70	−0.13	13 0 38.22	1.11
	6115	−30 25 31.84						
	6267	17 52 18.47	−13 5 55.88	+ 5 22.23	−1.70	+0.09	−13 0 35.26	1.85
	6325	− 8 19 33.29						

Latitude of observing station, Bahia, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 24	6267	−17 52 18.47	−13 7 53.68	+ 7 19.38	−1.70	+0.12	−13 0 35.88	1.23
	6367	8 23 28.89						
	6388	4 52 28.78	13 11 25.42	+10 44.64	+1.80	+0.18	13 0 38.80	1.69
	6432	21 30 22.05						
Aug. 28	5844	20 58 59.43	12 58 51.83	− 1 53.70	8.25	−0.03	13 0 37.31	0.20
	5890	4 58 44.23						
	5866	21 19 42.77	13 9 13.50	+ 8 28.01	8.15	+0.14	13 0 37.20	0.09
	5890	− 4 58 44.23						
	5941	+12 39 3.46	13 9 32.42	+ 8 49.36	4.90	+0.18	13 0 37.98	0.87
	5970	−38 58 8.29						
	5996	+ 4 37 12.10	12 48 37.78	−12 5.15	3.70	−0.24	13 0 39.47	2.36
	6074	−30 14 27.65						
	5996	+ 4 37 12.10	12 54 9.91	6 31.28	2.25	0.13	13 0 39.07	1.96
	6115	−30 25 31.92						
	6155	+ 3 58 32.96	12 57 5.82	− 3 34.24	+3.60	−0.07	13 0 36.53	0.58
	6209	−29 52 44.60						
	6267	17 52 18.45	13 5 55.81	+ 5 23.18	−3.95	+0.09	13 0 36.49	0.62
	6325	8 19 33.17						
	6267	17 52 18.45	13 7 53.58	− 7 21.72	4.25	0.12	13 0 35.99	1.12
	6367	8 23 28.70						
	6388	4 52 28.61	13 11 25.36	+10 47.44	0.35	+0.18	13 0 38.09	0.98
	6432	−21 30 22.12						
	6460	+ 4 2 59.40	13 0 2.90	− 0 33.52	−2.00	−0.01	13 0 38.43	1.32
	6489	−30 3 5.20						
	6501	3 52 14.33	12 53 35.90	− 7 3.16	+2.00	−0.12	13 0 37.18	0.07
	6507	21 54 57.48						
	6526	5 3 38.10	13 8 12.56	+ 7 37.70	−1.95	+0.13	13 0 36.68	0.43
	6548	21 12 47.01						
	6618	1 6 46.91	12 55 35.68	− 5 3.54	+3.15	−0.10	13 0 36.17	0.94
	6633	24 44 24.45						
	6704	24 58 50.97	12 56 20.62	− 4 15.83	−1.00	−0.09	13 0 37.54	0.43
	6736	0 53 50.27						
	6706	25 8 50.60	13 1 20.44	+ 0 44.05	1.00	+0.01	13 0 37.38	0.27
	6736	− 0 53 50.27						
	6811	+ 0 41 58.30	12 57 30.63	− 3 5.00	1.30	−0.06	13 0 36.99	0.12
	6823	−26 36 59.56						
	6811	+ 0 41 58.30	12 54 35.83	5 59.70	−1.10	0.12	13 0 36.75	0.36
	6842	−26 31 9.96						
	6894	13 0 9.70	−12 52 29.90	− 8 16.53	+9.00	−0.14	−13 0 37.57	0.46
	6935	−12 44 50.10						

Latitude of observing station, Bahia, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 28	6894	−13 0 9.70	−12 59 6.16	− 1 40.90	+9.00	−0.03	−13 0 38.09	0.98
	6938	12 58 2.61						
	6894	13 0 9.70	12 51 9.73	9 37.33	9.60	0.16	13 0 37.62	0.51
	6956	12 42 9.76						
	6894	13 0 9.70	12 56 24.00	4 21.98	9.60	0.07	13 0 36.45	0.66
	6972	12 52 38.30						
	6894	13 0 9.70	12 57 31.74	3 14.87	9.60	0.05	13 0 37.06	0.05
	6974	12 54 53.79						
	6935	12 44 50.54	12 56 27.22	− 4 14.48	5.10	−0.07	13 0 36.67	0.44
	6991	13 8 3.91						
	6938	12 58 2.77	13 3 3.34	+ 2 21.14	5.10	+0.04	13 0 37.06	0.05
	6991	13 8 3.91						
	6956	12 42 9.76	12 55 6.84	− 5 35.28	5.70	−0.10	13 0 36.52	0.59
	6991	13 8 3.91						
	6972	12 52 38.30	13 0 21.10	− 0 19.94	5.70	0.00	13 0 35.34	1.77
	6991	13 8 3.91						
	6974	12 54 53.79	13 1 28.85	+ 0 47.18	5.70	+0.01	13 0 35.96	1.15
	6991	13 8 3.91						
Aug. 29	5303	22 16 44.92	12 49 51.63	−10 52.77	8.30	−0.22	13 0 36.32	0.79
	5414	3 22 58.34						
	5844	20 58 59.42	12 58 51.80	− 1 55.12	9.70	−0.03	13 0 37.25	0.14
	5890	4 58 44.19						
	5866	21 19 42.76	13 9 13.48	+ 8 26.46	9.70	+0.14	13 0 37.18	0.07
	5890	− 4 58 44.19						
	5941	+12 39 3.52	13 9 32.33	+ 8 51.83	3.25	+0.19	13 0 37.06	0.05
	5970	−38 58 8.18						
	5996	+ 4 37 12.13	12 48 37.77	−12 0.58	0.80	−0.24	13 0 37.79	0.68
	6074	−30 14 27.67						
	5996	+ 4 37 12.13	12 54 9.90	6 29.47	1.90	0.13	13 0 37.60	0.49
	6115	−30 25 31.94						
	6123	+ 2 31 48.82	12 58 11.96	2 31.74	6.55	0.05	13 0 37.20	0.09
	6127	−28 28 12.74						
	6155	+ 3 58 33.03	12 57 5.80	− 3 37.93	6.20	−0.07	13 0 37.60	0.49
	6209	−29 52 44.63						
	6267	17 52 18.45	13 5 55.80	+ 5 12.22	7.50	+0.09	13 0 35.99	1.12
	6325	8 19 33.14						
	6267	17 52 18.45	13 7 53.55	7 10.40	7.95	0.12	13 0 35.08	2.03
	6367	8 23 28.65						
	6388	4 52 28.56	−13 11 25.34	+10 42.54	+4.05	+0.18	−13 0 38.57	1.46
	6432	−21 30 22.13						

Latitude of observing station, Bahia, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 29	6460	+ 4 2 59.46	−13 0 2.89	−0 39.08	+4.50	−0.01	−13 0 37.48	0.37
	6489	−30 3 5.24						
	6501	3 52 14.28	12 53 35.89	−7 0.50	0.60	−0.12	13 0 35.91	1.20
	6507	21 54 57.50						
	6526	5 3 38.07	13 8 12.55	+7 32.11	3.25	+0.13	13 0 37.06	0.05
	6548	21 12 47.03						
	6618	1 6 46.85	12 55 35.67	−5 1.23	0.75	−0.10	13 0 36.25	0.86
	6633	−24 44 24.49						
	6811	+ 0 41 58.36	12 57 30.62	3 12.37	5.20	0.06	13 0 37.85	0.74
	6823	−26 36 59.61						
	6811	+ 0 41 58.36	12 54 35.82	6 7.49	5.20	0.12	13 0 38.23	1.12
	6842	−26 31 10.01						
	6894	13 0 9.69	12 56 24.00	4 21.32	9.45	0.07	13 0 35.94	1.17
	6972	12 52 38.30						
	6894	13 0 9.69	12 57 31.74	3 13.95	+9.45	−0.05	13 0 36.29	0.82
	6974	12 54 53.78						
	6972	12 52 38.30	13 0 21.10	−0 13.03	−0.95	0.00	13 0 35.08	2.03
	6991	13 8 3.90						
	6974	12 54 53.78	−13 1 28.84	+0 54.35	−0.95	+0.01	−13 0 35.43	1.68
	6991	−13 8 3.90						
Mean (76 determinations)							−13 0 37.11	±0.09

Latitude of observing station, Rio de Janeiro, Brazil, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 17	5758	−21 23 49.36	−22 46 28.99	−7 49.61	−3.52	−0.14	−22 54 22.26	2.22
	5827	24 9 8.62						
	5907	23 52 9.75	22 51 18.30	3 1.74	3.02	0.05	22 54 23.11	1.37
	5954	21 50 26.84						
Aug. 18	5758	21 23 49.35	22 46 28.98	−7 53.96	1.00	−0.14	22 54 24.08	0.40
	5827	24 9 8.62						
	5844	20 58 59.31	22 55 51.66	+1 31.96	3.92	+0.02	22 54 23.60	0.88
	5851	24 52 44.01						
	5876	24 3 48.30	22 50 37.36	−3 46.15	0.61	−0.06	22 54 24.18	0.30
	5987	21 37 26.42						
	5907	23 52 9.75	−22 51 18.30	−3 7.32	−0.40	−0.05	−22 54 26.07	1.59
	5954	−21 50 26.84						

Latitude of observing station, Rio de Janeiro, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 18	6078	— 9 45 27.08	—22 49 52.92	— 4 24.38	— 7.14	—0.08	—22 54 24.52	0.04
	6103	35 54 18.76						
	6078	9 45 27.08	22 53 37.26	— 0 38.76	7.14	—0.01	22 54 23.17	1.31
	6117	36 1 47.43						
	6343	23 36 30.40	23 3 48.69	+ 9 31.73	7.54	+0.16	22 54 24.34	0.14
	6380	22 31 6.98						
	6343	23 36 30.40	22 57 15.87	+ 2 58.84	— 7.54	+0.05	22 54 24.52	0.04
	6407	22 18 1.34						
Aug. 19	5758	21 23 49.33	22 46 28.97	— 8 5.59	+ 9.05	—0.14	22 54 25.65	0.17
	5827	24 9 8.61						
	5844	20 58 59.30	22 55 51.66	+ 1 18.59	8.35	+0.02	22 54 24.70	0.22
	5851	24 52 44.01						
	5876	24 3 48.30	22 57 7.57	+ 2 37.85	6.04	+0.04	22 54 23.64	0.84
	5954	21 50 26.84						
	5907	23 52 9.75	22 51 18.30	— 3 15.11	6.04	—0.05	22 54 27.42	2.94
	5954	21 50 26.84						
	6078	9 45 27.04	22 49 52.92	4 36.28	2.62	0.08	22 54 26.66	2.18
	6103	35 54 18.80						
	6078	9 45 27.04	22 53 37.26	— 0 49.32	2.62	—0.01	22 54 23.97	0.51
	6117	36 1 47.48						
	6179	20 45 48.67	23 7 33.46	+13 8.39	2.82	+0.22	—22 54 22.03	2.45
	6263	25 29 18.26						
	6179	20 45 48.67	23 2 55.05	8 28.17	2.82	0.13	22 54 23.93	0.55
	6273	25 20 1.43						
	6247	20 36 21.92	23 2 50.09	8 24.70	2.92	0.14	22 54 22.33	2.15
	6263	25 29 18.26						
	6247	20 36 21.92	22 58 11.68	3 44.41	2.92	0.06	22 54 24.29	0.19
	6273	25 20 1.43						
	6343	23 36 30.42	23 3 48.71	9 21.10	2.52	0.16	22 54 24.93	0.45
	6380	22 31 7.00						
	6343	23 36 30.42	22 57 15.89	+ 2 48.81	2.52	+0.05	22 54 24.51	0.03
	6407	22 18 1.36						
Aug. 20	5758	21 23 49.32	22 46 28.96	— 8 8.13	15.59	—0.14	22 54 21.64	2.84
	5827	24 9 8.61						
	5876	24 3 48.30	22 57 7.57	+ 2 32.37	11.07	+0.04	22 54 24.09	0.39
	5954	21 50 26.84						
	5876	24 3 48.30	22 50 37.36	— 4 0.86	11.07	—0.06	22 54 27.21	2.73
	5987	21 37 26.42						
	6078	9 45 27.01	—22 49 52.93	— 4 46.03	+13.08	—0.09	—22 54 25.97	1.49
	6103	—35 54 18.85						

Latitude of observing station, Rio de Janeiro, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Correctinos.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Aug. 20	6078	— 9 45 27.01	−22 53 37.27	−1 0.55	+13.08	−0.02	−22 54 24.76	0.28
	6117	36 1 47.53						
	6343	23 36 30.44	23 3 48.72	+9 18.00	5.03	+0.16	22 54 25.53	1.05
	6380	22 31 7.01						
	6343	23 36 30.44	−22 57 15.90	+2 45.34	+ 3.82	+0.04	−22 54 26.70	2.22
	6407	−22 18 1.37						
	Mean (29 determinations)						−22 54 24.48	±0.19

Latitude of observing station, Montevideo, Uruguay, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
July 8	4377	−47 48 59.68	−35 8 38.62	+13 59.79	−1.76	+0.28	−34 54 40.31	3.40
	4395	22 28 17.56						
	4430	19 17 43.55	34 58 22.57	3 42.14	1.51	0.07	34 54 41.87	1.84
	4443	50 39 1.59						
	4435	19 17 57.86	34 58 29.72	+ 3 50.30	1.51	+0.08	34 54 40.85	2.86
	4443	50 39 1.59						
	4586	35 38 50.04	34 44 50.85	− 9 54.25	1.36	−0.16	34 54 46.62	2.91
	4603	33 50 51.66						
	4603	33 50 51.66	34 50 21.95	− 4 22.17	2.46	−0.07	34 54 46.65	2.94
	4620	35 49 52.24						
	4657	24 22 55.78	34 56 31.22	+ 1 50.24	1.81	+0.04	34 54 42.75	0.96
	4734	45 30 6.65						
	4657	24 22 55.78	34 59 7.18	4 27.32	2.67	0.09	34 54 42.44	1.27
	4779	45 35 18.59						
	4663	24 25 13.03	34 57 39.84	2 58.44	1.81	0.06	34 54 43.15	0.56
	4734	45 30 6.65						
	4663	24 25 13.03	35 0 15.81	+ 5 35.49	2.67	+0.11	34 54 42.88	0.83
	4779	45 35 18.59						
	4734	45 30 6.65	34 52 47.38	− 1 57.35	2.06	−0.04	34 54 46.83	3.12
	4767	24 15 28.12						
	4767	24 15 28.12	−34 55 23.36	+ 0 39.76	−2.92	+0.01	−34 54 46.51	2.80
	4779	−45 35 18.59						
	Mean (11 determinations)						−34 54 43.71	±0.50

Latitude of observing station, Buenos Ayres, Argentine Republic, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
July 19	4767	−24 15 27.81	−34 28 0.78	−8 31.85	+ 3.65	−0.14	−34 36 29.12	0.52
	4768	44 40 33.74						
	4767	24 15 27.81	34 31 12.13	−5 21.72	4.02	−0.09	34 36 29.92	0.28
	4793	44 46 56.45						
	5023	21 57 7.67	34 43 13.87	+6 40.98	5.65	+0.11	34 36 27.13	2.51
	5049	47 29 20.07						
July 20	4882	26 8 23.54	34 36 29.06	−0 8.62	8.20	0.00	34 36 29.48	0.16
	4892	43 4 34.58						
	4911	27 51 16.68	34 44 15.16	+7 34.84	9.66	+0.13	34 36 30.53	0.89
	4928	41 37 13.64						
	5089	16 17 40.06	34 38 52.16	+2 14.20	8.60	+0.04	34 36 29.32	0.32
	5159	53 0 4.27						
	5100	16 11 38.72	34 35 51.50	−0 46.71	8.80	−0.01	34 36 29.42	0.22
	5159	53 0 4.27						
	5112	16 26 31.60	34 43 17.94	+6 39.18	8.85	+0.13	34 36 29.78	0.14
	5159	53 0 4.27						
	5222	45 1 55.08	34 36 8.32	−0 29.81	9.55	−0.01	34 36 28.59	1.05
	5253	24 10 21.57						
	5229	44 55 15.59	34 32 48.58	3 51.10	9.55	0.07	34 36 30.20	0.56
	5253	24 10 21.57						
	5253	24 10 21.57	34 35 38.54	0 58.28	9.05	0.02	34 36 27.79	1.85
	5360	45 0 55.50						
	5383	19 8 8.16	34 27 4.84	9 34.34	7.04	0.19	34 36 32.33	2.69
	5404	49 46 1.53						
	5383	19 8 8.16	34 29 53.06	6 45.52	7.04	0.13	34 36 31.67	2.03
	5425	49 51 37.97						
	5382	19 8 45.76	34 27 23.64	9 13.49	7.04	0.18	34 36 30.27	0.63
	5404	49 46 1.53						
	5382	19 8 45.76	34 30 11.77	6 24.67	7.04	0.12	34 36 29.52	0.12
	5425	49 51 37.78						
	5442	49 17 5.41	34 31 10.00	−5 33.21	11.17	−0.11	34 36 32.15	2.51
	5467	19 45 14.59						
July 21	5089	16 17 40.03	34 38 52.19	+2 18.74	4.22	+0.04	34 36 29.19	0.45
	5159	53 0 4.35						
	5100	16 11 38.68	34 35 51.52	−0 47 92	7.54	−0.01	34 36 31.91	1.27
	5159	53 0 4.35						
	5112	16 26 31.56	34 43 17.96	+6 39.98	8.45	+0.13	34 36 29.40	0.24
	5159	53 0 4.35						
	5222	45 1 55.13	34 45 27.28	8 47.82	8.55	0.17	34 36 30.74	1.10
	5286	24 28 59.42						
	5286	24 28 59.42	−34 44 57.50	+8 16.68	+ 8.55	+0.16	−34 36 32.11	2.47
	5360	−45 0 55.58						

Latitude of observing station, Buenos Ayres, Argentine Republic, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1878.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
July 21	5442	−49 17 5.50	−34 31 10.04	−5 32.75	+12.27	−0.11	−34 36 30.63	0.99
	5467	19 45 14.58						
	5498	26 9 49.33	34 39 32.42	+2 56.70	5.43	+0.05	34 36 30.24	0.60
	5542	43 9 15.52						
	5498	26 9 49.33	34 30 22.08	−6 11.24	3.97	−0.11	34 36 29.46	0.18
	5649	42 50 54.84						
	5513	26 16 28.52	34 42 52.02	+6 15.18	5.53	+0.11	34 36 31.20	1.56
	5542	43 9 15.52						
	5513	26 16 28.52	34 33 41.68	−2 52.76	4.07	−0.05	34 36 30.42	0.78
	5649	42 50 54.84						
July 22	4882	26 8 23.48	34 36 29.04	0 12.56	11.87	0.00	34 36 29.73	0.09
	4892	43 4 34.60						
	4928	41 37 13.65	34 36 4.25	0 31.94	7.59	−0.01	34 36 28.61	1.03
	4940	27 34 54.85						
	5100	16 11 38.65	34 35 51.54	−0 48.58	8.45	−0.01	34 36 31.68	2.04
	5159	53 0 4.43						
	5112	16 26 31.52	34 43 17.98	+6 37.57	8.75	+0.13	34 36 31.53	1.89
	5159	53 0 4.43						
	5197	24 20 8.28	34 41 1.74	4 17.83	15.09	0.08	34 36 28.74	0.90
	5222	45 1 55.19						
	5222	45 1 55.19	34 45 27.30	+8 52.64	5.43	+0.17	34 36 29.06	0.58
	5286	24 28 59.42						
	5229	44 55 15.71	34 32 48.64	−3 55.11	15.49	−0.07	34 36 28.33	1.31
	5253	24 10 21.56						
	5253	24 10 21.56	34 35 38.61	−1 5.43	15.39	−0.02	34 36 28.67	0.97
	5360	45 0 55.66						
	5286	24 28 59.42	34 39 58.21	+3 12.34	16.80	+0.06	34 36 29.01	0.63
	5323	44 50 57.00						
	5383	19 8 8.12	34 27 4.91	−9 32.87	11.37	−0.19	34 36 26.60	3.04
	5404	49 46 1.70						
	5383	19 8 8.12	34 29 53.04	6 46.53	11.47	0.13	34 36 28.23	1.41
	5425	49 51 37.97						
	5382	19 8 45.72	34 30 11.84	6 26.81	11.47	0.12	34 36 27.30	2.34
	5425	49 51 37.97						
	5442	49 17 5.59	34 31 10.08	−5 28.27	6.84	−0.11	34 36 31.62	1.98
	5467	19 45 14.57						
	5498	26 9 49.34	34 39 32.46	+2 48.01	17.40	+0.05	34 36 27.00	2.64
	5542	43 9 15.58						
	5513	26 16 28.53	−34 42 52.06	+6 7.77	+17.40	+0.11	−34 36 26.78	2.86
	5542	−43 9 15.58						
Mean (41 determinations)							−34 36 29.64	±0.16

Latitude of observing station, Para, Brazil, from zenith telescope observations, Lieut. J. A. Norris, U. S. N., observer.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	v.
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Oct. 2	6653	+ 0 6 7.34	−1 28 0.85	+ 1 7.54	− 1.75	+0.02	−1 26 55.04	2.58
	6679	− 3 2 9.04						
	6805	+10 7 7.82	1 25 27.64	− 1 23.99	6.40	−0.02	1 26 58.05	0.43
	6938	−12 58 3.09						
	6805	+10 7 7.82	1 22 45.34	4 7.64	6.40	0.08	1 26 59.46	1.84
	6972	−12 52 38.50						
	6805	+10 7 7.82	1 23 53.06	3 0.10	6.40	0.06	1 26 59.62	2.00
	6974	−12 54 53.94						
	7134	−18 33 34.69	1 25 55.50	0 57.18	− 5.50	0.02	1 26 58.20	0.58
	7200	+15 41 43.70						
Oct. 3	6574	+21 21 23.28	1 25 16.17	− 1 50.68	+ 6.35	−0.03	1 27 0.53	2.91
	6636	−24 11 55.62						
	6589	+21 10 57.26	1 30 29.18	+ 3 25.33	6.45	+0.07	1 26 57.33	0.29
	6636	−24 11 55.62						
	6744	+17 12 6.38	1 25 23.78	− 1 39.52	6.55	−0.03	1 26 56.78	0.84
	6760	−20 2 53.93						
Oct. 4	7200	+15 41 44.12	1 20 25.78	− 6 40.33	9.15	−0.13	1 26 57.09	0.53
	7249	−18 22 35.68						
	7302	+ 6 42 41.62	1 33 31.40	+ 6 24.83	10.70	+0.10	1 26 55.77	1.85
	7415	− 9 49 44.41						
	7618	−14 6 51.80	1 18 8.12	− 8 57.95	9.10	−0.15	1 26 57.12	0.50
	7641	+11 30 35.55						
	7723	+ 5 36 36.82	1 23 2.87	4 2.28	7.10	0.07	1 26 58.12	0.50
	7773	− 8 22 42.56						
	7723	+ 5 36 36.82	1 24 19.98	2 45.22	7.15	0.04	1 26 58.09	0.47
	7784	− 8 25 16.78						
	7795	1 59 22.26	1 18 37.26	8 30.61	8.55	0.14	1 26 59.46	1.84
	7832	0 37 52.25						
	7795	1 59 22.26	1 21 40.99	5 26.86	8.50	0.09	1 26 59.44	1.82
	7868	− 0 43 59.72						
	7943	+11 33 38.51	1 19 53.72	− 7 13.56	8.75	−0.12	1 26 58.65	1.03
	7954	−14 13 25.94						
Oct. 8	7031	−18 36 13.46	1 33 20.32	+ 6 14.73	10.80	+0.12	1 26 54.67	2.95
	7149	+15 29 32.81						
	7031	−18 36 13.46	1 27 14.67	+ 0 8.29	10.85	0.00	1 26 55.53	2.09
	7200	+15 41 44.12						
	7042	−18 12 32.44	1 21 29.82	− 5 37.72	10.80	−0.11	1 26 56.85	0.77
	7149	+15 29 32.81						
	7042	−18 12 32.44	−1 15 24.16	−11 44.16	+10.85	−0.23	−1 26 57.70	0.08
	7200	+15 41 44.12						

Latitude of observing station, Para, Brazil, &c.—Continued.

Date.	B. A. C. No. of star.	Apparent declination.	Half sum of declinations.	Corrections.			Latitude.	*v.*
				Micrometer.	Level.	Ref.		
1879.		° ′ ″	° ′ ″	′ ″	″	″	° ′ ″	″
Oct. 8	7134	−18 33 34.93	−1 32 1.06	+5 2.15	+2.70	+0.10	−1 26 56.11	1.51
	7149	+15 29 32.81						
	7134	−18 33 34.93	1 25 55.40	−1 4.29	2.75	−0.02	1 26 56.96	0.66
	7200	+15 41 44.12						
	7723	+ 5 36 37.04	1 23 2.81	3 59.15	3.40	0.07	1 26 58.63	1.01
	7773	− 8 22 42.66						
	7723	+ 5 36 37.04	1 24 19.92	2 40.16	+3.40	0.04	1 26 56.72	0.90
	7784	− 8 25 16.88						
	7795	1 59 22.22	1 18 37.22	8 19.75	−0.05	0.14	1 26 57.16	0.46
	7832	0 37 52.22						
	7795	1 59 22.22	1 21 40.96	5 16.24	−0.35	0.09	1 26 57.64	0.02
	7868	− 0 43 59.70						
	7943	+11 33 38.80	1 19 53.69	7 8.29	+5.80	0.14	1 26 56.32	1.30
	7954	−14 13 26.18						
	8002	−30 6 20.88	−1 20 6.58	−6 59.84	+6.25	−0.15	−1 27 0.32	2.70
	8032	+27 26 7.73						
	Mean (28 determinations)						−1 26 57.62	±0.20

RECAPITULATION OF RESULTS.

LISBON.

The latitude of the Royal Observatory, as determined by the director, Capt F. A. Oom, R. P. N., is N. 38° 42′ 31″.25.

The telegraphic difference between the prime meridian at Greenwich and the Lisbon station (page 63) is $36^m\ 44^s.64$.

The center of the dome of the observatory is 13.7 meters west of the meridian of the observing station.

To reduce the longitude of the observing station to that of the observatory, a correction of $+0^s.04$ is necessary.

Applying this correction, the result is—

Center of dome of Royal Observatory,* { Lat., N. 38° 42′ 31″.25 / Lon., W. $0^h\ 36^m\ 44^s.68$ / Or in arc, 9° 11′ 10″.2

FUNCHAL, MADEIRA.

The latitude of the Funchal station, as determined from 37 observations of pairs of stars, is N. 32° 38′ 4″.45.

Longitude of the Lisbon station	$36^m\ 44^s.64$
Telegraphic difference of longitude between the Lisbon and Madeira stations	$30^m\ 50^s.95$
Longitude of the Madeira station	$1^h\ 7^m\ 35^s.59$

The flag-staff of Fort St. Jago was 37 feet south and 32.5 feet east of the transit pier.

To reduce the position of the station to that of the flag-staff, a correction to the latitude of — 0″.37 and to the longitude of $-0^s.03$ is required.

Applying these corrections, the result is—

Flag-staff of Fort St. Jago,† { Lat., N. 32° 38′ 4″.08 / Lon., W. $1^h\ 7^m\ 35^s.56$ / Or in arc, 16° 53′ 53″.4

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* The accepted longitude of the observatory heretofore has been 9° 9′ 2″.1.

† The flag-staff at Fort St. Jago is 3,100 feet east of the meridian of the English consulate at Funchal, the point to which the longitude has been generally referred. Applying the difference in longitude, 36″.4 would make the longitude of the consulate 16° 54′ 30″, instead of 16° 54′ 45″, adopted by the English and determined, in 1823, by Dr. Tiarks, who measured by transporting 26 chronometers from Falmouth, England, and back again.

PORTO GRANDE, CAPE VERDE ISLANDS.

The latitude of the station at Porto Grande, as determined from 42 observations of pairs of stars, is N. 16° 53′ 19.″74.

Longitude of the Madeira station	1h 7m 35s.59
Telegraphic difference of longitude between the Madeira and Porto Grande stations	32m 21s.98
Longitude of Porto Grande station	1h 39m 57s.57

The flag-staff of the Brazilian Submarine Company's station is 38 feet north and 125 feet east of the center of the transit pier, necessitating a correction of + 0″.38 to the latitude and — 0s.09 to the longitude to reduce the position of the station to that of the flag-staff.

Applying these corrections, the result is—

Flag-staff in front of Brazilian Submarine Tel. Co.'s office, { Lat., N. 16° 53′ 20″.12 / Lon., W. 1h 39m 57s.48 / Or in arc, 24° 59′ 22″.3

PERNAMBUCO.

The latitude of the Pernambuco station, as determined by 42 observations of pairs of stars, is S. 8° 3′ 43″.29.

Longitude of the Porto Grande station	1h 39m 57s.57
Telegraphic difference of longitude between the Porto Grande and Pernambuco stations	39m 31s.39
Longitude of the Pernambuco station	2h 19m 28s.96

From the transit pier the center of the light-house near Fort Picão bore north 40° 11′ 30″ east (true), distant 2,779.5 feet.

To reduce the position of the station to that of the light-house, a correction of — 21″.08 to the latitude and of — 1s.19 to the longitude is required.

Applying these corrections, the result is—

Light-house near Fort Picão,* { Lat., S. 8° 3′ 22″.21 / Lon., W. 2h 19m 27s.77 / Or in arc, 34° 51′ 56″.55

* Admiral Mouchez, of the French Navy, determined the position of this light-house as—
Lat., S. 8° 3′ 31″
Lon., W. 34° 51′ 38″

BAHIA.

The latitude of the Bahia station, as determined from 76 observations of pairs of stars, is S. 13° 0′ 37″.11.

Longitude of the Pernambuco station	$2^h\ 19^m\ 28^s.96$
Telegraphic difference of longitude between the Pernambuco and Bahia stations	$14^m\ 39^s.45$
Longitude of the Bahia station	$2^h\ 34^m\ 8^s.41$

From the transit pier the center of the San Antonio light-house bore south 62° 32′ 30″ east, distant 59.74 feet.

To reduce the position of the station to the center of the light-house, a correction to the latitude of + 0″.27 and to the longitude of — 0ˢ.04 is required.

Applying these corrections, the result is—

San Antonio light-house,* { Lat., S. 13° 0′ 37″.38 / Lon., W. $2^h\ 34^m\ 8^s.37$ / Or in arc, 38° 32′ 5″.55

RIO DE JANEIRO.

The latitude of the station at Rio de Janeiro, as determined from 29 observations of pairs of stars, is S. 22° 54′ 24″.48.

Longitude of the Bahia station	$2^h\ 34^m\ 8^s.41$
Telegraphic difference of longitude between the Bahia and Rio de Janeiro stations	$18^m\ 33^s.00$
Longitude of the Rio de Janeiro station	$2^h\ 52^m\ 41^s.41$

The transit pier was 74 feet south (true) of the center of the dome of the observatory.

To reduce the position of the station to that of the observatory, a correction of — 0″.73 to the latitude is necessary, the longitude being the same.

Applying this correction, the result is—

Center of dome of Imperial Observatory,† { Lat., S. 22° 54′ 23″.75 / Lon., W. $2^h\ 52^m\ 41^s.41$ / Or in arc, 43° 10′ 21″.15

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*The position of this light-house, as determined by Admiral Mouchez, French Navy, is—

Lat., S. 13° 0′ 47″ Lon., W. 38° 31′ 30″

†The longitude of few places has been so uncertain and so much in dispute as Rio de Janeiro. The point to which the longitude has usually been referred is the flag-staff of Fort Villagagnon which is S. 65° 31′ 51″ east, distant 5,327 feet from the dome of the observatory, or 21″.86 south, and 51″.91 east, making Fort Villagagnon flag-staff—

Lat., S. 22° 54′ 45″.61 Lon., W. 43° 9′ 29″.24

The *Connaissance des Temps* for 1867 publishes the longitude, as determined by M. E. Liais, as — 43° 0′ 43″.5

The same publication for 1878 and subsequent years states it as — 43° 7′ 42″

The English Hydrographic Office have adopted as a secondary meridian — 43° 9′ 0″

MONTEVIDEO.

The latitude of the Montevideo station, as determined from 11 observations of pairs of stars, is S. 34° 54′ 43″.71.

Longitude of the Rio de Janeiro station	$2^h\ 52^m\ 41^s.41$
Telegraphic difference of longitude between the Rio de Janeiro and Montevideo stations	$52^m\ 7^s.53$
Longitude of the Montevideo station	$3^h\ 44^m\ 48^s.94$

From the center of the transit pier the southeast tower of the cathedral bore north 6° 16′ 40″ west (true), distant 1,088 feet.

To reduce the position of the station to that of the cathedral, a correction of — 10″.71 to the latitude and + 0ˢ.08 to the longitude is required.

Applying these corrections, the result is—

Southeast tower of the cathedral,* { Lat, S. 34° 54′ 33″.0 / Lon., W. $3^h\ 44^m\ 49^s.02$ / Or in arc, 56° 12′ 15″.3

BUENOS AYRES.

The latitude of the station at Buenos Ayres, as determined from 41 observations of pairs of stars, is S. 34° 36′ 29″.64.

Longitude of the Montevideo station	$3^h\ 44^m\ 48^s.94$
Telegraphic difference of longitude between the Montevideo and Buenos Ayres stations	$8^m\ 40^s.25$
Longitude of the Buenos Ayres station	$3^h\ 53^m\ 29^s.19$

From the center of the transit pier the vane on the cupola of the custom-house bore S. 87° 2′ 30″ east (true), distant 304 feet.

To reduce the position of the station to that of the custom-house, a correction of + 0″.15 to the latitude and —0ˢ.24 to the longitude is required.

Admiral Mouchez, after an exhaustive discussion of every attainable determination (*Annales Hydrographiques*, 1867), decides upon	$3^h\ 1^m\ 57^s.00$ W. of Paris.	
Paris E. of Greenwich (U. S. Coast Survey Report of 1874, p. 180)	$9^m\ 20^s.97$	
	$2^h\ 52^m\ 36^s.03 =$	43° 9′ 0″.5

* The longitude of the cathedral was determined by M. E. Fleuriais, in 1867, from moon culminations, as 56° 11′ 3″.5.

Applying these corrections, the result is—

Cupola of custom-house,* { Lat., S. 34° 36′ 29″.79 / Lon., W. 3h 53m 28s.95 / Or in arc, 58° 22′ 14″.25

PARA.

The latitude of the station at Para, as determined from 28 observations of pairs of stars, is S. 1° 26′ 57″.62

Longitude of the Pernambuco station - - - - - - - - - - - -	2h 19m 28s.96
Telegraphic difference of longitude between the Pernambuco and Para stations -	54m 31s.21
Longitude of the Para station - - - - - - - - - - - - - - - -	3h 14m 0s.17

From the center of the transit pier the center of the roof of the portico on the north side of the custom-house bore S. 29° 5′ west, distant 225.4 feet.

To reduce the position of the station to the portico of the custom-house, a correction of + 1″.09 to the latitude and — 0s.13 to the longitude is required.

Applying these corrections, the result is—

Portico of custom-house, { Lat., S. 1° 26′ 58″.71 / Lon., W. 3h 14m 0s.04 / Or in arc, 48° 30′ 0″.6

* The position occupied by Dr. Gould during his measurement from Buenos Ayres to Cordova is now covered by the new post-office building, but its site bears from our pier S. 39° 18′ west (true), distant 228 feet.

This bearing and distance would give its position—

Lat., S. 34° 36′ 31″.07
Lon., W. 58° 22′ 17″.74